The V13 Physical Account

V14 · Part I

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Assembly and exposition with Astra

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Sixteen sections and a closing synthesis. Part I states the V13 physical account at its current supported scope.

Abstract

Solvency Field Theory proposes that open causal propagation, persistent recurrent matter, interaction, and geometry can be understood through complete physical configurations of common causal support. This volume presents the current V13 account as one continuous theory: from causal support and the first local standard, through the maintained electron, electromagnetic and quantum relations, formation, species and composites, atoms and chemistry, collective response, gravity, expansion, and physical measurement.

The exposition distinguishes what each sector actually establishes: exact mathematical consequences under stated premises, numerical tests of defined models, effective applications with disclosed inputs, and physical connections that remain open. V13.13 Revision B governs the ontology without replacing the technical sector results. Literal write-address machinery is not part of the current physical account. New dynamical construction beyond the V13 results belongs to V14 Part II.

Reading this volume

The argument follows physical continuity rather than research chronology. Sections 1–7 establish the proposed causal support, maintained particle, local clock and ruler, electromagnetic and quantum relations, and formation. Sections 8–12 develop species, composition, transitions, atoms, chemistry, and collective response. Sections 13–15 carry the same source-and-response discipline into gravity, expansion, propagation, uncertainty, and measurement. Section 16 closes the account and marks the boundary with Part II.

The mathematical status of a statement matters throughout. A definition fixes a quantity. A conditional theorem states what follows from declared premises. An effective application uses stated masses, couplings, or response laws. A numerical test belongs to the representation it actually evolves. An open connection is named where the current V13 sources have not yet supplied the missing dynamics.

The reader is therefore meant to answer two questions at once: **what does SFT think is physically happening?** and **what has the present mathematics actually earned?** The first gives the theory its ontology; the second keeps the claims testable.

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1. One physical world

A particle can remain the same kind of thing while moving, interacting, and becoming part of a larger structure. A clock can repeat without repeating an event. Two participants can possess a relationship that neither possesses alone. Solvency Field Theory begins by asking one question in several forms: **what is the physical configuration, what keeps it persistent, and what changes when it participates in something larger?**

At its particle scope, SFT proposes that light and massive matter are different organizations of common causal support. Open support propagates. A maintained massive particle is a recurrent organization of that support. Its mass, current, framing, response, and clock are readouts and structures of the organized state, not separate mechanisms attached to a passive object. The wider proposal is that this same explanatory order can carry us from particles to relationships and from relationships to the structures that perform measurements. [1, abstract and §§2–4]

This is the governing move of the account: identify the physical configuration first, then ask which facts belong to one participant and which arise only when the configuration contains more than one.

1.1 Persistence is something the configuration does

A particle does not finish forming and then stop doing the work that makes it a particle. Its support continues to propagate; its response continues to maintain the organization; its state continues through its history. Persistence is sustained activity of the configuration.

The maintained-electron model gives that proposal a concrete realization. Its local flow, finite support, source and current, geometry, charged dressing, spin and framing, and recurrence belong to one state. Framing names orientation information carried by the structure; its mathematics comes later. These descriptions do not require a smaller photon traveling inside an independently existing electron container. **The maintained organization is the particle.** [1, §§3.1–3.3]

A collective state can be at rest even while its local support remains active. In the electron model, the complete recurrent route has vanishing integrated translational momentum in its rest description while its local support is modeled as null. The locally propagating support and the history of the whole maintained object are therefore different geometrical objects. A point describing the motion of the whole is not a claim that all of its physical structure occupies that point. [1, §3.3]

This puts the difficult question in the right place. We do not first draw a particle-shaped object and then add a clock, a charge, and a stabilizing container. We ask which complete configuration carries those properties together and can continue to maintain them. Section 3 develops that answer at Electron V1 scope; here we establish the physical thing whose relations the rest of the book will examine.

1.2 What one thing has, and what a relationship adds

A persistent particle can possess invariant content: a mass, a charge or framing class, a spin magnitude, and a local recurrent history. These are properties of the physical state under the transformations allowed by the model. They are not created by another object observing them. [1, §§3.4–3.5, 4.1]

Separation is different. It concerns participants together. So do relative motion, relative orientation, and comparison of their clock readings. An analyzer-relative spin component belongs to the particle and analyzer in their actual arrangement; the particle's spin magnitude and that measured component are different kinds of fact.

One thing can have magnitudes. Two things have relationships. A single recurrent process can still possess a tally and dimensionless invariants. Nor does a relationship wait for a human observer: participants can interact, establish correlations, and alter one another's continuation without anyone watching. Measurement is a particular physical comparison inside that world. [1, §§4.1, 5.5; 2, §3]

Consider two clocks. Each has its own continuing process before any comparison is made. A comparison adds facts about the clocks together and about the connection between them: their paths, exchanged signals, or a reunion at which their records can be placed side by side. No private number carried by one isolated clock already answers every possible comparison.

The rule is simple: **assign a physical quantity to the configuration that makes the quantity defined.** An equation for a particle, its surroundings, and an apparatus together should not be reinterpreted as a hidden property of the isolated particle alone.

1.3 Interaction is one physical problem

When two particles interact, we no longer have two unchanged isolation problems. Their isolated states remain useful references, but the actual continuation belongs to the participants and the response they share. Real fields and geometric response may be part of that configuration; calling force a relational readout does not make those structures unreal. [2, §§1–4]

The V13 interaction architecture expresses this through a complete state X, a lawful domain $\mathcal{D}_J$, and an action S_J:

$$X \in \mathcal{D}_J, \qquad \delta S_J[X] = 0. \tag{1.1}$$

The state contains the variables participating in the problem. The domain specifies their constraints, preparation, and boundaries. The action supplies the dynamics. This architecture is shared; the constitutive terms are supplied by the physical model being solved. [2, §7; 3, §2]

For a separated pair, let $X_A^\star$ and $X_B^\star$ be isolated reference solutions and R their physical separation. A lawful joint branch must approach those same references when the interaction becomes negligible:

$$X_{AB}^\star(R) \longrightarrow X_A^\star \oplus X_B^\star \qquad (R \longrightarrow \infty). \tag{1.2}$$

Here $\oplus$ denotes the separated reference configuration, not a claim that the finite-separation nonlinear state is obtained by simply adding two isolated solutions. At finite separation, source, shared field, support, and motion can respond while the particles retain their identities. Surplus energy can appear as relative motion, field response, radiation, excitation, or an available reaction channel. It need not redefine the intrinsic particle root. [2, §4]

This is why the whole configuration matters. If an external preparation really holds one participant fixed, that preparation belongs to the problem. Otherwise a model cannot freeze a particle at its isolated solution while simultaneously claiming that the joint response changes it.

1.4 A shared energy makes the interaction tangible

A conservative pair gives the most direct example. Evaluate the joint state and the isolated references with the same energy functional and define the interaction by subtraction:

$$\Delta E_{AB}(R) = E[X^{\star}_{AB}(R)] - E[X^{\star}_{A}] - E[X^{\star}_{B}],$$
$$F_R = -\frac{d\Delta E_{AB}}{dR}. \tag{1.3}$$

Energy divided by length has force units. With increasing R chosen outward, the derivative states how the joint energy changes as the participants are separated. The interaction law determines ΔE_{AB}; its slope supplies the conservative force readout. [2, §§2, 5]

The shared field is not counted again merely because its mutual contribution is called an interaction. In the electromagnetic example, one total field responds to the combined current. The isolated self-fields are reference solutions of that same sector, and baseline subtraction isolates the mutual contribution. A complete model must distinguish genuinely additional energy from a second description of energy already counted. [2, §5; 1, §9.4]

Force is not a third substance inserted between two particles. It is a measurable change in how each continues as part of the joint physical configuration. Equation (1.3) is the conservative version of that statement. Scattering, radiation, and reactions require the evolving configuration and its exchanges. Gravitational free fall will later show why the broader joint-continuation idea is more general than proper force alone. [2, §1]

Two named participants also need not be independent. Their joint state can contain restrictions or phase-bearing structure that separate local descriptions do not recover. Classical conservation already provides simple examples of restricted joint possibilities; the quantum section will add the stronger phase-bearing case without confusing the two. [1, §§6.1–6.4]

1.5 A repeated state is not a repeated event

A clock must be repeatable, yet its returns must be successive occurrences. Otherwise it would have no accumulated history to report. SFT therefore distinguishes recurrence of organization from identity of events.

Let $q(s)$ describe a recurrent physical organization, with s an ordering parameter and $P > 0$ a return interval. Write $\sim$ for the physical equivalence appropriate to that state description. If e_n and e_{n+1} label successive returns, then

$$q(s + P) \sim q(s), \qquad e_{n+1} \neq e_n. \tag{1.4}$$

The notation states the distinction; it does not select the period or supply the recurrence dynamics. Returning to an equivalent organization is not returning to the same occurrence. The process has continued between the two returns. [1, §§2.2, 5.1–5.3]

The current meaning of *no-overwrite* follows directly: what occurred cannot later become a different realized event. It does not forbid later activity at the same spatial location, and it does not require a cosmic database, memory surface, or operator that appends reality to the world. Mathematical counts represent continuing history; they are not machines that make events real. [3, §§2–3]

A record is another present physical state. It may preserve a useful distinction about an earlier event, lose that distinction, or be altered later. Those changes affect the record, not the identity of the earlier occurrence. Realized history and accessible evidence are therefore related but different physical ideas.

V13's time proposal gives recurrence a stronger role: normalized local continuation supplies the clock's elapsed-history standard rather than sampling an external universal flow. That proposal will be developed in Section 4. Comparisons between clocks still belong to the physical relation between them. [1, §§5.1–5.3]

1.6 The instruments belong to the same world

A duration or length becomes operational through a physical standard. V13's cones-and-ruler argument separates causal structure from that standard. A causal cone tells us which local directions permit influence; by itself it does not assign a dimensional length or duration to every interval. Persistent recurrent structure supplies the physical process from which a local standard can be made. [4, §§1, 4–6, 9]

For a formed particle of positive rest mass m, the stated mass–cadence correspondence gives the Compton-associated time and length units

$$T_C = \frac{h}{mc^2}, \qquad \ell_C = cT_C = \frac{h}{mc}. \tag{1.5}$$

Here h is Planck action in the cycle convention and c is the causal-speed constant. These relations provide a unit dictionary for the formed branch. A particle's support width, an atomic separation, and a laboratory rod remain different physical lengths whose relations must be supplied by their own configurations. [4, §4; 1, §§5.2–5.5]

Two persistent systems can compare standards and histories. Many systems can maintain an extended arrangement, making rods, detectors, and macroscopic apparatus possible. A ruler's spacing is maintained by the matter that constitutes it; its endpoints do not come from a pre-installed grid outside that matter. The apparatus is therefore part of the theory's explanatory task, not an observer exempted from it. [1, §§4.2, 8.1; 4, §§5–6]

This ladder describes physical dependencies, not a rigid cosmic timetable. Open supports can interact before stable particles exist; the formation program contains a temporary joint regime before a persistent daughter has its own clock. What matters is the direction of explanation: the mature clock, ruler, and particle identity are consequences of persistence and cannot be fed backward as the cause of the same formation. [1, §§3.1, 4, 5.2]

The route through the account is now visible. **Causal support propagates. Some configurations persist through recurrent organization. Persistent participants form physical relationships and larger structures. Those structures eventually make the instruments with which the world is measured.** The model-specific mathematics changes with the physical problem, but the order of explanation does not: the physical configuration comes first.

The next question is what causal propagation supplies before a persistent structure supplies a ruler. That takes us from the physical picture to the first mathematical distinction: the difference between a causal cone and a metrical scale.

Sources for this section

[1] Shaun Fosmark, GPT-5.6 SOL, and Fable. *SFT V13.13: What Exists, What Is Relational, and What Is Bookkeeping — A Provenance-Governed Ontology Constitution for the V13 Stack.* Live Revision B,

September 2026. Main sources here: §§2–9; its status and inheritance rules govern the interpretation. This separately recovered revision is retained alongside, not substituted into, the original folder's first-public draft.

[2] Shaun Fosmark and GPT-5.6 SOL. *SFT V13.4-V2: Relational Interactions — Persistent Particle Roots and Complete Joint Configurations.* Final v1.0, August 2026. Main sources here: §§1–5 and the action architecture in §7. The original V13 folder names this PDF `SFT_V13_4_V2_FORCES_Relational_Interactions.pdf`.

[3] Shaun Fosmark and GPT-5.6 SOL. *SFT V13.8-V2: There Is No Write Operator — From Projector-Compressed Bookkeeping to Lawful Physical Evolution.* Final v1.0, August 2026. Main sources here: §§2–3 and §6.

[4] Shaun Fosmark. *SFT V13.11: The Cones and the Ruler — Where the Scale of the Metric Comes From.* v1.0, August 2026. Main sources here: §§4–6 and §9, read with Revision B's local-scale and proposal distinctions. The original credits Claude/Opus for narrative and argument construction and GPT-5.6 SOL for derivation and audit canon.

Scope. This section states the governing V13 physical interpretation and its opening mathematical architecture. It introduces no additional particle, gravity, cosmological, or detector model.

2. Causal support, cones, and the ruler

A light signal can connect an emission to a reception. A clock can assign a duration to the interval between two events on its history. Those are different physical resources. **Causal structure tells us what can influence what. A physical standard tells us how much interval has elapsed.**

That distinction is the center of V13's cones-and-ruler argument. A causal cone supplies local shape without fixing the scale of non-null intervals. A persistent recurrent process adds a rest description and a repeatable local unit. Comparing such processes makes their standards usable; arranging many of them makes extended apparatus possible. [1, §§1, 4–6; 2, §§4–5]

The argument is local and concrete. Once a Lorentzian description is available, conformal rescaling shows exactly what the cones leave undetermined. The physical proposal is that clocks and rulers need not be primitive ingredients of that description: maintained structures can supply the standards with which the geometry is measured.

2.1 What a cone tells us

An event is a specified occurrence. A causal cone describes the local directions in which influence can continue from it. Take metric signature $(-,+,+,+)$ and length-valued coordinates X^μ, with $X^0 = ct$ in a local inertial chart. For a nonzero infinitesimal displacement,

$$\begin{aligned} &ds^2 = g_{\mu\nu}\, dX^\mu dX^\nu, \\ &ds^2 < 0 \quad \text{timelike}, \\ &ds^2 = 0 \quad \text{null}, \\ &ds^2 > 0 \quad \text{spacelike}. \end{aligned} \tag{2.1}$$

Repeated indices are summed. Timelike directions lie inside the cone, null directions on its boundary, and spacelike directions outside it. A future orientation distinguishes allowed forward continuation from the corresponding past directions. The classification supplies causal type; it does not by itself place a ruler throughout the region. [1, §§1, 3]

In the local inertial form, a null displacement satisfies

$$ds^2 = -c^2 dt^2 + d\mathbf{x}\cdot d\mathbf{x} = 0 \quad \Longrightarrow \quad \frac{|d\mathbf{x}|}{dt} = c, \qquad dt > 0. \tag{2.2}$$

This is the primitive propagation role SFT assigns to causal support. A maintained particle can organize locally null support into a recurrent whole whose integrated motion admits a rest description. The support has not become still; local propagation and the continuation of the complete object are different geometrical statements about one physical configuration. [2, §§2.1, 3.1–3.3]

A cone therefore constrains possible continuation without supplying the complete dynamics. The actual path or field evolution still depends on the governing equations, state, and initial or boundary data.

2.2 The same cones can carry different interval scales

The missing scale appears immediately under a positive conformal rescaling. Retain the same events and coordinate differentials, and replace the metric by

$$\tilde{g}_{\mu\nu} = \Omega^2 g_{\mu\nu}, \qquad \Omega > 0, \qquad d\tilde{s}^2 = \Omega^2 ds^2. \tag{2.3}$$

Because Ω^2 is positive, zero remains zero and the signs of nonzero intervals are unchanged. The local timelike, null, and spacelike classification is therefore identical. The cone structure survives while the scale of non-null intervals can change. [1, §1; 2, §5.4]

For their respective timelike and spacelike displacements,

$$\begin{aligned} d\tau &= \frac{1}{c}\sqrt{-g_{\mu\nu}dX^\mu dX^\nu}, \quad d\tilde{\tau} = \Omega\, d\tau, \\ d\ell &= \sqrt{g_{\mu\nu}dX^\mu dX^\nu}, \qquad d\tilde{\ell} = \Omega\, d\ell. \end{aligned} \tag{2.4}$$

The causal type is unchanged, but proper durations and spatial lengths acquire the scale factor. A null interval stays zero and therefore cannot select that factor for us.

A constant rescaling makes the point visible without any additional geometry:

$$\begin{aligned} d\tilde{s}^2 &= \Omega_0^2(-c^2dt^2 + d\mathbf{x}^2), \\ \text{null direction}: \quad & |d\mathbf{x}|/dt = c, \\ d\mathbf{x} = 0: \quad & \Delta\tilde{\tau} = \Omega_0 \Delta t. \end{aligned} \tag{2.5}$$

with $\Omega_0 > 0$. The coordinate slope of the null ray is unchanged while the stationary proper duration changes. Equation (2.5) compares two metric assignments; it does not say that one already specified physical clock would report both. Its point is narrower and stronger: **the cone fixes causal shape without fixing the physical scale of timelike and spacelike intervals.** [1, §§1, 9–10]

A complete theory must therefore connect metric scale to the physical standards that report it. Changing units consistently is a change of description; changing the metric while holding an independently specified clock law fixed can change a prediction. The cone identity alone does not decide which physical completion is realized.

2.3 Why open light is not its own rest standard

Light participates in measurement, but a freely propagating photon cannot provide the massive rest standard contemplated here. In a local inertial description, write $p^\mu = (E/c, \mathbf{p})$ with $E > 0$. Null momentum gives

$$p_\mu p^\mu = -\frac{E^2}{c^2} + |\mathbf{p}|^2 = 0, \qquad E = c|\mathbf{p}|. \tag{2.6}$$

A photon rest frame would require $\mathbf{p} = 0$, and then (2.6) would require $E = 0$. An invertible Lorentz transformation cannot take a nonzero four-vector to the zero vector. A nonzero photon therefore has no rest frame. [1, §3]

The same distinction appears in proper time. Along a null path, $d\tau = 0$. A parameter may order successive points along the ray, and an observer can measure the ray's energy, frequency, polarization, and phase relations, but that parameter is not the proper-time tally of a massive recurrent clock. [1, §§3–4; 2, §§3.1, 5.3]

This does not make light unreal or historyless. V13 retains open propagation and phase-bearing relations as physical. The missing ingredient is specifically the **resting recurrent standard**. The useful contrast is not light versus matter-as-reality; it is open continuation versus an organization that can return and persist. [2, §§2.1, 12, J-04]

2.4 A persistent process supplies a local unit

A maintained massive structure has the additional possibility of a rest description for the whole while its internal support remains active. For the complete massive branch,

$$P_\mu P^\mu = -m^2c^2 < 0, \qquad P^\mu_{\text{rest}} = (mc, \mathbf{0}), \qquad E_0 = mc^2, \qquad m > 0. \tag{2.7}$$

Here P^μ belongs to the whole object, not to the locally null support tangent. V13's maintained-electron interpretation assigns rest to the complete recurrent configuration and its maintaining response. No separate stationary substance is added to make active support appear at rest. [1, §4; 2, §§3.2–3.4]

Once a formed branch has positive invariant mass, the stated mass-cadence correspondence supplies the local Compton-associated units

$$\boxed{T_C = \frac{h}{mc^2}, \qquad \nu_C = \frac{1}{T_C} = \frac{mc^2}{h}, \qquad \ell_C = cT_C = \frac{h}{mc}.} \tag{2.8}$$

Planck action h uses the cycle convention, so T_C is a period and ν_C is its cycle frequency in local proper time. Multiplying by c gives the associated length unit. These equations translate a formed mass into a clock-and-length dictionary; they do not determine the mass or prove which full-state return realizes the cycle. Those are particle and clock questions developed in the next two sections. [1, §4; 2, §§5.2–5.4]

This is the constructive step in the ruler argument. We have not assigned equal marks to a null ray and called them a clock. We have identified a repeatable physical process that can carry a standard. V13's stronger proposal is that continued recurrence supplies the clock's own elapsed history rather than sampling a separate universal flow. The cone/scale distinction does not require that stronger interpretation to be a theorem. [2, §5.3]

The associated length ℓ_C is likewise a unit, not a rigid electron diameter. Support width, scattering radius, atomic spacing, and laboratory-rod length belong to their own configurations and can be compared in this unit without becoming the same physical length.

2.5 A second standard makes comparison possible

A local unit and a comparison of units are different facts. For two formed species with positive masses m_A, m_B and the same h, c dictionary, (2.8) gives

$$\frac{\nu_{C,B}}{\nu_{C,A}} = \frac{m_B}{m_A}, \qquad \frac{T_{C,B}}{T_{C,A}} = \frac{\ell_{C,B}}{\ell_{C,A}} = \frac{m_A}{m_B}. \tag{2.9}$$

These are local unit ratios. A remotely received frequency additionally depends on emission, propagation, motion, gravity, and reception. Likewise, raw cycle counts become comparable durations only after each is interpreted through its own period. [1, §§4–5; 2, §§4.1, 5.1–5.5]

The intrinsic-versus-relational distinction remains intact. A particle can possess its mass, spin magnitude, or local tally. A comparison adds another standard and the physical operation connecting them. Calibration reports a shared number; it does not create the intrinsic magnitude or explain the particle's formation. [2, §5.5]

Many maintained participants add a further resource: their interactions can preserve separations and produce rods, resonators, reflectors, and detectors. An extended ruler is therefore a collective structure. A constituent process can supply a unit in which its length is reported, but the interactions among the constituents must explain the ruler's spacing and persistence. [1, §§5–6; 2, §4.2]

The apparatus belongs to the same physical world as the thing it measures. Its standards and comparisons are maintained configurations and relations, not an external grid supplied by the observer.

2.6 A local ruler does not determine the metric everywhere

A physical standard can anchor scale where it is used without determining an entire function $\Omega(X)$ across a region. Different scale fields can agree at the anchor and differ elsewhere. Extending the standard requires the actual transport of clocks and signals, together with the source and geometry through which that transport occurs. [1, §9]

This does not make geometry unreal. A metric can describe a vacuum region, and a source can possess local metric response before a second clock arrives. The claim is about how metrical scale becomes physically anchored and compared, not about requiring a material ruler at every point or treating geometry as a deformable medium. [1, §9; 2, §§8.1–8.3]

Another admitted model can contain a dimensional scale directly. For a positive cosmological constant,

$$\Lambda > 0, \qquad [\Lambda] = L^{-2}, \qquad R_\Lambda = \sqrt{\frac{3}{\Lambda}}. \tag{2.10}$$

This supplied de Sitter length does not come from the cones alone. Nor does its existence imply that cosmological and matter-based scales share one microscopic origin. The lesson needed here is simpler: causal shape and dimensional scale are distinct pieces of physical information. [1, §§7, 9–10]

Persistent structure makes a local physical standard available. Consistent geometry across separated standards requires their actual connection. A coordinate convention or a period assigned at one location cannot replace that transport-and-response problem.

2.7 The standard must itself be formed

V13 organizes these dependencies as a ladder. Level 0 contains open null causal support without a massive recurrent standard. Level 1 adds persistent closure and intrinsic content. Level 2 adds relations among persistent participants, including comparison and interaction. Level 3 contains collective arrangements capable of sustaining extended apparatus and macroscopic response. [1, §5; 2, §4]

The ladder is not a rule that open supports cannot interact. Revision B explicitly allows a temporary joint configuration in the transition from Level 0 to Level 1. Such an encounter can respond before a daughter particle owns a persistent rest frame or clock. The direction of explanation is what matters: **formation must produce the organization that later supplies the standard; the mature standard cannot be used as the cause of its own formation.** [2, §§3.1, 4, 5.2]

The route proposal gives that idea a simple kinematic form. Let $\mathcal{L}_{\mathrm{ret}}$ be the length of one complete traversal in an idealized fixed-route rest description where local propagation along that route has speed c. Then

$$T_{\mathrm{ret}} = \frac{\mathcal{L}_{\mathrm{ret}}}{c}, \qquad T_{\mathrm{ret}} = T_C \quad \Longrightarrow \quad \mathcal{L}_{\mathrm{ret}} = \ell_C. \tag{2.11}$$

The first equality is the traversal-time relation in this fixed-route picture. The second identifies that traversal with the Compton-associated return. Neither equation selects a circle, a radius, a support law, or the dynamics that keep the route recurrent. A nonuniform metric or timing law would require its own travel-time relation. [1, §8; 2, §§3.1–3.3, 5.2]

The particle model must therefore contribute more than a closed curve. It must supply a maintained organization whose propagation, source, support, and response continue together. Only then does recurrence become a physical standard rather than a geometric sketch.

The cones supply causal shape. A maintained structure supplies something with which to measure it. The next section asks how that structure holds together.

Sources for this section

[1] Shaun Fosmark. *SFT V13.11: The Cones and the Ruler — Where the Scale of the Metric Comes From.* v1.0, August 2026, physical pp. 1–5. Original credits: Shaun Fosmark, originator and lead author; Claude/Opus, narrative and argument construction; GPT-5.6 SOL, derivation and audit canon. This section develops its local cone/scale argument, formed-state unit, relational ladder, and route proposal, with the limits in §§9–10 retained.

[2] Shaun Fosmark, GPT-5.6 SOL, and Fable. *SFT V13.13: What Exists, What Is Relational, and What Is Bookkeeping — A Provenance-Governed Ontology Constitution for the V13 Stack.* Live Revision B, September 2026. Main sources: §§2–5, §§8.1–8.3, and the null-transfer and scale boundaries in §12. Revision B supplies the current physical interpretation and temporary formation regime; it does not replace technical dynamics.

Scope. This final edit preserves equations (2.1)–(2.11) and their original scientific reach. The conformal examples establish local cone/scale underdetermination; the Compton relations are formed-state unit dictionaries; the de Sitter length is a supplied alternative dimensional input; and the route equation is the stated fixed-route proposal under its timing assumptions. No new physical model, parameter selection, or experimental result is introduced. The local-standard argument does not select a unique global metric or derive a particle's mass.

3. The maintained particle

A closed line on a drawing is not yet a particle. The physical state must remain on a recurrent route, retain finite support, carry its source and framing, and respond to disturbance without losing the organization that makes it the same thing. **The particle is the maintained configuration itself.**

V13.12 turns that claim into a concrete numerical model: **Electron V1**. Its finite source, local geometry, electromagnetic dressing, framing, recurrence, and transverse response are evaluated on one frozen state. The central question here is maintenance: once the configuration exists, what keeps it a persistent particle? Formation must eventually produce that state through its own continuous dynamics; it cannot be used to excuse a failure of the state after formation. [1, pp. 2, 11]

3.1 One state, not a particle plus a container

The source paper collects the electron's physical roles in one state notation:

$$\mathcal{S}_e = \{k^\mu, f_\perp, q_\perp, g_{\mu\nu}, A_\mu, D, \mathcal{H}\}. \tag{3.1}$$

The locally null flow is represented by k^μ. The transverse carrier/source distribution is $f_\perp$; $q_\perp$ is a distinct quadratic response. The metric $g_{\mu\nu}$ describes the geometry in which the state continues, A_μ its electromagnetic connection, and D the transported orientation or *director*. The final entry, $\mathcal{H}$, records recurrence and realized history. It is bookkeeping for what the configuration does, not a separate physical memory device. These entries describe one electron, not seven objects. [1, pp. 4–5]

The model uses coordinates carried with its recurrent worldtube, the region swept out by its finite support. A route coordinate θ labels progress around the spatial circuit so that corresponding cross-sections and orientations can be compared. It is not an independently supplied clock.

Local support continues to propagate while its directional contributions cancel in the complete rest-frame momentum readout. Rest therefore belongs to the organization as a whole, not to microscopic stillness. The same state must satisfy the source, support, metric, electromagnetic, and framing conditions together. The frozen representative has finite numerical residuals: it is a practical numerical solution in the declared representation, not an exact analytic global solution. [1, p. 5]

That is the one-state challenge V13.12 accepts. Mass, spin, charge response, recurrence, and stability are not calculated on separately retuned objects and then declared to describe one electron.

3.2 A finite source with two different profiles

The carrier occupies a finite cross-section rather than an infinitely thin line. V13.12 begins from the nodeless G12U source profile. With dimensionless radius r in the unit ball,

$$f_3(r) = \frac{\sin(\pi r)}{4r}, \qquad 0 < r < 1, \qquad f_3(0) = \frac{\pi}{4}. \tag{3.2}$$

The central value is the continuous limit and the profile vanishes at the outer edge. In the current interpretation this is a finite source profile, not an elementary storage cell. The mature recurrent route is not made by stacking copies of the three-dimensional ball; the profile is projected onto the cross-section transverse to that route. [1, pp. 5, 11]

For dimensionless transverse radius $0 < u < 1$, the projection gives

$$f_\perp(u) = \frac{1}{2}\int_u^1 \frac{\sin(\pi r)}{\sqrt{r^2-u^2}}\,dr,$$
$$q_\perp(u) = 2\int_u^1 \frac{r f_3(r)^2}{\sqrt{r^2-u^2}}\,dr. \tag{3.3}$$

The first distributes the carrier/current source. The second projects the *quadratic* source response, called the settlement margin in the technical model, and is more centrally concentrated. Squaring the already-projected $f_\perp$ would reverse the order of the operations and would not produce the retained $q_\perp$. The amount of source and its density-dependent response therefore need not have the same spatial weighting. [1, pp. 5, 12]

The two normalizations follow directly. Interchanging the radial integrals uses $\int_0^r u\,du/\sqrt{r^2-u^2} = r$; substituting (3.2) gives

$$2\pi\int_0^1 u f_\perp(u)\,du = 4\pi\int_0^1 r^2 f_3(r)\,dr = \pi\int_0^1 r\sin(\pi r)\,dr = 1,$$
$$2\pi\int_0^1 u q_\perp(u)\,du = 4\pi\int_0^1 r^2 f_3(r)^2\,dr = \frac{\pi}{4}\int_0^1 \sin^2(\pi r)\,dr = \frac{\pi}{8}. \tag{3.4}$$

The first fixes the dimensionless amount of compact source; it is not a prediction of charge in coulombs. The second keeps the quadratic response at its own integrated weight instead of renormalizing it into a duplicate carrier profile. [1, p. 12]

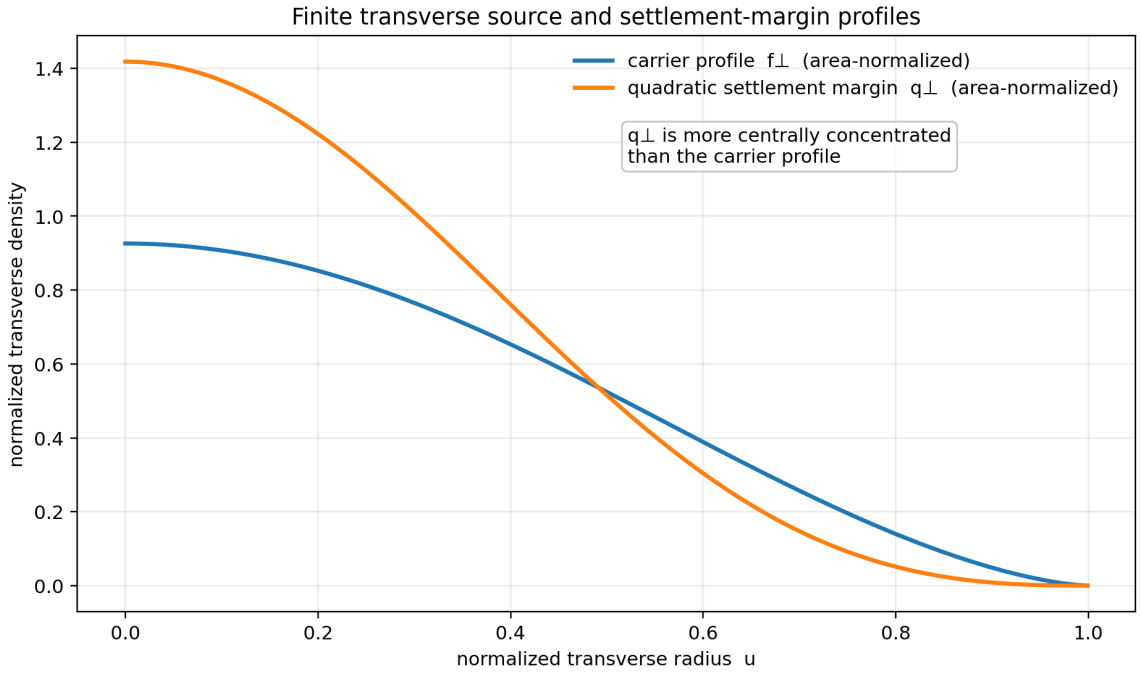


Figure 1: Transverse source profiles reproduced from V13.12, Figure 2. Each curve is separately area-normalized for this shape comparison; their actual integrated weights are the different values in (3.4).

The density response changes with the material geometry. For fixed material measure μ_{mat} and physical-volume Jacobian $J_3 > 0$,

$$n = \frac{\mu_{\mathrm{mat}}}{J_3}, \qquad \epsilon_{\mathrm{margin}} = p_{\mathrm{margin}} = \frac{K_w}{2} n^2 = \frac{K_w \mu_{\mathrm{mat}}^2}{2 J_3^2}. \tag{3.5}$$

Here n is the model's material density and K_w supplies the corresponding energy-density normalization. The quadratic response scales as J_3^{-2}, so compression changes it faster than the linear density. V13.12 checked this law in the solved curved geometry without replacing the profile to improve a residual. [1, p. 12]

3.3 Geometry and support maintain the route together

The finite source participates in the geometry it follows. V13.12 calls the compact turning and branch-maintaining response **closure gravity**, distinguishing it from the dilute macroscopic gravitational readout developed later. The maintained state includes both the local metric and the support reaction. Its metric is solved on a finite correction domain matched to a reference tube; the exterior gravitational field remains a separate matching problem. [1, p. 12]

Support reaction is not a rigid wall around a smaller particle. In the constrained model, the support constraint vanishes on the legal state and its multiplier contributes no independent on-shell action. But the constraint depends on the metric, so metric variation can still produce support stress. A term can therefore help maintain the state without becoming a second positive energy reservoir. Source and reaction belong to the same configuration. [1, p. 13]

The connected scalar-support branch selects a round cross-section, $r_{\mathrm{eq}} = 0$, where r_{eq} is an ellipticity parameter rather than the radius r of (3.2). Roundness concerns the scalar support; the internal director can still turn around it.

For the frozen representative, the supplied support normalization and the resulting chart-level support ratio are

$$\Gamma_\star = 0.1856234976\ldots, \qquad s = 0.2496557386\ldots. \tag{3.6}$$

$\Gamma_\star$ is a formation/support input and s a readout of this maintained representative. The stability calculation asks what happens around that chosen normalization. Selecting it dynamically during formation is a separate problem; the chart parameterization is not an absolute electron-size prediction. [1, pp. 13, 15–16]

The decisive maintenance question is now sharp: **does the coupled response keep a small disturbance bounded?**

3.4 How a rotating response can confine

The confinement mechanism has an analytic control that can be understood before the numerical electron. The C17A companion studies small transverse deviations $\mathbf{u}(\theta) \in \mathbb{R}^2$ from a recurrent route, with primes denoting differentiation in the dimensionless route angle. Its minimal rotating-saddle system is

$$\mathbf{u}'' + K(\theta)\mathbf{u} = 0, \qquad K(\theta) = a_s\, \mathsf{R}(\nu_s\theta) D_0 \mathsf{R}(\nu_s\theta)^T. \tag{3.7}$$

The coefficients a_s and ν_s are in route-angle units; they are not an SI acceleration or clock frequency. The rotation and saddle matrices are

$$\mathsf{R}(\beta) = \begin{pmatrix} \cos\beta & -\sin\beta \\ \sin\beta & \cos\beta \end{pmatrix}, \qquad D_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \nu_s = \frac{1}{2}. \tag{3.8}$$

With fixed axes, positive a_s focuses one transverse direction and defocuses the other: $u_1'' + a_s u_1 = 0$ while $u_2'' - a_s u_2 = 0$. The latter grows exponentially. In the recurrent model, however, the physical support axes turn through π during one 2π spatial circuit. Because the director is headless, opposite representatives describe the same axis. **The defocusing direction does not remain fixed.** [2, §§3–5]

Set $\mathbf{u} = \mathsf{R}(\nu_s\theta)\mathbf{y}$ and $J = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Differentiating twice gives

$$\mathbf{y}'' + 2\nu_s J\mathbf{y}' + (a_s D_0 - \nu_s^2 I_2)\mathbf{y} = 0. \tag{3.9}$$

For $\mathbf{y} = \mathbf{y}_0 e^{\lambda\theta}$,

$$\begin{aligned} 0 &= \det \begin{pmatrix} \lambda^2 + a_s - \nu_s^2 & -2\nu_s\lambda \\ 2\nu_s\lambda & \lambda^2 - a_s - \nu_s^2 \end{pmatrix} \\ &= (\lambda^2 + \nu_s^2)^2 - a_s^2. \end{aligned} \tag{3.10}$$

The two positive oscillation frequencies and the strict bounded interval are therefore

$$\boxed{\omega_-^2 = \nu_s^2 - a_s, \qquad \omega_+^2 = \nu_s^2 + a_s, \qquad 0 < a_s < \nu_s^2.} \tag{3.11}$$

Inside that interval both mode pairs oscillate. At $a_s = \nu_s^2$ one frequency reaches zero; above it a growing mode appears. Rotating coordinates alone cannot create this confinement: if the physical saddle is removed, the original system is free transverse drift. The analytic control therefore isolates the mechanism that the completed electron map is tested against. [2, §§3–5]

3.5 Bounded support without a wall

Stability concerns the complete transverse departure, including position and conjugate momentum. Collect those variables in $\mathbf{z}$, and let the one-cycle monodromy map M advance the perturbation from one return to the next. Its eigenvalues are the Floquet multipliers.

Within the strict rotating-saddle interval, the companion establishes a positive quadratic invariant:

$$\mathbf{z}_{n+1} = M\mathbf{z}_n, \qquad G_\perp = G_\perp^T > 0, \qquad M^T G_\perp M = G_\perp, \qquad I_\perp = \mathbf{z}^T G_\perp \mathbf{z}. \tag{3.12}$$

$G_\perp$ defines the perturbation norm; it is not the spacetime metric or Newton's constant. Applying the map repeatedly gives

$$I_\perp(\mathbf{z}_n) = \mathbf{z}_0^T (M^n)^T G_\perp M^n \mathbf{z}_0 = I_\perp(\mathbf{z}_0). \tag{3.13}$$

If $g_{\min} > 0$ is the smallest eigenvalue of $G_\perp$ in the fixed normalized phase-space coordinates, then

$$g_{\min}\|\mathbf{z}_n\|^2 \le I_\perp(\mathbf{z}_n) = I_\perp(\mathbf{z}_0), \qquad \|\mathbf{z}_n\|^2 \le \frac{I_\perp(\mathbf{z}_0)}{g_{\min}}. \tag{3.14}$$

A finite invariant level therefore encloses a bounded set of transverse positions and momenta. **The confinement is in the dynamics of the occupied state, not in a material wall placed at its edge.** Common translation of the whole maintained structure remains available. [2, §§3–5, 9]

Free transport does not have the same property. For one transverse direction over nonzero route interval $\Delta\theta$,

$$M_{\mathrm{free}} = \begin{pmatrix} 1 & \Delta\theta \\ 0 & 1 \end{pmatrix}, \qquad M_{\mathrm{free}}^{T} G_{\mathrm{free}} M_{\mathrm{free}} = G_{\mathrm{free}} \implies (G_{\mathrm{free}})_{11} = (G_{\mathrm{free}})_{12} = 0. \tag{3.15}$$

Only a semidefinite invariant remains; it cannot bound displacement as well as momentum. This is a successful exclusion of confinement by free transport alone. The maintained model needs a physical responding support, not merely a trajectory drawn closed. [2, §§3–5]

3.6 The frozen electron survives the stability tests

The final V1 audit evaluates the reaction-completed map at the frozen electron state without retuning its structure. Its support readout is

$$\frac{a_s}{\nu_s^2} = 0.0864525425\ldots, \qquad \left(\frac{a_s}{\nu_s^2}\right)^{-1} \simeq 11.567. \tag{3.16}$$

The ratio lies well inside the strict interval. Holding the twist normalization fixed, the inverse is the mean-saddle multiplication needed to reach the analytic boundary. The state therefore has a definite stability margin, not merely a visually closed orbit. [3, §1]

The published one-cycle multiplier pairs are

$$\lambda_{1,2} \simeq 0.99117056 \pm 0.13259307i, \qquad \lambda_{3,4} \simeq 0.99037325 \pm 0.13842265i. \tag{3.17}$$

The displayed values are rounded. In the retained full-precision calculation, the maximum departure of their moduli from one is 6.48×10^{-13}; the positive-invariant residual is 1.72×10^{-13}. Because the perturbation coordinates mix displacement and momentum, the audit tracks the invariant norm rather than an arbitrary Euclidean amplitude. [3, §§2–3]

Direct powers through **100,000 cycles** reach a maximum invariant-norm amplitude of 1.0000000463. Across **105 additional threefold-symmetric profile perturbations**, every tested cycle retains a positive invariant and no unstable cycle is found. This is the transverse stability certificate of the stationary V1 dynamics at its declared scope. [3, §§3–5, 8]

The control also finds instability when the saddle is deliberately overdriven. Multiplication by 11.5 remains inside the interval; multiplication by 12 gives

$$12\frac{a_s}{\nu_s^2} \simeq 1.03743 > 1, \qquad \rho(M) \simeq 1.83639. \tag{3.18}$$

Here $\rho(M)$ is the largest multiplier modulus. Its value above one signals growth. The same stability diagnostic therefore distinguishes the stable branch from the overdriven unstable one. [3, §6]

A second control removes the maintained support/reaction description and launches a free null test geodesic in the frozen metric. It misses the route return, with reduced transverse return norm about

0.79403. **The stable object is the coupled recurrent support, not the free trajectory left after its reaction is removed.** [3, §7]

3.7 Round support can carry a twisted frame

The scalar profile describes where support is distributed; framing describes how its internal orientation is transported. A round cross-section therefore need not have a trivial frame. In V13.12 the director makes a half-twist around the spatial loop, while the oriented spinorial description returns over its 4π cycle. The signed branches $n_g = \pm 1$ reverse framing chirality while retaining conjugate-equal scalar mass-like readouts. [1, p. 13]

The rest-ground assignment is the Gordon/Pauli branch, not a classical charged orbit carrying all of the spin. Its retained leading readouts are

$$L_{\text{ordinary}} = 0, \qquad s_{\text{spin}} = \frac{1}{2}, \qquad \frac{\mu}{q\hbar/(2m)} = 1, \qquad g_{\text{tree}} = 2. \tag{3.19}$$

L_{ordinary} is the ordinary charge-orbital contribution, s_{spin} the dimensionless spin label, q signed electric charge, and μ the leading magnetic-moment readout. The tree-level factor comes from the model's first-order Pauli/Clifford coupling; its operator derivation belongs in the later charge-and-framing section. The point here is source ownership: the leading magnetic response is not supplied by appending a second current to an already complete spin source. [1, pp. 13–14]

The electromagnetic dressing is solved on the curved maintained configuration rather than attached afterward as an unchanged flat-space field. Its normalization uses the external correspondence value $\alpha_{\text{EM}} = 1/137$, which the frozen maintenance tests do not select. The charged field participates in the same source and angular-momentum account as the rest of the state. [1, pp. 14–15]

3.8 Activity without a source that radiates away every lap

A stationary circulating current distribution is not the same source as an accelerating point charge. Electron V1 is stationary in material worldtube coordinates: its profiles recur as one maintained source. Using a time-direction generator ξ,

$$\mathcal{L}_\xi J^\mu = 0, \qquad \mathcal{L}_\xi T_{\mu\nu} = 0. \tag{3.20}$$

J^μ is electric current, $T_{\mu\nu}$ the source stress-energy, and $\mathcal{L}_\xi$ the Lie derivative along ξ. At the maintained finite-domain scope, the ordinary radiative multipole time derivatives vanish. Extending this to a whole-space outgoing electromagnetic field requires the exterior matching named by the source paper. [1, pp. 9, 17]

Stationarity does not mean local support has stopped. It means the organized source is not being replaced each lap by a changing macroscopic charge pattern. Recurrence likewise does not add another copy of the source energy every cycle. The organization persists while its realized history grows. [1, p. 18]

3.9 A clock becomes a readout of the achieved structure

The frozen state also supplies route timing and momentum readouts. In its reference chart,

$$\frac{cT_{\text{cycle}}}{R_{\text{coord}}} = 12.07620577648\ldots, \qquad \frac{L_{\text{proper}}}{R_{\text{coord}}} = 12.06354587887\ldots, \qquad \frac{c|\mathbf{P}|}{E} \simeq 3.45 \times 10^{-15}. \tag{3.21}$$

The factor c in the last ratio restores momentum and energy to common units; the source reports it with $c = 1$. The net spatial momentum is essentially zero at the numerical quadrature precision. R_{coord} is a chart scale. The proper spatial path length differs slightly from cT_{cycle} because lapse and shift enter the null-cycle timing. The clock readout comes from the actual metric traversal, not from substituting a Euclidean circumference. [1, p. 17]

Only after the recurrent solution is specified does the paper apply its mature energy-cycle correspondence:

$$ET_{\text{cycle}} = h, \qquad \frac{ER_{\text{coord}}}{\hbar c} = \frac{2\pi}{cT_{\text{cycle}}/R_{\text{coord}}} = 0.52029465409\ldots. \tag{3.22}$$

The middle equality uses $h = 2\pi\hbar$. This is a post-closure relation in the stated chart, not an invariant-radius prediction or a formation selector. A later dimensional anchor can express the solution in laboratory units. The maintenance result is that the **same state** supplying the finite source and stable transverse response also supplies a common rest/clock readout. [1, pp. 6–7]

3.10 What the maintained model brings forward

Electron V1 places several physical jobs in one configuration. It has a finite current profile and a distinct density-dependent response. Its scalar support is round while its internal frame twists. Local geometry and support reaction participate in recurrence. Its transverse dynamics has a positive invariant and survives the frozen long-cycle and profile tests; the same state supplies the rest, cycle, leading spin, and charged readouts without observable-specific retuning. **That is the one-state result of V13.12.** [1, pp. 17, 19]

Its present inputs and boundaries are equally concrete: the support normalization is supplied, the electromagnetic strength is externally normalized, the dimensional anchor is downstream, and the solved dynamics is finite-domain. Formation must reach the maintained branch; finite-overlap interaction and exterior matching must connect it to other participants and distant fields. A probe-dependent scattering size, for example, must be calculated through an interaction rather than identified with the internal support width by definition. [1, pp. 19–20]

Those are next problems, not reasons to erase the maintained result. **V13.12 has an explicit recurrent particle model in which source, support, framing, charged response, transverse stability, rest, and a clock readout coexist on one frozen state.** The next section follows that continuing structure into local elapsed history: what a cycle counts, how phase becomes duration, and how another clock can compare it.

Sources for this section

[1] Shaun Fosmark and GPT-5.6 Sol; external review by Claude / Fable. *SFT V13.12: The Maintained Electron*, public draft v0.2, August 2026. Complete 24-page paper. Main locations: §§1–10, 11–14 and Appendix A. The original V13-folder PDF and the reproduction pack's public v0.2 PDF are byte-identical in this recovered snapshot. The numerical quantities in this chapter are source-reported C17G19 reference values, not new electron solves.

[2] *IEMM-C17A: Recurrent-Support Optical Invariant and Carrier-Operator Provenance*, v0.1. Technical development companion, especially §§3–5 and 9. Used only for the analytic transverse-transport control and positive-invariant argument. Its historical branch-wide claims are not substituted for the later Electron V1 state.

[3] *IEMM-C17G16: Final Stability Audit of the Frozen Electron V1*, v1.0, §§1–9. Exact copy recovered inside the C17G19 authority ZIP in the V13.12 reproduction pack. Owns the frozen-map, long-cycle, profile, instability-boundary, and free-geodesic numerical reports used here.

Interpretation. V13.13 Revision B and V13.8-V2 govern the reading of recurrent structure, physical support, and realized history without a literal write device.

Equation and computation note. Equations (3.1)–(3.6) present the V13.12 state, source, and parameterization. Equations (3.7)–(3.15) develop the analytic transverse control and its invariant bound. Equations (3.16)–(3.18) are reported frozen-state audit values. Equations (3.19)–(3.22) give the named state's leading branch, stationarity, and post-closure readouts. The displayed algebra was checked in the assembly; this final edit introduces no new electron solve, metric solve, or 100,000-cycle run.

4. Local history and the clock

The maintained particle does not finish its activity when it becomes stable. It keeps returning. Its source and support can recover the same organization while each return is a new occurrence. **The same thing persists, but it has done more.** V13 places the physical meaning of elapsed time in that continuation.

A recurrent structure therefore does more than sit inside an external time coordinate. It supplies a repeatable local process whose completed history can be counted. Another clock can compare that history with its own, but neither needs a universal master clock to authorize the next cycle. This is the local-history proposal of V13.10 under Revision B. [1, §§3–5] [2, §5]

Section 3 supplied the maintained configuration and its cycle readout. Here we make the connection among recurrence, phase, duration, motion, gravity, and measurement explicit. The mass–cadence relation is a post-closure correspondence; the relativistic calculations use the admitted metric. No second time mechanism is added to the particle.

4.1 What returns, and what does not

A repeated position is not enough to define the particle's cycle. The current, support, reaction, and transported frame must return in the sense required by the maintained model. Describe those internal data in a comoving representation by $\mathcal{C}(s)$, where s orders the continuation. If $P > 0$ is the selected return interval and $\mathcal{G}$ contains the model's genuine gauge equivalences,

$$[\mathcal{C}(s+P)]_{\mathcal{G}} = [\mathcal{C}(s)]_{\mathcal{G}}, \qquad e_{n+1} \neq e_n. \tag{4.1}$$

The brackets identify physically equivalent state descriptions. The labels e_n identify successive occurrences. **The state can return while the event does not.** This is the complete-state recurrence used by the maintained-particle account. [3, §1.6] [2, §§2.2–3.3]

A clock hand can point to the same mark after another turn; the position alone does not tell us how many turns occurred. Likewise, Section 3's stationary source can coexist with continuing internal propagation and an increasing history. Stationarity of the source does not erase the activity between returns. [3, §§10.1–10.3, 13]

This distinction matters when the state contains several periodic handles. A director can return as an axis after a half-twist while an oriented representation has a longer return. The scalar clock phase and the signed charge/framing phase therefore remain separate. Their periods cannot be identified merely because 2π or 4π appears in both. The physical state determines what constitutes the clock return. [3, §§1.7, 6] [2, §3.4]

4.2 A period becomes a unit of elapsed history

Take a formed branch with fixed invariant rest mass $m > 0$. In the local rest-energy correspondence,

$$E_0 = mc^2, \qquad T_C = \frac{h}{E_0}, \qquad \nu_C = \frac{1}{T_C} = \frac{mc^2}{h}, \qquad \omega_C = 2\pi\nu_C = \frac{mc^2}{\hbar}. \tag{4.2}$$

Planck action satisfies $h = 2\pi\hbar$. The cycle frequency ν_C has inverse-time units, while ω_C is the same cadence in radians per unit proper time. A faster cadence means a shorter local unit; it does not mean that an external universal time is passing faster. [1, §4, item 3]

The ordering matters. V13.12 first specifies the recurrent state and its cycle timing, then uses the mature relation $ET_{\text{cycle}} = h$. Its lapse and shift contribute to that timing. Here T_C is the local proper-time unit of the formed-clock correspondence. Identifying one complete return with that unit requires the stated rest normalization and cycle marker; a coordinate circumference is not substituted for the clock. [3, §1.6 and §10.1]

Use an unwrapped clock angle ϕ_C increasing along the chosen future direction. From starting event τ_0 and phase $\phi_{C,0}$,

$$\begin{aligned} \phi_C(\tau) - \phi_{C,0} &= \omega_C(\tau - \tau_0), \\ N_C = \frac{\phi_C - \phi_{C,0}}{2\pi} &= \nu_C(\tau - \tau_0), \\ \boxed{\tau - \tau_0 = \frac{N_C}{\nu_C} = T_C N_C.} & \end{aligned} \tag{4.3}$$

The phase and count are dimensionless; the final quantity is a duration. Integer N_C marks completed turns. Between them, the unwrapped phase supplies a continuous description whenever that phase is physically accessible. This is the normalization already contained in V13.10's $\tau = N/\nu_0$, made explicit rather than promoted into a new mass or recurrence law. [1, §4, item 3] [3, §1.6]

Mass and the local clock are two readouts of the maintained branch. The same rest energy fixes the cadence in this correspondence; no separate clock substance is required to make the state age.

4.3 A count, a phase, and a reading

A phase reduced modulo one turn forgets how many turns came before it. With a marker-aligned start and monotone future phase,

$$\begin{aligned} n_C = \lfloor N_C \rfloor, \qquad f_C = N_C - n_C, \qquad 0 \le f_C < 1, \\ \tau - \tau_0 = T_C(n_C + f_C). \end{aligned} \tag{4.4}$$

A counter that reports only n_C discards the within-cycle phase. A phase-sensitive instrument can resolve more, provided its coupling and resolution make that information accessible. A recurrence period therefore supplies a natural unit without forcing every measurable interval to be an integer multiple of it. This is the phase/count distinction retained in place of V13.10's stronger grain-size claim. [5, C02.S01, phase/count clarification]

A physical clock also needs a readable marker. An isolated global action phase is not automatically a flashing signal. A relative phase, transition, current response, or other marker must couple to an apparatus. For a simple calibrated readout with constant angular frequency $\omega_{\text{read}} > 0$ and additive phase error ε_ϕ,

$$\widehat{\Delta\phi} = \omega_{\text{read}}\Delta\tau + \varepsilon_\phi, \qquad \widehat{\Delta\tau} = \frac{\widehat{\Delta\phi}}{\omega_{\text{read}}}, \qquad \widehat{\Delta\tau} - \Delta\tau = \frac{\varepsilon_\phi}{\omega_{\text{read}}}. \tag{4.5}$$

The hats denote reported quantities. This illustrative readout assumes reliable phase unwrapping, and its frequency need not equal ω_C. It shows how a definite physical history can coexist with an imperfect measurement of that history. A missed turn, a drifting cadence, or a disturbed clock requires the corresponding apparatus model. [5, C02.S02, readout clarification]

4.4 More turns need not mean more time

Two species can have different natural cadences. For fixed masses $m_A, m_B > 0$,

$$\frac{\nu_{C,B}}{\nu_{C,A}} = \frac{m_B}{m_A}, \qquad \frac{T_{C,B}}{T_{C,A}} = \frac{m_A}{m_B}, \qquad \Delta N_A T_{C,A} = \Delta N_B T_{C,B} = \Delta\tau \tag{4.6}$$

when the ideal clocks accumulate the same proper duration. A hypothetical branch with twice the mass completes twice as many scalar clock turns in that interval, each assigned half the period. The normalized durations still agree. **A finer cadence is not faster aging.** [1, §4, item 3]

This is also why a larger object need not synchronize every constituent into one giant microscopic cycle. A macroscopic clock can use a selected transition or collective mode while its constituents maintain their own local processes. Elapsed duration is not multiplied by the number of particles present. [2, §§4–5]

The next effect is different: identical clocks can accumulate different normalized durations when they follow different histories.

4.5 Motion changes the compared rate

Use the flat metric from Section 2, with $X^\mu = (ct, \mathbf{x})$ and signature $(-,+,+,+)$. Along a future-directed timelike history,

$$ds^2 = -c^2dt^2 + d\mathbf{x}^2 = -c^2d\tau^2, \qquad v = \left|\frac{d\mathbf{x}}{dt}\right| < c. \tag{4.7}$$

Here $d\tau$ belongs to the moving clock, while t is the inertial comparison coordinate. Dividing by $-c^2dt^2$ gives

$$\frac{d\tau}{dt} = \sqrt{1 - \frac{v^2}{c^2}} = \frac{1}{\gamma}, \qquad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}}, \qquad \boxed{\frac{dN_C}{dt} = \frac{\nu_C}{\gamma}.} \tag{4.8}$$

The local proper-time cadence remains ν_C on the ideal fixed-mass branch. What changes is the number of returns accumulated per unit laboratory coordinate time. This is the relativistic clock correspondence used by V13.10, not a derivation of a boosted finite-electron solution from a center trajectory. [1, §§2–4] [5, C02.S01, motion dictionary]

The same interval can be written

$$v^2 + \left(c\frac{d\tau}{dt}\right)^2 = c^2. \tag{4.9}$$

The second term has speed-squared units but is not motion through a hidden spatial direction. It is the proper-history rate expressed in the comparison coordinate. The locally active support and the timelike motion of the whole object remain different parts of the same physical description. [2, §§3.3, 5.3] [5, C02.S01, speed-budget scope]

For example, if $v = 3c/5$,

$$\gamma = \frac{5}{4}, \qquad \Delta\tau = \frac{4}{5}\Delta t, \qquad \Delta N_C = \frac{4}{5}\nu_C\Delta t. \tag{4.10}$$

A clock at rest in that comparison frame accumulates Δt during the same coordinate interval. A real journey uses the actual path, including changes in motion.

4.6 Moving energy and internal cadence are different derivatives

The moving state has laboratory energy $E = \gamma mc^2$, so $E/h = \gamma\nu_C$. Equation (4.8), however, gives ν_C/γ for recurrence per laboratory time. The two frequencies differ because they belong to different derivatives. [5, C02.S01, observer energy and recurrence]

For a free massive phase correspondence, take

$$\varphi(t, \mathbf{x}) = \frac{Et - \mathbf{p}\cdot\mathbf{x}}{\hbar}, \qquad E = \gamma mc^2, \qquad \mathbf{p} = \gamma m\mathbf{v}, \qquad \left.\frac{\partial\varphi}{\partial t}\right|_{\mathbf{x}} = \frac{E}{\hbar}. \tag{4.11}$$

The partial derivative follows phase at a fixed laboratory position. Following the particle instead gives

$$\begin{aligned} \frac{d\varphi(t, \mathbf{x}(t))}{dt} &= \frac{E - \mathbf{p}\cdot\mathbf{v}}{\hbar} \\ &= \frac{\gamma m(c^2 - v^2)}{\hbar} = \frac{mc^2}{\gamma\hbar} = \omega_C\frac{d\tau}{dt}. \end{aligned} \tag{4.12}$$

Dividing by 2π recovers the recurrence rate. This is a phase dictionary, not an extra wave trapped inside the electron. The proper-time clock, the laboratory energy frequency, and a received signal frequency are different readouts and should not be interchanged.

The same warning applies to composites: a sum of observer energies divided by h is not automatically a sum of constituent proper recurrence rates per observer time.

4.7 Two paths, two completed histories

Let two ideal clocks separate at event A and reunite at event B. In one flat chart, with common endpoint times t_A, t_B,

$$\Delta\tau_i = \int_{t_A}^{t_B}\sqrt{1 - \frac{v_i(t)^2}{c^2}}\,dt, \qquad \Delta N_i = \nu_{C,i}\Delta\tau_i, \qquad i = 1, 2. \tag{4.13}$$

Each clock must remain on its intended branch during transport. Under that ideal-clock assumption, the different readings belong to the different paths. Reunion makes the comparison local again; no distant simultaneity convention is needed to compare the two accumulated records. [1, §§2, 5] [5, C02.S01, path comparison]

For identical fixed-mass standards,

$$\Delta N_1 - \Delta N_2 = \nu_C\int_{t_A}^{t_B}\left[\sqrt{1 - \frac{v_1^2}{c^2}} - \sqrt{1 - \frac{v_2^2}{c^2}}\right]dt. \tag{4.14}$$

If the rates differ, the histories accumulated between the shared endpoints differ. **There is no missing time to store elsewhere.** One clock simply completes fewer normalized returns along its path. [1, §5]

Remote comparison needs an actual signal. For future null momentum p^μ received by an observer with four-velocity U^μ,

$$p_\mu p^\mu = 0, \qquad U_\mu U^\mu = -c^2, \qquad E(U) = -p_\mu U^\mu > 0, \qquad \nu(U) = \frac{E(U)}{h}. \tag{4.15}$$

The photon connects emitter and receiver without carrying the massive rest clock of its own. Its null proper-time interval is zero, while its propagation, polarization, and phase-bearing relations remain physical. Emission, transport, and reception are still distinct events in realized history. [2, §§2–3, J-04] [5, C02.S01, null-observer dictionary]

4.8 Gravity compares the same local process through a different geometry

A gravitational clock comparison changes the geometry through which local clocks and signals are related. To display the roles without selecting a particular source solution, take a static zero-shift metric with positive lapse $N(\mathbf{x})$ and positive spatial metric h_{ij}:

$$ds^2 = -N(\mathbf{x})^2 c^2 dt^2 + h_{ij}(\mathbf{x}) dx^i dx^j, \qquad v_{\mathrm{loc}}^2 = \frac{h_{ij} dx^i dx^j}{N^2 dt^2}. \tag{4.16}$$

A static observer has proper time $N dt$, so v_{loc} is the local spatial speed measured by that observer. The timelike interval gives

$$\frac{d\tau}{dt} = N\sqrt{1 - \frac{v_{\mathrm{loc}}^2}{c^2}} = \frac{N}{\gamma_{\mathrm{loc}}}, \qquad \frac{dN_C}{dt} = \nu_C \frac{N}{\gamma_{\mathrm{loc}}}. \tag{4.17}$$

The lapse and the motion factor act on the same recurrence comparison but describe different physics. A stationary clock has $v_{\mathrm{loc}} = 0$ and can still tick at a different rate relative to the reference coordinate. Gravitational dilation is therefore not the particle center simply moving faster. The gravitational model must supply the lapse; recurrence counting alone does not. [4, §4]

For stationary emitter A and receiver B, suppose successive pulses follow the same time-independent propagation and apparatus delays do not change. Then

$$\delta\tau_A = N_A \delta t, \qquad \delta\tau_B = N_B \delta t, \qquad \boxed{\frac{\nu_{\mathrm{rec,B}}}{\nu_{\mathrm{emit,A}}} = \frac{N_A}{N_B}.} \tag{4.18}$$

Receiving at a larger lapse gives a smaller pulse frequency. The emitter's proper cadence, the receiver's own cadence, and the received signal rate remain distinct. One source-to-geometry law must ultimately account for all of them together with the spatial response seen by light. [5, C02.S02, stationary signal comparison]

The static chart above assumes $N > 0$. A limiting coordinate rate approaching zero does not imply that an infalling clock's local history stops. V13.1-V4 keeps continued infalling recurrence distinct from the distant-time description. [4, §4]

4.9 Lifetime is a distribution, not a preset last lap

An unstable particle sharpens the distinction between a realized history and a population law. Expressing lifetime in cycle units does not determine the cycle on which each individual leaves its maintained branch. We therefore retain the population interpretation rather than V13.10's earlier fixed-expiry-count picture. [1, §3] [5, C02.S02, lifetime clarification]

For an illustrative constant decay rate $\lambda_d > 0$ per proper time, let $D \geq 0$ be the random proper lifetime of a prepared member. Then

$$\Pr(D > \tau) = e^{-\lambda_d \tau}, \qquad \mathbb{E}[D] = \frac{1}{\lambda_d}, \qquad \mathbb{E}[N_D] = \frac{\nu_C}{\lambda_d}, \qquad N_D = \nu_C D. \tag{4.19}$$

The expectation belongs to the stipulated population, and N_D may include a fractional final turn. The rate is supplied for this example; it does not follow from the Compton frequency or the electron's transverse stability. A physical species needs its own departure channels and dynamics. [5, C02.S02, supplied exponential example]

At constant laboratory speed with Lorentz factor γ,

$$D_{\text{lab}} = \gamma D, \qquad \Pr(D_{\text{lab}} > t) = e^{-\lambda_d t/\gamma}, \qquad \mathbb{E}[D_{\text{lab}}] = \frac{\gamma}{\lambda_d}. \tag{4.20}$$

This is the same proper-time lifetime law expressed in laboratory time. The population distribution stretches while the stipulated proper-time rate and proper cadence remain unchanged. Once an individual history occurs, its duration is definite; the population law does not make every endpoint identical.

4.10 The direction of history and the reading of a clock

Every completed return adds another occurrence to the maintained thing's history. Revision B's no-overwrite statement concerns that actuality: what occurred does not later become a different event. Returning to the same position or orientation, or erasing a later record, does not undo the intervening history. [2, §2.2]

A record is a present physical state that preserves accessible information about some of that history. It can be altered, lost, or read imperfectly. The clock's continuation and the apparatus that retains its count are therefore related processes, not the same thing by definition. For the same reason, historical irreversibility alone does not supply a thermodynamic entropy-production or heat law; those require a specified state, exchange, and coarse description. [2, §§2.2, 9] [5, C01.S02, event/record distinction]

The stationary maintained particle also pays no new rest-energy bill simply for completing another turn. Another recurrence is not another copy of the particle's energy. When an interaction changes the particle or its surroundings, the exchanged energy belongs to the changed joint configuration. Persistence and accumulating history are therefore compatible with a stationary source. [3, §§10.3, 13]

One persistent structure is enough for the proposed local clock. A second makes histories comparable; many make the instruments through which those comparisons become usable. A larger structure can contain many clocks without manufacturing time again at its own scale. The central move has already occurred: a recurrent physical process supplies a duration standard from within the world. [2, §§4–5]

V13.12 gives that proposal a maintained-particle setting; the phase dictionary connects recurrence to duration; the metric correspondence connects histories and signals. The remaining clock-specific tasks are physical ones: identify the actual cycle marker, show how an apparatus reads it, and keep that standard intact through the comparison being made.

The particle now carries a local history, but its mass cadence is not its signed charge. **The next section follows the other handle of the same maintained state: framing, charge, and electromagnetic response when another participant is present.** [2, §3.4]

Sources for this section

[1] Shaun Fosmark; narrative by Claude/Opus; derivation and audit canon credited to GPT-5.6 SOL. *SFT V13.10: What a Clock Is Actually Doing — Time as the count of completed history*, August 2026. The local-history proposal, mass normalization, path comparison, and distinction between duration and a universal flow supply the narrative basis.

[2] *SFT V13.13: Live Ontology Constitution*, Revision B, especially §§2.2–3.4, 4–5, and 8–9. It governs the current interpretation of local history, null propagation, distinct mass/charge handles, and no-overwrite without restoring literal recording hardware.

[3] Shaun Fosmark and GPT-5.6 Sol, with external review by Claude/Fable. *SFT V13.12: The Maintained Electron*, public draft v0.2, especially §§1.6–1.7, 6, 10, and 13. Its reference-chart cycle readouts and post-closure $ET_{\mathrm{cycle}} = h$ relation are used at their stated scope.

[4] Claude and Astra; ontology/program by Shaun Fosmark. *Gravity as a Relation Between Two Magnitudes*, V13.1-V4, v1.4, 8 September 2026, §§4–4.2. Its distinction among static lapse, local motion, reference traversal time, and continuing infall supplies the limited gravitational comparison here; its older counting/occupancy mechanism is not adopted in this section.

[5] *SFT Composed Manuscript*, v0.1, C01.S02 and C02.S01–S02, used narrowly as the correction record for phase versus completed turns, apparatus accuracy, observer-energy versus recurrence frequency, and population lifetime.

Scope. This final edit preserves equations (4.1)–(4.20) and the scientific reach of the original assembly. The fixed-mass clock dictionary is post-closure; the readout and exponential lifetime laws are illustrative supplied models; the special-relativistic and static-gravity relations use admitted metrics. No new clock, detector, boost, gravitational source, or lifetime calculation is introduced. The chapter does not retain the older claims of a universal smallest measurable interval, a fixed individual expiry count, or cessation of local infalling history.

5. Charge, framing, and electromagnetic response

The particle that makes a local clock can also enter an electromagnetic encounter. Its charge, spin, current, and response belong to the maintained organization already built; they are not switched on as a second object when another particle approaches. **The same particle that persists in isolation becomes a source inside a larger joint configuration.**

V13.12 and V13.4-V2 make that connection concrete. The frozen electron supplies a normalized charge source and a leading spin/magnetic response. Two unchanged parent specifications then reproduce the leading long-range electric, magnetic, motion-dependent, and spin-orbit interaction classes through one electromagnetic response. No separate magnetic, velocity, or spin-orbit coupling is fitted for the pair. This is the particle-to-interaction result developed here at long-range, nonoverlap scope. [2, §§6–7; 3, §§4–6; 4, §73]

The first step is to keep two periodic properties of the same state distinct: the mass-associated cadence used as a clock, and the signed framing through which the state participates electromagnetically.

5.1 The clock does not determine the charge

The mass cadence of Section 4 advances along a future-directed history. Charge can have either sign. Equal-mass particle-antiparticle branches therefore cannot identify the two variables. With positive charge normalization q_0,

$$m_{e^-} = m_{e^+} = m, \qquad \nu_{C,e^-} = \nu_{C,e^+} = \frac{mc^2}{h}, \qquad q_{e^-} = -q_0, \quad q_{e^+} = +q_0. \tag{5.1}$$

Both branches have positive rest energy and the same cadence. Reversing charge does not reverse elapsed history, and charge alone does not determine mass. V13 therefore keeps mass/cadence and charge/framing as independent handles of the same maintained state. We denote their phases by ϕ_C and θ_g, respectively. [1, §3.4; 2, §§1.8, 6.1]

The framing phase is compact: adding a complete turn returns the same phase. On a specified closed internal route C, choose a continuous lifted angle $\tilde{\theta}_g$ in one framing description. Its degree is

$$\theta_g \sim \theta_g + 2\pi, \qquad n_g = \frac{\tilde{\theta}_g(\text{end}) - \tilde{\theta}_g(\text{start})}{2\pi} \in \mathbb{Z}. \tag{5.2}$$

The start and end are the same point of the internal route, not the same spacetime event. V1 uses signed branches $n_g = \pm 1$ with conjugate-equal scalar mass-like readouts. Equation (5.2) gives integer degree for the declared compact map; it does not select the realized winding or determine q_0. Once frames are compared through a connection, the physical transported phase includes that connection as well. [2, §§1.8, 6.1; 3, §11]

5.2 Framing carries the spin branch

A round scalar distribution can carry nontrivial orientation. In the maintained electron, the director turns around the route while the scalar cross-section remains round. Its leading spinorial framing uses a two-component complex representation with Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{5.3}$$
$$\sigma_i \sigma_j = \delta_{ij} I_2 + i \epsilon_{ijk} \sigma_k.$$

Here $i^2 = -1$, I_2 is the identity, and $\epsilon_{123} = 1$. The matrices specify the first-order representation used by V1. The physical point is that the maintained framing carries the spin branch; spin is not being manufactured by deforming the scalar outline of the support. [2, §§1.7, 6.1–6.3]

For unit axis $\hat{\mathbf{n}}$ and rotation angle β,

$$U(\beta) = e^{-i\beta \hat{\mathbf{n}} \cdot \sigma/2} = I_2 \cos\frac{\beta}{2} - i \hat{\mathbf{n}} \cdot \sigma \sin\frac{\beta}{2}, \tag{5.4}$$
$$U(2\pi) = -I_2, \qquad U(4\pi) = I_2.$$

The exponential follows from $(\hat{\mathbf{n}} \cdot \sigma)^2 = I_2$. A spinorial representative changes sign after one full spatial rotation and returns after two. This rotation is not the clock phase of Section 4 and not a choice of electromagnetic phase zero. Whether the sign becomes observable depends on a phase-sensitive relation with another part of the physical configuration.

The associated spin operators obey

$$\mathbf{S} = \frac{\hbar}{2}\sigma, \qquad \mathbf{S}^2 = \frac{3\hbar^2}{4} I_2, \qquad \operatorname{spec}(\hat{\mathbf{n}} \cdot \mathbf{S}) = \left\{ -\frac{\hbar}{2}, +\frac{\hbar}{2} \right\}. \tag{5.5}$$

Spin one-half, the invariant magnitude encoded by $\mathbf{S}^2$, and a component along an analyzer are different statements. The last requires a relation with the chosen direction. Individual outcome selection belongs to the later measurement problem. [1, §4.1; 2, §6]

5.3 Comparing charge frames

A charge/framing angle at one event cannot be compared with another until the local angle conventions are related. The electromagnetic connection A_μ supplies that comparison. V13 treats it as the response associated with the compact charge/framing sector, not as a physical grid of addresses. [3, §10; 6, R262F, read with 1, §§2–3]

For equations (5.6)–(5.13), use $c = \hbar = 1$, signature $(-,+,+,+)$, and length-valued coordinates. The coefficient q_0 is dimensionless and A_μ has inverse-length units. One consistent convention is

$$\theta'_g = \theta_g + \beta(x), \qquad A'_\mu = A_\mu + \frac{1}{q_0}\partial_\mu \beta, \tag{5.6}$$
$$\partial_\mu \theta'_g - q_0 A'_\mu = \partial_\mu \theta_g - q_0 A_\mu.$$

A smooth change of phase zero therefore changes the connection so that the transported comparison stays invariant. For a representative with charge $q = q_0 Q$,

$$D_\mu = \partial_\mu - iqA_\mu, \qquad \psi'_Q = e^{iQ\beta}\psi_Q, \qquad D'_\mu \psi'_Q = e^{iQ\beta} D_\mu \psi_Q, \tag{5.7}$$
$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \qquad [D_\mu, D_\nu]\psi_Q = -iqF_{\mu\nu}\psi_Q.$$

The curvature is unchanged by the relabeling because mixed derivatives of smooth β commute. It measures the local noncommutativity of transport. Holonomy is the accumulated transport around a closed route. Neither operation should be confused with the spatial spin rotation in (5.4). [3, §10; 6, R262F]

5.4 Curvature has a response law

Gauge freedom constrains a response law but does not select its strength. V13.4-V2 adds the stated premises of smooth local Lorentz-invariant weak-field response, a stable parity-even bulk law, and leading derivative/power order. Under those premises the unbroken connection sector has the Maxwell form

$$S_{\text{field+source}} = \int d^4x \left[-\frac{K_{\text{EM}}}{4} F_{\mu\nu}F^{\mu\nu} + J^\mu A_\mu \right], \qquad K_{\text{EM}} > 0. \tag{5.8}$$

K_{EM} is the dimensionless field stiffness in this natural-unit normalization. J^μ has inverse-length-cubed units. The particle and support terms of the full joint problem accompany this field sector when those variables are varied. Equation (5.8) fixes a leading form, not the numerical stiffness. [3, §10]

Varying A_ν gives

$$\delta S_{\text{field+source}} = \int d^4x \left[\partial_\mu (K_{\text{EM}} F^{\mu\nu}) + J^\nu \right] \delta A_\nu, \tag{5.9}$$

$$\boxed{\partial_\mu (K_{\text{EM}} F^{\nu\mu}) = J^\nu.}$$

The index order follows from the positive $J^\mu A_\mu$ pairing and the chosen metric convention. With $A_\mu = (-\Phi, \mathbf{A})$, $J^\mu = (\rho_q, \mathbf{j})$, $E_i = F_{i0}$, and $B_i = \frac{1}{2}\epsilon_{ijk}F_{jk}$, the electric and magnetic pieces are components of the same response. [4, §11.2; 6, R262B]

Antisymmetry then gives charge continuity:

$$\partial_\nu J^\nu = 0, \qquad \frac{d}{dt}\int_V \rho_q \, d^3x = -\int_{\partial V} \mathbf{j} \cdot d\mathbf{S}. \tag{5.10}$$

Charge leaving a fixed region crosses its boundary. A prescribed current can source a field, while reciprocal recoil and energy-momentum exchange require the corresponding particle equations. [4, §11.2; 6, R262C]

In a flat constant-stiffness region, Lorenz gauge gives

$$\partial_\mu A^\mu = 0, \qquad \Box A^\nu = -\frac{J^\nu}{K_{\text{EM}}}, \qquad \Box = -\partial_t^2 + \nabla^2. \tag{5.11}$$

The vacuum characteristics are null. Retarded initial and boundary conditions select propagation from the past source, together with any independently prepared radiation. The static energy treatment used below is the conservative limit of that field problem, not an instantaneous update rule for moving charges. [6, R262B; 3, §§5–6]

5.5 One electromagnetic strength

A field-coordinate rescaling cannot choose the physical interaction strength. Canonical normalization carries the source coefficient with it:

$$A_\mu^{\rm can} = \sqrt{K_{\rm EM}}A_\mu, \qquad q_0^{\rm can} = \frac{q_0}{\sqrt{K_{\rm EM}}}, \qquad \boxed{\alpha(\mu_0) = \frac{q_0^2}{4\pi K_{\rm EM}} = \frac{(q_0^{\rm can})^2}{4\pi}.} \tag{5.12}$$

The reference scale μ_0 specifies where this normalization applies. Compact framing and the Maxwell form do not determine the ratio. Electron V1 uses $\alpha_{\rm EM} = 1/137$ as a supplied reference, so its electromagnetic amplitude remains an input. [2, §7; 3, §10]

What matters for integration is that the pair receives no second electromagnetic coefficient:

$$\alpha_{\text{electron dressing}}(\mu_0) = \alpha_{\text{pair interaction}}(\mu_0) \equiv \alpha(\mu_0). \tag{5.13}$$

This is the V13.4-V2 particle-interaction matching condition. A changed coupling belongs to a correspondingly changed and resolved particle family; one cannot keep an exact dressed parent fixed while independently retuning the force it sources. **The self-field and the pair field must speak the same electromagnetic language.** [3, §5; 4, §73.6]

5.6 The magnetic factor comes from the same spin response

V1's leading magnetic response comes from its first-order Pauli branch. Restore $\hbar$ and use dimensional potentials for which $q\mathbf{A}$ has momentum units and $q\Phi$ energy units:

$$\Pi_i = -i\hbar\partial_i - qA_i, \qquad [\Pi_i, \Pi_j] = iq\hbar\epsilon_{ijk}B_k, \qquad \mathbf{B} = \nabla \times \mathbf{A}. \tag{5.14}$$

Combining this commutator with the Pauli algebra gives

$$\begin{aligned}(\sigma\cdot\Pi)^2 &= \Pi^2 + \frac{i}{2}\epsilon_{ijk}\sigma_k[\Pi_i, \Pi_j] \\ &= \Pi^2 - q\hbar\sigma\cdot\mathbf{B}.\end{aligned} \tag{5.15}$$

The contraction uses $\epsilon_{ijk}\epsilon_{ij\ell} = 2\delta_{k\ell}$. This is the first-order identity retained and numerically checked by the electron program. It is an operator result of the maintained spin branch, not a classical charge bead orbiting inside the particle. [2, §§1.7, 6.3]

The leading Pauli Hamiltonian is therefore

$$\begin{aligned}H_P &= \frac{(\sigma\cdot\Pi)^2}{2m} + q\Phi = \frac{\Pi^2}{2m} + q\Phi - \frac{q}{m}\mathbf{S}\cdot\mathbf{B}, \\ \mu &= g\frac{q}{2m}\mathbf{S} = \frac{q}{m}\mathbf{S}, \qquad \boxed{g_{\rm tree} = 2.}\end{aligned} \tag{5.16}$$

Tree level is the leading response before higher-order dressing corrections. V13.12 obtains this magnetic moment without assigning the electron's spin to an ordinary charge orbit. Its finite-operator check retains the first-order identity to relative numerical precision about 1.22×10^{-16}. The anomalous magnetic moment is a separate higher-order problem. [2, §§6.2–6.4; 5, §1]

The finite source has a corresponding magnetization current. Let n_{V1} be the compact normalized density and $\hat{\mathbf{s}}$ a uniform spin-polarization direction in the reduced rest-frame example:

$$\begin{aligned} \int n_{\mathrm{V1}}\, d^3x = 1, \qquad \rho_q = q n_{\mathrm{V1}}, \\ \mathbf{M} = \frac{q\hbar}{2m} n_{\mathrm{V1}} \hat{\mathbf{s}}, \qquad \mathbf{j}_{\mathrm{mag}} = \nabla \times \mathbf{M}. \end{aligned} \tag{5.17}$$

For a regular localized profile with vanishing surface terms,

$$\nabla \cdot \mathbf{j}_{\mathrm{mag}} = 0, \qquad \mu = \frac{1}{2}\int \mathbf{x} \times \mathbf{j}_{\mathrm{mag}}\, d^3x = \int \mathbf{M}\, d^3x = \frac{q\hbar}{2m}\hat{\mathbf{s}}. \tag{5.18}$$

The current adds no net charge. The curved maintained-state solve incorporates this selected branch and counts its field contribution in the same angular-momentum/source account. Equivalent stress or angular-momentum descriptions are not extra stocks to append afterward. [5, §§2–3; 2, §§6.4–7.1]

5.7 One field for the pair

The isolated electron already has electromagnetic dressing. Adding a second electron does not mean adding a third field on top of two immutable self-fields. The joint current sources one field configuration, with the isolated dressed electrons supplying the separated references:

$$\delta S_J[X_A, X_B, Y] = 0, \qquad X^\star_{AB}(R) \longrightarrow X^\star_A \oplus X^\star_B \quad (R \longrightarrow \infty). \tag{5.19}$$

Y contains the shared response and a star denotes a solved state. Finite separation can deform the joint field and response while the branch identities of the particles remain intact. [3, §§4–5; 4, §§73.3, 73.7]

The linear electrostatic limit makes the accounting transparent. Let G_C be the symmetric Coulomb response on the fixed boundary problem and $\mathcal{W}$ its quadratic source energy:

$$\begin{aligned} \mathcal{W}[\rho] &= \frac{1}{2}\langle \rho, G_C \rho\rangle, \qquad G_C^\dagger = G_C, \\ \Delta\mathcal{W}_{AB} &= \mathcal{W}[\rho_A + \rho_B] - \mathcal{W}[\rho_A] - \mathcal{W}[\rho_B] \\ &= \frac{1}{2}\big(\langle \rho_A, G_C\rho_B\rangle + \langle \rho_B, G_C\rho_A\rangle\big) = \langle \rho_A, G_C\rho_B\rangle. \end{aligned} \tag{5.20}$$

The mutual energy is the cross term of the same field energy after subtracting the two isolated references. **Self-dressing and interaction are two readings of one field sector.** This static reduction does not replace the causal initial-value problem for changing currents. [3, §5; 4, §73.4]

5.8 The long-range pair

At separations large compared with the finite supports, integrated charge controls the leading electric term. Restore dimensions with $C_{\mathrm{EM}} = \alpha\hbar c$ and write $q_i = q_0 Q_i$, $\mathbf{R} = \mathbf{x}_A - \mathbf{x}_B$:

$$\begin{aligned} U_{AB}(R) &\simeq \frac{C_{\mathrm{EM}} Q_A Q_B}{R}, \\ \mathbf{F}_A = -\nabla_{\mathbf{x}_A} U_{AB} &\simeq \frac{C_{\mathrm{EM}} Q_A Q_B}{R^2}\widehat{\mathbf{R}}, \qquad \mathbf{F}_B = -\mathbf{F}_A. \end{aligned} \tag{5.21}$$

Two electrons have $Q_A Q_B = +1$, so the leading force is repulsive; conjugate charge reverses the sign. The coefficient comes from the integrated sources and the same electromagnetic normalization already used by the parents. No new pair parameter appears. Higher multipoles probe more of the finite source structure. [3, §6; 4, §§31.2, 73.5]

The same frozen Pauli moment generates a direction-sensitive spin-spin interaction:

$$U_{ss} = \frac{C_{\mathrm{EM}}}{m^2 c^2 R^3}\left[\mathbf{S}_A \cdot \mathbf{S}_B - 3(\mathbf{S}_A \cdot \widehat{\mathbf{R}})(\mathbf{S}_B \cdot \widehat{\mathbf{R}})\right]. \tag{5.22}$$

For a product of fully polarized spin states, $\langle \mathbf{S}_i \rangle = (\hbar/2)\hat{\mathbf{s}}_i$. Relative to $U_C = C_{\mathrm{EM}}/R$,

$$\begin{gathered}\frac{\langle U_{ss} \rangle}{U_C} = \frac{\bar{\ell}_{C,e}^{\,2}}{4R^2}\left[\hat{\mathbf{s}}_A \cdot \hat{\mathbf{s}}_B - 3(\hat{\mathbf{s}}_A \cdot \widehat{\mathbf{R}})(\hat{\mathbf{s}}_B \cdot \widehat{\mathbf{R}})\right], \\ \bar{\ell}_{C,e} = \frac{\hbar}{mc}.\end{gathered} \tag{5.23}$$

The bar marks the reduced Compton length. The common coupling cancels from this relative correction. Parallel polarizations along $\widehat{\mathbf{R}}$ give bracket -2; parallel transverse polarizations give $+1$. A genuinely joint spin state requires its pair correlations rather than two substituted mean directions. [4, §31.3]

5.9 Motion and framing modify that same field

Moving sources change both spatial current and retarded response. In the conservative slow-motion expansion of the same Maxwell pair problem, V13.4 obtains the Darwin interaction through order v^2/c^2:

$$L_{AB} = -\frac{C_{\mathrm{EM}}}{R} + \frac{C_{\mathrm{EM}}}{2c^2 R}\left[\mathbf{v}_A \cdot \mathbf{v}_B + (\mathbf{v}_A \cdot \widehat{\mathbf{R}})(\mathbf{v}_B \cdot \widehat{\mathbf{R}})\right]. \tag{5.24}$$

L_{AB} is the pair-interaction part of the Lagrangian, not the complete energy or a radiation-reaction law. The velocity correction contains both current/magnetic and longitudinal-retardation pieces. Its checked transverse and longitudinal boost limits use the same C_{EM}; no velocity-specific coupling is introduced. [3, §6; 4, §31.4]

Transporting spin between moving rest frames adds the Thomas/Wigner correction. With the same leading $g = 2$ moment and the equal-mass center-of-momentum convention of the audit,

$$H_{\mathrm{SO}}^{\mathrm{COM}} = -\frac{3C_{\mathrm{EM}}}{2m^2 c^2 R^3}\mathbf{L}_{\mathrm{rel}} \cdot (\mathbf{S}_A + \mathbf{S}_B). \tag{5.25}$$

$\mathbf{L}_{\mathrm{rel}} = \mathbf{R} \times \mathbf{p}$ is relative orbital angular momentum; it is not the electron's internal charge-orbital term. The source obtains the factor 3/2 from the partner-current contribution and the Thomas/Wigner half correction under the stated kinematics. No additional spin-orbit strength is fitted. Contact terms and fully relativistic scattering remain finite-joint problems. [4, §31.5]

The point is not four unrelated correspondences. **The same frozen charge source, the same leading spin response, and one electromagnetic normalization generate the leading electric, magnetic, motion, and spin-orbit classes together.** That makes the integration test stronger than matching Coulomb's law alone. [3, §6; 4, §73.5]

5.10 Close contact asks for more of the finite structure

Long-range fields are governed first by integrated source quantities. Once finite supports overlap, the relative arrangement of charge, current, and framing can matter directly. The older gear intuition survives only at this local scope: as a candidate framing-compatibility cost, not as invisible teeth acting across a Coulomb separation.

Let Δ_g be the relative compact framing angle in the common overlap description. The retained exchange-even periodic cost has expansion

$$\begin{aligned} V_{\mathrm{ov}}(\Delta_g) &= \sum_{n\geq 1} \kappa_n [1-\cos(n\Delta_g)], \\ V_{\mathrm{ov}}^{(1)} &= \kappa_g[1-\cos\Delta_g] = \frac{\kappa_g}{2}\Delta_g^2 + O(\Delta_g^4), \qquad \kappa_g > 0. \end{aligned} \tag{5.26}$$

For a reduced energy cost, the κ_n have energy units. The even cosine form expresses exchange symmetry and the small-angle limit gives a restoring candidate. The actual overlap law must determine its coefficient and transported reference angle. V13.4 preserves this term as a finite-overlap candidate; it is not the origin of α, the cause of long-range Coulomb response, or an extra isolated-electron mass spring. [3, §18]

Rotating one particle's spinorial frame is also not the same operation as exchanging two identical particles. The long-range electromagnetic results require no choice of exchange sign. The physical composition and statistics problem remains separate. [3, §18; 4, §§32–33]

5.11 One particle, one electromagnetic sector

The V13 result can now be stated compactly. The maintained electron supplies its charge, leading Pauli moment, and electromagnetic dressing. Two unchanged copies source one joint field whose cross response produces the leading long-range interaction classes without parent retuning or separate pair couplings. [3, §6; 4, §73]

Finite overlap, radiation reaction, and probe-dependent form factors require deeper joint calculations. A scattering radius must be obtained from an actual interaction and readout rather than identified by definition with the internal support width. Those are extensions of the same physical account, not conditions for recognizing what the long-range integration has already established. [3, §19; 2, §12]

The same maintained organization can supply a clock when followed locally and an electromagnetic interaction when it participates jointly. Mass cadence, signed framing, spin, and magnetic response retain their different meanings while belonging to one state. The force is the participant-side consequence of the shared response, not another substance inserted between the particles.

A pair can also retain phase-bearing structure after direct mechanical overlap ends. Section 6 asks what can remain in that relationship: **quantum relations, distinguishability, and measurement.** [1, §6; 3, §3]

Sources and scope

[1] Shaun Fosmark, GPT-5.6 SOL, and Fable. *SFT V13.13 — What Exists, What Is Relational, and What Is Bookkeeping*, Revision B. Governing physical interpretation, especially §§2–4 and 6–9.

[2] Shaun Fosmark and GPT-5.6 SOL. *SFT V13.12 — The Maintained Electron*, Public Draft v0.2. Especially §§1.7–1.8, 6–7, and 12. The first-order identity and charged-state results retain the source's model scope and inputs.

[3] Shaun Fosmark and GPT-5.6 SOL. *SFT V13.4-V2 — Relational Interactions: Persistent Particle Roots and Complete Joint Configurations*, Final v1.0. Primary authority for the electromagnetic integration, especially §§4–6, 10–11, and 18–19.

[4] *SFT V13.4 — Relational Interaction Update Blend v0.11.* Controlling technical companion named by [3], especially §§11.2, 31–33, and 73.

[5] *IEMM-C17G8 — Native Pauli–Belinfante Worldtube Lift, Gordon Partition, and Local Charged Root*, v0.2. Used only for the retained current and angular-momentum partition in §§1–3; earlier reduced charged-energy thresholds are not imported into frozen V1.

[6] *SFT V13.4 — Forces Full Derivation.* R261F, R262F, R262B, and R262C provide ancestry for the two-phase, connection, and field arguments. The replacement [3] governs present scope. Literal address/write machinery is not retained, and the positive source-coupling convention used in (5.8)–(5.11) is kept internally consistent.

Section scope. This section edits and exposes the V13 particle-to-electromagnetic integration. No new electron, finite-overlap pair, scattering, or force solve is claimed. The algebraic identities and the inherited long-range results retain their original evidence and inputs.

6. Quantum relations, distinguishability, and measurement

Two particles can separate until their direct mechanical interaction is negligible and still retain physical structure that neither particle possesses alone. **Ending the force is not necessarily ending the relation.** In SFT, that is where the distinction between a thing and what belongs to things together becomes a quantum question.

V13.9-V3 develops this idea in a deliberately small domain: a normalized binary pure response, clean local transport, branch-preserving environmental marking, ideal binary analyzers, and an optional completion-rate law. Within that domain it gives exact relations among local and joint response capacities, an exact redistribution identity when alternatives become distinguishable, and the singlet angular response with Bell-strength correlations and remote-setting-independent marginals. [2, §§1–2; 3, §§6–14]

The maintained particles remain physical participants. The new content belongs to their joint configuration. A detector enters the same way: as another physical participant that supplies a comparison, interacts, and can retain a record. No human observer is required to make the relation or the interaction real. [1, §§4.1, 6–7]

6.1 What the pair possesses

An isolated particle can possess invariant mass, signed charge, and spin magnitude. A spin component along an analyzer axis is different because the axis is supplied by another physical structure. A pair can likewise possess constraints and phase-bearing structure that are absent from either local description taken alone.

A fixed total momentum already gives a classical example of joint constraint: the two descendant momenta are not freely assignable independently. Quantum structure requires more than that. V13.9 therefore keeps both the lawful joint possibilities Ω_{AB} and their relative phase information Φ_{AB}:

$$\mathfrak{C}_{AB} = (\Omega_{AB}, \Phi_{AB}), \qquad \Delta\phi = \frac{S[\Gamma_1] - S[\Gamma_2]}{\hbar} \pmod{2\pi}. \tag{6.1}$$

The second expression is the retained action-phase correspondence for massive alternative histories Γ_1, Γ_2. Transport and holonomy can also contribute to relative phase. These phases compare alternatives of the joint process; they are not a substance stretched between the particles and do not assign a proper-time clock to light. [2, §5; 3, §§4–5]

A common origin can explain how a relation was prepared, but shared past information is not automatically enough to reproduce every later joint response. V3 treats the present phase-bearing relation as part of the physical configuration that reaches the analyzers. [2, §§3, 14]

6.2 A small chart for a genuinely joint state

Take two response alternatives at each participant, with orthonormal local bases $|0\rangle_A, |1\rangle_A$ and $|0\rangle_B, |1\rangle_B$. The four complex coefficients of the pure joint response are arranged as

$$|\Psi_Z\rangle = \sum_{i,j=0}^{1} Z_{ij} |i\rangle_A \otimes |j\rangle_B, \qquad Z = \begin{pmatrix} Z_{00} & Z_{01} \\ Z_{10} & Z_{11} \end{pmatrix}, \qquad \mathrm{Tr}(ZZ^\dagger) = 1. \tag{6.2}$$

The tensor product describes paired alternatives; the dagger is complex-conjugate transpose. The trace condition is simply $\sum_{ij} |Z_{ij}|^2 = 1$. The matrix is a coordinate for the relation, not a hidden material sheet connecting the particles. [2, §§5–6; 3, §§4, 8]

A pure pair factorizes exactly when its coefficient matrix has rank one:

$$Z_{ij} = u_i v_j \quad \Longleftrightarrow \quad Z = \mathbf{u}\mathbf{v}^T \quad \Longleftrightarrow \quad \operatorname{rank} Z = 1 \quad \Longleftrightarrow \quad \det Z = 0. \tag{6.3}$$

A rank-two matrix therefore cannot be replaced by one local state for A multiplied by one local state for B. This is the model's precise binary notion of a nonfactorizable pure relation. [2, §§6–7]

The local descriptions alone do not determine which such relation is present. For example,

$$Z_{\pm} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & 1 \\ \pm 1 & 0 \end{pmatrix}, \qquad G_A^{(+)} = G_A^{(-)} = \frac{I_2}{2}, \qquad G_A = ZZ^\dagger. \tag{6.4}$$

Both states give the same local Gram matrix on either wing, yet they differ by a relative sign in the joint amplitudes. A phase-sensitive pair comparison can distinguish them while either local description cannot. **The particles can each be complete physical objects without their separate descriptions being a complete description of the pair.** [2, §§5–6, 11]

6.3 One finite response budget

Write the local Gram matrix and three nonnegative readouts as

$$G_A = ZZ^\dagger = \begin{pmatrix} a & b \\ b^* & d \end{pmatrix}, \qquad a + d = 1, \qquad P = |a - d|, \quad V = 2|b|, \quad C = 2|\det Z|. \tag{6.5}$$

P measures local bias in the chosen basis, V local phase accessibility, and C the pure binary factorization defect. Local basis changes can move weight between P and V; their relation to C is fixed by the determinant identity:

$$\begin{aligned} P^2 + V^2 + C^2 &= (a-d)^2 + 4|b|^2 + 4(ad - |b|^2) \\ &= (a+d)^2 = 1, \end{aligned} \tag{6.6}$$

$$\boxed{P^2 + V^2 + C^2 = 1.}$$

This is the binary pure-response triality. A rank-one pair has $C = 0$ and its normalized budget is entirely local. A maximally nonfactorizable pair has $C = 1$ and $P = V = 0$: neither wing carries a local preference even though the pair has a definite relation. The source's random-chart check found a largest numerical residual of 1.8×10^{-15}; the algebra explains the identity itself. [2, §6]

At this stage the norm is a response capacity, not yet an observed event probability. The detector bridge comes later.

6.4 Separation does not erase a relation

After preparation, suppose each participant undergoes clean local transport. V3 writes

$$Z_{ab} = L_A Z_J L_B^T, \qquad L_A, L_B \text{ invertible} \quad \Longrightarrow \quad \operatorname{rank} Z_{ab} = \operatorname{rank} Z_J. \tag{6.7}$$

Invertible local maps preserve rank, so a nonfactorizable pure relation remains nonfactorizable even after direct mechanical overlap becomes negligible. The relation needs no persistent force or material cable to remain part of the joint description. [2, §7; 3, §6]

A stronger numerical invariance requires a stronger premise. Taking determinants gives

$$\det Z_{ab} = \det L_A \, \det Z_J \, \det L_B, \qquad L_i^\dagger L_i = I_2 \quad \Longrightarrow \quad C_{ab} = C_J. \tag{6.8}$$

Unitary local transport preserves both the norm and the numerical value of C. General invertibility preserves rank but need not preserve normalized capacity after renormalization. Keeping those statements separate prevents distance from being treated as an automatic decoherence law.

A relation changes when some physical process changes it. Elapsed time by itself is not that process. The model can be parameterized by laboratory time, proper times, or another valid evolution coordinate without introducing a universal present. [1, §§5.1, 6.2; 2, §7]

6.5 Distinguishing alternatives changes accessibility

Now let an environment respond differently to the two alternatives. In V3's diagonal pure-root chart,

$$\begin{aligned} Z_0 &= \begin{pmatrix} \alpha_0 & 0 \\ 0 & \beta_0 \end{pmatrix}, \qquad |\alpha_0|^2 + |\beta_0|^2 = 1, \\ |\Psi_0\rangle &= \alpha_0|00\rangle + \beta_0|11\rangle, \\ |\Psi_{ABE}\rangle &= \alpha_0|00\rangle|e_0\rangle + \beta_0|11\rangle|e_1\rangle. \end{aligned} \tag{6.9}$$

The environment states $|e_0\rangle, |e_1\rangle$ may represent a scattered photon, a recoiling atom, or detector degrees of freedom. The enlarged state remains coherent; the marking simply correlates different physical responses with the alternatives. No observer has to inspect them. [2, §§2, 8; 3, §9]

Their overlap controls how much pair-only phase accessibility remains:

$$\gamma_E = \langle e_1|e_0\rangle_G, \qquad 0 \le |\gamma_E| \le 1, \qquad C_0 = 2|\alpha_0\beta_0|, \qquad A_{AB} = C_0|\gamma_E|. \tag{6.10}$$

If the environment responses differ only by phase, $|\gamma_E| = 1$ and no branch distinction has been created. If they are orthogonal, $\gamma_E = 0$ and the pair-only phase accessibility vanishes. What matters is not "observation" in the abstract but whether the interaction leaves physically distinguishable responses. [2, §8]

6.6 The inaccessible part belongs to the larger configuration

V3 also tracks the complementary response retained when the environment is included. Its weighted environment Gram matrix is

$$G_E^{(w)} = \begin{pmatrix} |\alpha_0|^2 & \alpha_0^*\beta_0\gamma_E \\ \alpha_0\beta_0^*\gamma_E^* & |\beta_0|^2 \end{pmatrix}, \qquad \det G_E^{(w)} = |\alpha_0\beta_0|^2(1-|\gamma_E|^2). \tag{6.11}$$

and the complementary enlarged-configuration readout gives

$$\begin{aligned} X_{AB:E} &= 2\sqrt{\det G_E^{(w)}} = C_0\sqrt{1-|\gamma_E|^2}, \\ A_{AB}^2 + X_{AB:E}^2 &= C_0^2|\gamma_E|^2 + C_0^2(1-|\gamma_E|^2), \\ \boxed{C_0^2 = A_{AB}^2 + X_{AB:E}^2.} \end{aligned} \tag{6.12}$$

This is the Binary Relational Accessibility Redistribution Theorem in its stated domain: a normalized binary pure root, branch-preserving marking, normalized environment responses, and coherent enlarged state. The quantities are dimensionless response capacities, not fragments of matter or energy transported from one particle to another. [2, §9; 3, §10]

At one endpoint the environment carries no distinguishing information and the full capacity remains directly accessible to the pair. At the other, direct pair accessibility vanishes while the complementary enlarged-configuration readout carries the same initial magnitude. For a maximal root with $|\gamma_E| = 3/5$, the two readouts are $3/5$ and $4/5$: their squares still sum to one.

The model therefore says something stronger than "coherence got smaller." It identifies where the inaccessible response appears in the enlarged configuration. It does not claim a universal entanglement measure for arbitrary mixed or multipartite states.

6.7 Repeated encounters, surviving correlations, and restoration

Independent branch-preserving marks multiply their overlaps:

$$A_N^2 = C_0^2\prod_{k=1}^{N}|\gamma_k|^2, \qquad X_N^2 = C_0^2\left(1-\prod_{k=1}^{N}|\gamma_k|^2\right). \tag{6.13}$$

The increment from the nth mark is

$$\Delta X_n^2 = A_{n-1}^2(1-|\gamma_n|^2), \qquad A_n^2 + \Delta X_n^2 = A_{n-1}^2, \qquad A_N^2 + X_N^2 = C_0^2. \tag{6.14}$$

so the squared capacities telescope exactly. Equal overlap magnitude $r < 1$ gives $A_N = C_0 r^N$; converting that sequence into a decay rate in physical time requires an encounter process and a clock. The reported single-mark and ten-mark numerical closures are implementation checks, not a microscopic collision law. [2, §9; 3, §11]

Loss of direct phase accessibility is not the same thing as complete statistical independence. At full marking, the equal-weight example reduces to

$$\begin{aligned} \rho_{AB}^{\mathrm{cl}} &= \tfrac{1}{2}(|00\rangle\langle 00| + |11\rangle\langle 11|), \\ \rho_A = \rho_B = I_2/2, &\qquad \rho_{AB}^{\mathrm{cl}} \neq \rho_A\otimes\rho_B = I_4/4. \end{aligned} \tag{6.15}$$

The local states are maximally mixed, yet the pair retains classical matching correlation. Zero direct coherence therefore does not imply a product state. A fully independent description requires the additional factorization itself. [4, C08.S01, zero-coherence correction]

If the enlarged marking interaction remains coherent and controllable, reversing it can restore pair accessibility. Merely ignoring the marker cannot. Recoherence changes the present state without erasing the earlier marking event; both belong to the realized history. [3, §11; 1, §§3.5, 6]

6.8 An analyzer asks a local question

Measurement begins with an actual apparatus relation. For the spin branch of Section 5, V3 uses the ideal sharp binary analyzer

$$D(\mathbf{n}) = \mathbf{n}\cdot\sigma, \qquad |\mathbf{n}| = 1, \qquad P_s(\mathbf{n}) = \frac{I_2 + sD(\mathbf{n})}{2}, \quad s = \pm 1. \tag{6.16}$$

where the unit vector $\mathbf{n}$ is supplied by the apparatus and the labels $s = \pm 1$ denote the two response sectors. The Pauli algebra gives

$$D(\mathbf{n})^2 = I_2, \quad P_s^\dagger = P_s, \quad P_s P_t = \delta_{st} P_s, \quad P_+ + P_- = I_2. \tag{6.17}$$

so the sectors are orthogonal and exhaustive. This defines the local question used in the joint calculation. A real detector still has to realize its coupling and readout. [2, §11; 3, §12]

The particle therefore need not carry a prewritten answer for every axis that might later be chosen. Its spin magnitude belongs to the maintained root; a component belongs to the particle-analyzer relation actually formed. [1, §4.1; 2, §§2, 4]

6.9 The singlet answers two local questions together

Select the maximal zero-total-spin binary preparation used in the Bell calculation:

$$Z_- = \frac{1}{\sqrt{2}}\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \Pi_0 = |\Psi_-\rangle\langle\Psi_-| = \frac{1}{4}\left(I_4 - \sum_{i=1}^{3}\sigma_i \otimes \sigma_i\right). \tag{6.18}$$

with $|\Psi_-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$. This is one declared preparation, not a claim that every formation event produces a singlet. The source state contains no future analyzer settings; the apparatus supplies axes $\mathbf{a}, \mathbf{b}$ only when the comparison is made. [2, §11]

In this orthonormal chart the joint sector capacity is

$$\begin{aligned} \mu_{st}(\mathbf{a},\mathbf{b}) &= \|P_s(\mathbf{a}) Z_- P_t(\mathbf{b})^T\|_F^2 \\ &= \mathrm{Tr}\{\Pi_0 [P_s(\mathbf{a}) \otimes P_t(\mathbf{b})]\}. \end{aligned} \tag{6.19}$$

Using the singlet identities then gives

$$\begin{aligned} \mu_{st} &= \frac{1}{4}\left[1 + s\langle D(\mathbf{a}) \otimes I_2\rangle_- + t\langle I_2 \otimes D(\mathbf{b})\rangle_- + st\langle D(\mathbf{a}) \otimes D(\mathbf{b})\rangle_-\right] \\ &= \boxed{\frac{1}{4}(1 - st\,\mathbf{a}\cdot\mathbf{b})}. \end{aligned} \tag{6.20}$$

The angular dependence comes from the prepared joint state and the two local analyzer geometries. No extra adjustable long-range interaction kernel is inserted to produce it. [2, §§11–12]

The four capacities are nonnegative and normalized. Their marginals and correlation are

$$\sum_t \mu_{st} = \frac{1}{2}, \qquad \sum_s \mu_{st} = \frac{1}{2}, \qquad E_{\mathrm{cap}}(\mathbf{a},\mathbf{b}) = \sum_{s,t} st\,\mu_{st} = -\mathbf{a}\cdot\mathbf{b}. \tag{6.21}$$

Thus each local marginal is one half and independent of the remote setting, while the joint correlation is $-\mathbf{a}\cdot\mathbf{b}$. The model combines strong joint correlation with no remotely controllable change in either wing's own capacity distribution. [2, §12]

6.10 Bell locality is a stronger requirement than local interaction

Bell-local hidden-answer models assume that, conditioned on shared source data λ, the two responses are independently complete:

$$p(s,t \mid \mathbf{a},\mathbf{b}) = \int d\lambda\, w(\lambda) p_A(s \mid \mathbf{a},\lambda) p_B(t \mid \mathbf{b},\lambda), \qquad w \geq 0, \quad \int w\, d\lambda = 1. \tag{6.22}$$

Rich private clocks, local randomness, or stored frames remain inside this class if the conditional response still factorizes this way. V3 rejects this factorized representation for the singlet response; it does not reject physical reality or local interactions in general. [2, §3]

For deterministic binary answers the CHSH combination obeys

$$S_\lambda = A_0(B_0 + B_1) + A_1(B_0 - B_1) \in \{-2, +2\}, \qquad |S| \leq 2. \tag{6.23}$$

and nonnegative averaging preserves $|S| \leq 2$. Choosing the standard singlet axes,

$$\begin{aligned} \mathbf{a}_0 &= \hat{\mathbf{z}}, & \mathbf{a}_1 &= \hat{\mathbf{x}}, \\ \mathbf{b}_0 &= (\hat{\mathbf{z}} + \hat{\mathbf{x}})/\sqrt{2}, & \mathbf{b}_1 &= (\hat{\mathbf{z}} - \hat{\mathbf{x}})/\sqrt{2}, \\ S_{\mathrm{cap}} &= E_{00} + E_{01} + E_{10} - E_{11} = -2\sqrt{2}, & |S_{\mathrm{cap}}| &= 2\sqrt{2} > 2. \end{aligned} \tag{6.24}$$

produces $|S_{\mathrm{cap}}| = 2\sqrt{2}$. This is the model's reproduction of the quantum-strength singlet response under the declared analyzer interface, not a new Bell experiment. [2, §§3, 12]

The local analyzer operations themselves commute:

$$(P_s \otimes I_2)(I_2 \otimes P_t) = P_s \otimes P_t = (I_2 \otimes P_t)(P_s \otimes I_2). \tag{6.25}$$

so the ideal representation requires no preferred first analyzer for spacelike-separated comparisons. The settings act locally; the prepared relation is joint. SFT therefore keeps causal local interactions and operational no-signaling while rejecting Bell-local factorization at this interface. A later-known remote record can alter a conditional description without becoming a mechanical command transmitted to the other particle. [2, §13; 1, §6]

6.11 From response capacity to recorded frequencies

The response capacities are not yet a microscopic event law. V3 adds a separate detector premise: eligible sectors have additive capacities μ_i and share one nonnegative rate profile in detector time u,

$$\lambda_i(u) = g_D(u)\mu_i, \qquad g_D(u) \geq 0, \qquad M_\mu = \sum_i \mu_i > 0. \tag{6.26}$$

where λ_i and g_D have inverse-time units. These are first-completion hazards. The common profile supplies timing; it is not contained in the squared norm itself. [2, §12; 3, §14]

For a finite observation window, the common profile integrates to

$$\begin{aligned} J_W &= \int_{u_0}^{u_1} S_0(u) g_D(u)\, du, \qquad & q_i(W) &= \mu_i J_W, \\ p(C_W) &= M_\mu J_W > 0, & p(i \mid C_W) &= \frac{\mu_i}{M_\mu}, \\ p(\emptyset_W) &= 1 - M_\mu J_W. \end{aligned} \tag{6.27}$$

so the shared timing factor cancels in the conditional ratio $p(i \mid C_W) = \mu_i / M_\mu$. For exhaustive normalized singlet sectors, $M_\mu = 1$ and the conditional event proportions equal the capacities in (6.20). This is a bridge from response to conditional completed-event frequencies, not a derivation of the detector's microscopic actualization mechanism. [2, §12; 4, C08.S05]

Unequal efficiencies, channel-dependent latency, and no-detection events can change the observed sample. A detected-coincidence table is therefore not automatically the distribution of all prepared trials.

The open microscopic question is specific: what physical continuation selects one event and amplifies it into a durable record? V3 leaves that to transition and detector dynamics. Decoherence by itself is not promoted into an outcome selector.

6.12 Relations can be prepared and extended in more than one way

The account is not limited to particles born together. V13.9 retains entanglement-swapping constructions in which a later local interaction joins alternatives from initially independent source pairs. The outer relation is conditioned on the middle record, and that record must be communicated before the corresponding subensemble is operationally identified. Shared birth is therefore one preparation route, not the definition of a relation. [2, §14; 4, C08.S01]

Nor must a larger configuration reduce to a list of independently maximal pairs. For one finite interface A, the source keeps the bounded statement

$$\mathcal{R}_{A|X_i} \subseteq \mathcal{H}_A, \qquad \dim \mathcal{H}_A = d_A < \infty, \qquad \dim \operatorname{span}_i \mathcal{R}_{A|X_i} \leq d_A. \tag{6.28}$$

The independent response directions available through that interface cannot exceed its dimension. This is not a universal monogamy formula; apparently separate demands may be redundant or belong to one larger multipartite relation. [2, §10; 3, §7]

The same binary structure can represent spin, polarization, path alternatives, or time bins when the corresponding two-state preparation and analyzer are supplied. What unifies the cases is not the

hardware but the requirement to specify both the phase-bearing relation and the physical operation that distinguishes it. [3, §17]

6.13 The relation remains part of the physical world

The V13 binary model now gives a connected result. A pure pair can contain structure absent from either local response. Clean invertible transport preserves its nonfactorization. Distinguishing interactions redistribute direct accessibility into the enlarged configuration with an exact squared-capacity balance. The same interface gives the singlet law, $2\sqrt{2}$ CHSH response, and remote-setting-independent marginals. A separate common-profile law then converts normalized capacities into conditional event frequencies. [2, §17; 3, §18]

None of this requires particles to carry future analyzer answer tables, a universal present to decide which wing acts first, or a material fluid called entanglement. The physical claim is simpler and stronger: **the relation belongs to the complete configuration just as surely as the participants do. A local description of each part can omit real structure of the whole.**

Measurement therefore stays inside the same ontology. The apparatus supplies a relation, interacts with the available alternatives, and can retain a record. Phase accessibility, statistical correlation, and an individual recorded event are distinct physical questions; each requires the process that owns it.

We now have a maintained participant, its clock, its electromagnetic response, and the joint structure it can share. The next section goes upstream: how open causal support can enter an encounter and acquire the maintained organization with which this account began. **Formation must make the particle; the finished particle cannot be its own construction instruction.** [1, §§4, 5.2; 2, §§2, 15]

Sources and scope

[1] Shaun Fosmark, GPT-5.6 SOL, and Fable. *SFT V13.13 — What Exists, What Is Relational, and What Is Bookkeeping*, Revision B. Especially §§4–7. Its Level-2 statements apply to persistent participants; they do not exclude the temporary open-support formation relation discussed in Revision B.

[2] Shaun Fosmark and GPT-5.6 SOL. *SFT V13.9 — How Two Things Become Independent: Relational Capacity, Distinguishability, and the Bell Singlet*, replacement v1.1, August 2026. The exact PDF in the original V13 folder is byte-identical to the replacement PDF in the V3 release bundle. The published narrative and equations, not the earlier V1/V2 narratives, supply this chapter's development.

[3] *SFT V13.9-V3 Working Relational Model v1.1 — SUPER FREEZE*, August 26, 2026. Governing model recovered from the original V13 reproduction/historical bundle. Its binary pure/branch-preserving restrictions and separate detector bridge are retained.

[4] *SFT Composed Manuscript* v0.1, C08.S01–S02 and C08.S05. Used only for the established zero-coherence/product-state correction, conditioned-network interpretation, and finite-window accounting of the existing completion bridge. The chapter's narrative and primary results are assembled from [1]–[3].

Section scope. This section is a source-grounded exposition of the V13 binary relational model. No new quantum, formation, or detector law is solved here. The common complex response chart, singlet preparation, ideal analyzers, and optional completion-rate model remain explicit premises. General mixed-state and multipartite measures, absolute event rates, and microscopic actualization remain separate work. The original numerical residuals are retained as source-reported checks, not rerun evidence.

7. Formation: from open support to a maintained particle

We began with the maintained electron because it tells us what persistence must accomplish. Formation asks the upstream question: **how can real open support acquire that maintained organization without changing into a different substance?** In SFT, the electron or positron is the recurrent organization the support achieves, not a material product into which light is converted. [1, §§2-3; 2, §2]

V13.14 gives that question a concrete entrance program. For the electron-positron branch it establishes the invariant-energy threshold, constructs a connected two-carrier response, produces mirror inward first turns in the carriers themselves, and generates nascent support with the same rotating trace-free saddle geometry used by the maintained electron. It also connects the measured first turn to a legal finite-source mean-loop family through one disclosed free normalization. [2, Abstract, §§5-8]

The remaining test is equally concrete: carry the outgoing sectors through uninterrupted, daughter-local, self-maintaining returns after the initiating overlap fades. That is unfinished physics, not a second engine waiting to receive a completed loop.

7.1 The maintained particle is the destination

V13.12 tells formation what a successful descendant must eventually possess: finite carrier support, reaction, compatible geometry, charged dressing, transported framing, and recurrence. Those are destination criteria. Feeding the mature route, support width, or Compton period into the incoming photons would instead use the answer as part of the preparation. [2, §3; 3, §§1-3]

During overlap the incoming photons participate in one encounter; they are not preassigned one-to-one to an eventual electron and positron. Two daughter sectors have to become separately identifiable through the evolution and then maintain themselves. The desired endpoint is two persistent particles, not one two-lobed object given two labels. [2, §2]

Maintenance and capture are also different problems. Section 3 showed bounded transverse motion around a specified recurrent background. That elliptic stability does not attract arbitrary open trajectories onto the background. **Formation must make the mean route that maintenance can then protect.** [2, §3; 7, §5]

A daughter clock need not exist before the encounter. Revision B allows a temporary joint configuration of open support between the open and recurrent regimes. Rest frame, ruler, and clock become available when persistent recurrence is achieved; they are downstream consequences of the transition. [1, §§3.1, 4, 5.2]

7.2 The energy belongs to the encounter

The first entrance condition belongs to the complete incoming four-momentum. With signature $(-,+,+,+)$, two ideal real photons of energies $E_1, E_2 > 0$ and propagation directions $\hat{\mathbf{n}}_1, \hat{\mathbf{n}}_2$ have

$$p_i^\mu = \left(\frac{E_i}{c}, \frac{E_i}{c}\hat{\mathbf{n}}_i\right), \qquad p_i^2 = 0, \qquad P^\mu = p_1^\mu + p_2^\mu, \qquad s_{\mathrm{inv}} = -c^2 P_\mu P^\mu. \tag{7.1}$$

Here s_{inv} has energy-squared units; its positive square root is the center-of-momentum energy when the total momentum is timelike. If $\hat{\mathbf{n}}_1 \cdot \hat{\mathbf{n}}_2 = \cos\theta$,

$$\begin{aligned} s_{\mathrm{inv}} &= (E_1 + E_2)^2 - |E_1\hat{\mathbf{n}}_1 + E_2\hat{\mathbf{n}}_2|^2 \\ &= 2E_1E_2(1 - \cos\theta). \end{aligned} \tag{7.2}$$

Thus co-propagating ideal photons have $s_{\mathrm{inv}} = 0$, while head-on propagation maximizes the available invariant energy at fixed E_1, E_2. A collective center-of-momentum frame can exist even though neither photon acquires a massive rest clock. [2, §4.1]

For the supplied electron branch,

$$\boxed{s_{\mathrm{inv}} \geq 4m_e^2c^4.} \qquad \begin{cases} \theta = 0: & s_{\mathrm{inv}} = 0, \\ \theta = \pi: & E_1E_2 \geq m_e^2c^4. \end{cases} \tag{7.3}$$

Equal head-on energies reach threshold at $E_1 = E_2 = m_ec^2$; using the conventional electron-mass anchor gives about 1.022 MeV of center-of-momentum energy. This establishes accessibility of the supplied branch. It does not select m_e and does not imply that every accessible encounter captures. A single isolated real photon has null total four-momentum and cannot supply this isolated massive two-body final state by itself. [2, §4.1]

Once the equal-mass two-body channel is selected, its center-of-momentum kinematics is

$$E_-^* = E_+^* = \frac{\sqrt{s_{\mathrm{inv}}}}{2}, \qquad \mathbf{p}_-^* = -\mathbf{p}_+^*, \qquad p_*c = \sqrt{\frac{s_{\mathrm{inv}}}{4} - m_e^2c^4}. \tag{7.4}$$

At threshold the relative momentum vanishes. Above threshold, surplus invariant energy appears in relative motion or other allowed outgoing channels while the daughter rest mass remains the selected branch mass. **More collision energy is not automatically more electron.** [2, §§4.1, 9]

7.3 Overlap and a direction of turning are different ingredients

Kinematic permission does not make a particle. Two source-free Maxwell beams can overlap and continue through one another. V13.14 therefore separates the strength of the joint encounter from the directional information that can steer it. [2, §4.2]

In the exact head-on null control, set $c = 1$ so electric and magnetic fields have matching units. With the second beam counter-propagating and both electric fields transverse, define

$$\begin{aligned} &\mathbf{B}_1 = \hat{\mathbf{n}} \times \mathbf{E}_1, && \mathbf{B}_2 = -\hat{\mathbf{n}} \times \mathbf{E}_2, \\ &\mathcal{I}_{12} = \mathbf{B}_1 \cdot \mathbf{B}_2 - \mathbf{E}_1 \cdot \mathbf{E}_2 = -2\mathbf{E}_1 \cdot \mathbf{E}_2, \\ &\mathcal{J}_{12} = \mathbf{E}_1 \cdot \mathbf{B}_2 + \mathbf{E}_2 \cdot \mathbf{B}_1 = 2\hat{\mathbf{n}} \cdot (\mathbf{E}_1 \times \mathbf{E}_2). \end{aligned} \tag{7.5}$$

The two mixed invariants satisfy

$$\begin{aligned} \mathcal{I}_{12}^2 + \mathcal{J}_{12}^2 &= 4\left[(\mathbf{E}_1 \cdot \mathbf{E}_2)^2 + |\mathbf{E}_1 \times \mathbf{E}_2|^2\right] \\ &= \boxed{4|\mathbf{E}_1|^2|\mathbf{E}_2|^2.} \end{aligned} \tag{7.6}$$

At fixed local field magnitudes, rotating one transverse polarization redistributes $\mathcal{I}_{12}$ and $\mathcal{J}_{12}$ while leaving their squared sum unchanged. **The scalar null-departure channel is therefore polarization-angle**

blind in this exact head-on limit. Framing, helicity, the spin-two response, and finite-packet structure can still carry directional information. [2, §4.2]

An impact parameter supplies another ingredient: a signed offset and orbital lever arm. Exact centering can maximize scalar overlap while giving this particular offset-steered channel no handed turning direction. A wide miss gives leverage but little overlap. The useful entrance can therefore lie between those extremes.

7.4 Why an intermediate offset turns more effectively

For identical translated packets in the ideal scalar limit, define the normalized transverse intensity overlap

$$O(\mathbf{b}) = \frac{\int I(\mathbf{r}) I(\mathbf{r}-\mathbf{b})\, d^2r}{\int I(\mathbf{r})^2\, d^2r}, \qquad O(\mathbf{0}) = 1. \tag{7.7}$$

For the illustrative Gaussian amplitude $A(\mathbf{r}) \propto e^{-r^2/(2w^2)}$, completing the square gives

$$r^2 + |\mathbf{r}-\mathbf{b}|^2 = 2|\mathbf{r}-\mathbf{b}/2|^2 + \frac{b^2}{2}, \qquad O(b) = e^{-b^2/(2w^2)}. \tag{7.8}$$

This is an analytic reference, not an identification of the actual production packet with a real-space Gaussian. The numerical packet uses its own compact spectral preparation. [2, §4.2]

For that packet, the measured overlap dependence is summarized by

$$x = \frac{b}{\sigma_{\mathrm{pkt}}}, \qquad O(x) \simeq e^{-\kappa_O x^2}, \qquad \kappa_O = 0.452792, \qquad T_{\mathrm{off}}(x) \propto x e^{-\kappa_O x^2}. \tag{7.9}$$

T_{off} combines even overlap with the signed offset lever arm. It is a turn diagnostic, not a temperature or probability. The fit reproduces the earlier live-turn variation with $R^2 \simeq 0.989$; κ_O belongs to this preparation rather than defining a universal constant. [2, §4.2]

Differentiation gives

$$\frac{d}{dx}\left(x e^{-\kappa_O x^2}\right) = e^{-\kappa_O x^2}(1 - 2\kappa_O x^2), \qquad x_{\mathrm{peak}} = \frac{1}{\sqrt{2\kappa_O}}, \qquad \kappa_O > 0. \tag{7.10}$$

The channel vanishes at exact centering and at large separation and peaks in between: leverage initially wins, then loss of overlap dominates. This explains the shape of the diagnostic without replacing the underlying formation dynamics.

7.5 The response follows the joint source

The formation model drives a complex spin-two response Y_2 with the **connected** carrier source Q_F: the part of the source belonging to the encounter rather than the sum of unrelated isolated responses. Spin two describes the rotational behavior of the transverse shear; it does not introduce another particle. [4]

On the admitted finite domain, B1B defines the response by the constrained solve

$$HY_2 + \mathsf{C}^\dagger \lambda = Q_F, \qquad \mathsf{C}Y_2 = 0, \qquad \lambda = (\mathsf{C}H^{-1}\mathsf{C}^\dagger)^{-1}\mathsf{C}H^{-1}Q_F. \tag{7.11}$$

Solving the first equation and enforcing the constraint gives the Schur response

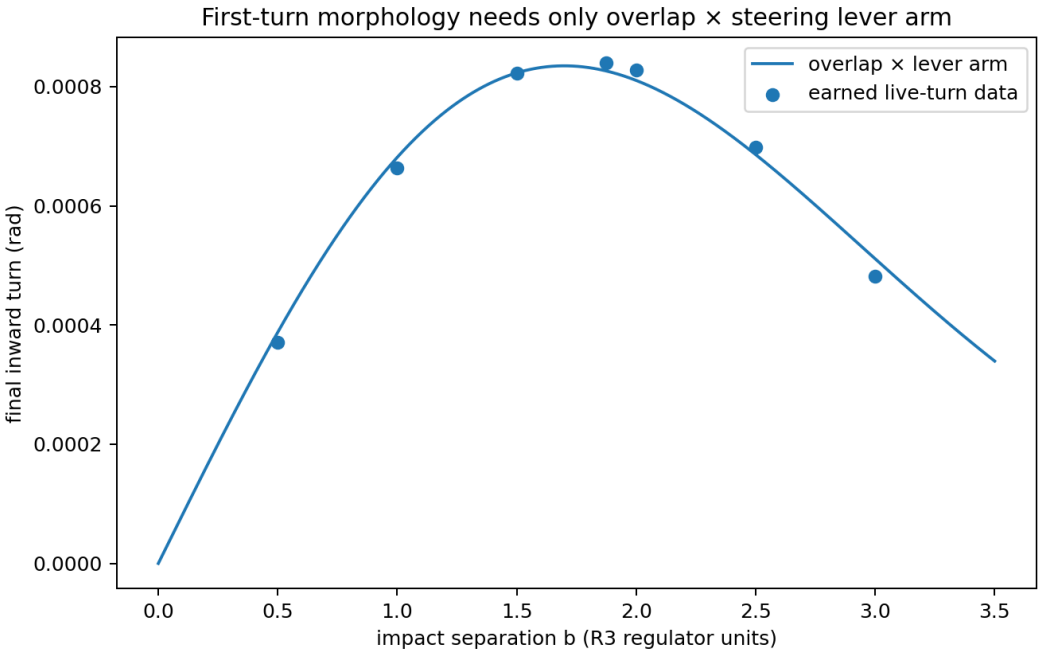


Figure 2: The original V13.14 offset diagnostic compares the overlap-times-lever-arm fit with its earlier live-turn data. These regulator-dependent angles are not the separately normalized B1C angles below. Reproduced unchanged from the formation package.

$$\begin{aligned} &\mathcal{R} = H^{-1} - H^{-1}\mathsf{C}^{\dagger}(\mathsf{C}H^{-1}\mathsf{C}^{\dagger})^{-1}\mathsf{C}H^{-1}, \\ &Y_2 = \mathcal{R}Q_F, \qquad \mathsf{C}\mathcal{R} = 0, \qquad Q_F = 0 \implies Y_2 = 0. \end{aligned} \tag{7.12}$$

The response vanishes with its connected source. It is therefore source-conditioned rather than a free reaction field left oscillating after the encounter disappears. The chosen spatial operator is

$$H(\mathbf{k}) = 1 + (|\mathbf{k}|/k_0)^2, \qquad k_0 > 0, \tag{7.13}$$

Identity and quartic alternatives were tested as controls. The surviving suppression tests support this response class while leaving the physical selection and normalization of H explicit. [4]

A scalar breathing/support response accompanies the shear:

$$\sigma_{\mathrm{br}} = -\frac{1}{2}\ln\left(1 - \frac{2}{15}|Y_2|^2\right), \qquad 1 - \frac{2}{15}|Y_2|^2 > 0. \tag{7.14}$$

Its dependence on $|Y_2|$ makes it insensitive to the phase of the spin-two response. The logarithmic bound marks the regularity domain of this chart, not a capture criterion. The scalar housing and rotating shear are concurrent responses of one forming configuration, not a separate container around it. [4; 6, §5]

7.6 The encounter creates the same kind of rotating saddle

The complex response has a direct transverse geometry:

$$\mathsf{S}_{\mathrm{TF}}(Y_2) = \begin{pmatrix} \operatorname{Re} Y_2 & \operatorname{Im} Y_2 \\ \operatorname{Im} Y_2 & -\operatorname{Re} Y_2 \end{pmatrix}, \qquad \operatorname{spec} \mathsf{S}_{\mathrm{TF}} = \{+|Y_2|, -|Y_2|\}. \tag{7.15}$$

Writing $Y_2 = |Y_2|e^{2i\psi}$ gives

$$\boxed{\mathsf{S}_{\mathrm{TF}} = |Y_2|\mathsf{R}(\psi) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \mathsf{R}(\psi)^T, \qquad \psi = \tfrac{1}{2} \arg Y_2 \pmod{\pi}.} \tag{7.16}$$

The principal axis is headless and rotates with half the complex phase. The retained formation source winds through more than nine complex turns during its active window, so the response is a rotating saddle rather than a single static inward shove. [2, §6; 7, §3]

Section 3's maintained transverse operator has the same geometric form:

$$K(\vartheta) = a_s \mathsf{R}(\nu_s \vartheta) \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \mathsf{R}(\nu_s \vartheta)^T, \qquad 0 < a_s < \nu_s^2. \tag{7.17}$$

This is a real structural bridge between formation and maintenance: both use opposite transverse principal responses carried by a rotating headless frame. It does **not** yet identify their amplitudes, time parameters, or rotation rates. A formation-time phase is not automatically the uniform route-angle twist of the maintained control. [2, §6; 7, §§4-5]

Formation has produced the right class of local support geometry. The next question is whether the mean route completes while retaining it.

7.7 The turn appears in the carrier itself

A response field matters only if it changes the propagating support. B1C inserts the constrained response into the Maxwell constitutive variation and compares the two signs

$$\begin{aligned} Y_2 &= -\mathcal{R}Q_F \quad \text{retained inward branch,} \\ Y_2 &= +\mathcal{R}Q_F \quad \text{outward comparison.} \end{aligned} \tag{7.18}$$

The selected minus sign is the inward branch of this lift; it remains distinct from B1B's positive response convention. The paired sign control shows that the choice has an observable geometric consequence. [5]

The physical readout adds the stored correction to the base carrier state:

$$\mathbf{X}_{\mathrm{phys}} = \mathbf{X}_0 + \Delta\mathbf{X}, \qquad \hat{\mathbf{k}}_{\mathrm{phys}} = \frac{\mathbf{k}_0 + \Delta\mathbf{k}}{|\mathbf{k}_0 + \Delta\mathbf{k}|}. \tag{7.19}$$

At the baseline read section the two daughter sectors turn inward by

$$\theta_R \simeq 0.01643321 \text{ rad}, \qquad \theta_L \simeq 0.01633254 \text{ rad}, \qquad \theta_{\mathrm{first}} = \frac{\theta_R + \theta_L}{2} \simeq 0.0163828762 \text{ rad}. \tag{7.20}$$

Both independently logged sectors bend inward in mirror-related directions while also carrying phase, frame, and support responses. Co-propagating and large-offset controls remain suppressed to about 1.16×10^{-7} and 0.00708 of the main case. These angles belong to this packet/regulator preparation. [2, §5; 5; 9]

The calculation has changed the carrier trajectory itself. The first turn is not merely an angular-momentum label drawn next to two otherwise unchanged rays.

7.8 A first turn is not yet a return

The forward continuation follows the corrected carriers as the connected source fades. At the later trusted section that shared source is only about 1.17% of peak, while both sectors still move outward along their original longitudinal directions. A local spatial return requires a reversal somewhere along each path:

$$\hat{\mathbf{k}}_R \cdot \hat{\mathbf{x}} < 0, \qquad \hat{\mathbf{k}}_L \cdot (-\hat{\mathbf{x}}) < 0 \quad \text{at some stage before a local spatial return.} \tag{7.21}$$

The unit-response forward calculation never reaches that necessary condition. Diagnostic gain increases also fail before the scalar chart approaches its regularity boundary; those high-gain points are not controlled nonlinear solutions. [6, §§2-7]

This is a useful negative result. It does not erase the measured first turn, and it is not a no-go theorem for formation. It identifies what the present implementation lacks: **after the connected entrance source fades, daughter-local response has to continue the turn.** Transient shared steering alone is insufficient.

A completed return is stricter than position closure. Each outgoing sector must return in position, tangent, phase, headless frame, and reaction/support state:

$$\mathfrak{d}_i = (\Delta\mathbf{x}_i, \Delta k_i, \Delta\phi_i, \Delta D_i, \Delta\mathcal{S}_{\text{reaction},i}) = 0, \qquad i = R, L. \tag{7.22}$$

These blocks carry different units and equivalences; they are not components of one arbitrary Euclidean distance. The condition demands two complete local state returns at successive events, not a closed causal curve through the same event. [5, parent contract; 8]

7.9 Two local returns require no physical handoff

There is a window in the retained evolution where the outgoing sectors are already distinguishable while the connected formation source is still nonzero. B1C2 uses that regime to separate two facts: daughter-local response can begin before a cycle is complete, but **ownership by a persistent particle begins only when that local state returns.** [8]

A finite graph control distinguishes two local cycles from one crossed cycle:

$$r_{\text{cyc}} = N_e - N_v + N_c, \qquad \begin{cases} \text{two local returns}: & (r_{\text{cyc}}, N_c) = (2, 2), \\ \text{two crossed returns}: & (r_{\text{cyc}}, N_c) = (1, 1). \end{cases} \tag{7.23}$$

Cross-connecting the two open chains produces one combined cycle, not two recurrent descendants. The graph count is only topology; lawful endpoint identification still has to satisfy the phase, framing, and support conditions of (7.22). [8]

The same report checks that the local response itself need not jump when the shared contribution vanishes. In the retained G12U mode decomposition,

$$\chi_{1n} = \frac{1}{n^2 - 1}, \quad n \geq 2, \qquad y_{i,n} = \chi_{1n}(h_{i,n}^{\mathrm{loc}} + h_{i,n}^{\mathrm{joint}}) \longrightarrow \chi_{1n} h_{i,n}^{\mathrm{loc}} \quad (h_{i,n}^{\mathrm{joint}} \to 0). \tag{7.24}$$

Across 4,096 source checks on modes $n = 2, \ldots, 6$, the report found zero reaction-state discontinuity when only the ownership classification changed. Conditional on a nonzero daughter-local source, the same response kernel continues smoothly as the shared term goes to zero. [8]

That is the **no-handoff** result. Nothing is reset, copied, or assigned a new coupling at the first exact return. What remains to be generated dynamically is the sustaining daughter-local source itself.

7.10 A finite-source loop gives a quantitative destination

The independent C17D control asks a narrower destination question: can the exact finite tangent-current source support a circular mean route in its reduced binding description? For a supplied unit-radius torus it uses

$$\mathcal{K}(\mathbf{X}, \mathbf{X}'; \mathbf{T}, \mathbf{T}') = \frac{[1 - \mathbf{T} \cdot \mathbf{T}']^2}{|\mathbf{X} - \mathbf{X}'|}, \qquad |\mathbf{T}| = |\mathbf{T}'| = 1. \tag{7.25}$$

The distributed current-current, current-margin, and margin-margin terms retain their distinct weights. This is a reduced null-current contraction kernel, not a completed exterior metric. [10, §5]

Across the tested support family,

$$\begin{aligned} &0.1057234868 \leq C_{\mathrm{required}} \leq 0.3480058363, \\ &0.1009796275 \leq C_A \leq 0.2960307130 < \tfrac{1}{2}. \end{aligned} \tag{7.26}$$

A legal finite-source circular family exists inside the compactness bound, but the family does not select one width. Because the route is supplied, this answers what can support the loop class, not how incoming packets find it. [2, §7.1; 10, §§5-6]

V13.14 connects the measured first turn to that independently normalized loop family through one explicit parameter:

$$\boxed{C_{\mathrm{loop}} = \eta_{\mathrm{turn}\to\mathrm{loop}}\, \theta_{\mathrm{first}}.} \tag{7.27}$$

Using the physical mean first turn gives

$$\eta_{\min} = \frac{C_{\min}}{\theta_{\mathrm{first}}}, \qquad \eta_{\max} = \frac{C_{\max}}{\theta_{\mathrm{first}}}, \qquad 6.4533 \lesssim \eta_{\mathrm{turn}\to\mathrm{loop}} \lesssim 21.2420. \tag{7.28}$$

At the minimum mapping the main event reaches the easiest legal loop edge while the centered, large-offset, and co-propagating controls remain suppressed. The source's ruling is therefore exactly what it says:

PASS WITH FREE PARAMETER at reduced first-mean-loop scope. The two independently normalized calculations admit a finite interface that preserves the controls. The normalization itself is not derived, and the interface does not override the failed forward-return test of §7.8. [2, §7.2; 9]

The next dynamical target is now sharply defined: derive the normalization and produce the mean loop through the same evolution that generates its sustaining local response.

7.11 The daughter earns its identity and clock at recurrence

A daughter becomes an electron-class participant when the initiating shared response can fade and a complete local recurrent state remains. Only then does its recurrence supply its clock and its invariant root readouts become properties of that persistent object.

The equal-mass conjugate pair should carry opposite charge labels with equal positive rest energies and cadences. The headless saddle angle in (7.16) does not by itself select the signed electromagnetic winding of Section 5; V13.14 leaves that conjugate framing selection open. [2, §§2, 8-11]

Absolute support scale remains open as well. The maintained representative uses $\Gamma_\star \simeq 0.1856235$ and stays elliptic across the reported support-ratio range 0.05 to 0.60. That establishes a maintenance family, not the formation selector. Neither $\Gamma_\star$ nor the first-loop conversion factor may be hidden in a later laboratory calibration and then counted as an upstream prediction. [2, §3]

The hierarchy is therefore clean. The invariant threshold makes the branch accessible. The connected response creates a real first turn and the right support geometry. Two complete self-maintaining returns establish the new persistent participants.

7.12 What formation has earned

V13.14 supplies more than kinematic permission. Its connected response acts on the carriers, produces mirror inward turns, exposes phase/frame/support readouts, and generates the rotating-saddle geometry needed by the maintained model. The finite-loop control gives that route a quantitative destination and shows exactly where an independent normalization still enters. [2, §§5-8]

The forward-return control is equally valuable: more of the same transient entrance steering does not finish the job. A completed execution must generate daughter-local feedback, close two lawful full states, and select their absolute support and conjugate signed framing. Production probability and cross section remain downstream of that completed transition and its preparation ensemble. [2, §§10-12]

Formation and maintenance are two regimes of one proposed physical process: support overlaps, responds, turns, returns, and continues. There is no substance conversion at the boundary and no new maintenance engine switched on after capture. The entrance and first turn are calculated. The self-maintaining return is the next decisive test.

With formation connected to the maintained branch, the next question is identity beyond one electron: **what makes repeated particles instances of one species, what other recurrent structures can exist, and how do complete particles combine without double-counting their shared response?**

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Sources and scope

[1] *SFT V13.13 — Live Ontology Constitution*, Revision B, especially §§3-5. Governs the open-support transition and the rule that mature clock/scale information is downstream of closure.

[2] Shaun Fosmark and GPT-5.6 SOL. *SFT V13.14 — Formation: How Light Enters a Persistent Particle Branch*, Public Research Draft v0.2, September 2026; second-bench review by Claude/Fable. Primary formation narrative and result owner.

[3] *SFT V13.12 — The Maintained Electron*, public v0.2, and Section 3 of this reader. Used as the maintained destination, not as upstream formation data.

[4] *F1-UNIFIED1-B1B — Constraint-Complete Slaved Two-Mode Response*, report v0.1, with its response contract. Owns the constrained Schur response, spatial-operator controls, and scalar support response.

[5] *F1-UNIFIED1-B1C — Two-Entity Physical Carrier Predefect Lift*, report v0.1. Owns the selected inward sign and the first-order carrier, phase, frame, and support readouts.

[6] *F1-UNIFIED1-B1C3A — Formation-Slaved Double-Return Accessibility*, report v0.1. Owns the late forward-return limitation, base-plus-correction readout, scalar regularity coefficient 2/15, and high-gain diagnostic boundary.

[7] *SFT First-Turn -> Recurrent-Branch Audit*, v0.1, with the C17A optical companion. Owns the rotating-saddle comparison and separation between mean-route closure and transverse stability.

[8] *F1-UNIFIED1-B1C2 — Two-Daughter Cycle Ownership Onset*, report v0.1. Owns the two-local-versus-crossed topology control and conditional response-continuity test.

[9] *SFT First Mean Loop — Physical First-Turn Interface Audit*, v0.2. Owns the physical first-turn interface, its free normalization, and the preserved suppression controls.

[10] *IEMM-C17D — Background/Deviation Separation, Exact G12U Tangent Source, and Liouville Nonuniqueness Audit*, v0.1, especially §§5-6. Used for the finite distributed source, reduced loop family, and nonselection of width.

Sources [4]-[10] remain preserved in the V13.14 handoff/reproduction package. No new native formation or maintenance solve is claimed here.

Section scope. Equations (7.1)-(7.28) preserve the original V13 core assembly displays exactly. The mixed field invariants retain their fixed normalization; s_{inv} remains distinct from the maintained support ratio; B1B's response sign and B1C's selected inward sign remain separate. Reported numerical values retain their original evidence. This pass is editorial and mathematical verification only.

8. Species, identity, and complete composites

A maintained electron can move, interact, bind, and acquire new relationships without ceasing to be an electron. Another electron can have an entirely different history while sharing the same intrinsic species properties. When several persistent participants form a bound system, the whole can then possess mass, charge, response, and admissibility conditions that do not belong to any constituent alone. **Identity persists through participation; composition creates new physical structure.**

SFT therefore treats a species as a lawful class of maintained organization and an individual particle as one continuing instance of that class. The electron makes this proposal concrete. The wider V13 program asks which other recurrent classes exist, which combinations of complete roots are allowed, and what new configurations those combinations can maintain. Revision B keeps structural identity as a physical proposal rather than promoting it to a uniqueness theorem before the dynamics has earned one. [1, §10]

Several V13 results make that program precise. One nonlinear root cannot be multiplied into two particles by amplitude scaling. Surplus energy can change motion and shared fields without changing the intrinsic branch. Exchange symmetry requires an actual composition law. And a lawful composite can contain projected fractional sectors that are not lawful isolated particles. [2, §§4, 11-12, 16, 18; 3, §§33, 38, 40, 44-45]

8.1 One class, many histories

Two electrons need not share position, motion, orientation, relationships, or biography. What the species proposal holds fixed is the intrinsic content of the maintained class. Let $\mathscr{R}_j$ denote one proposed recurrent class and let $\mathcal{I}$ collect its intrinsic readouts:

$$X \in \mathscr{R}_j \quad \Longrightarrow \quad \mathcal{I}[X] = \mathcal{I}_j = (m_j, q_j, s_j, ...). \tag{8.1}$$

Here m_j is invariant rest mass, q_j signed charge, and s_j the dimensionless spin label. The label j names the class, not a hidden serial number. Equation (8.1) states what species identity would mean; it does not generate the members of $\mathscr{R}_j$ or prove that only one branch has those readouts. [1, §§4.1, 10.2]

The maintained electron supplies the first concrete example because its source, support, framing, geometry, and readouts already coexist on one frozen state. But its support normalization and dimensional anchor remain explicit. A family of electron-like numerical solutions does not become physically unique by being given one name. Formation and the complete source law must select the realized branch. [5, §§4, 11-12]

The candidate explanation for identical intrinsic properties is shared lawful organization, not shared history.

8.2 Two electrons are not twice one electron

Electron V1 is a nonlinear maintained configuration. Its current sources a field, the field contributes stress, and support and geometry respond together. Scaling one piece while freezing the rest generally leaves the solution class. The electromagnetic part already shows the mismatch:

$$J \mapsto \lambda J, \qquad F \mapsto \lambda F, \qquad T_{\mathrm{EM}}[\lambda F] = \lambda^2 T_{\mathrm{EM}}[F]. \tag{8.2}$$

At fixed geometry, Maxwell response is linear in its source while electromagnetic stress is quadratic in the field. This does not state the scaling of every term in the complete electron; it shows why doubling one frozen root is not a second root. A genuine two-electron configuration requires its own joint solution. V13.4 calls this **primitive-root non-scalability**. [3, §38.2]

Multiplicity is therefore different from amplitude. Energy variation is different again. For a formed root moving freely in the relativistic correspondence,

$$E = \gamma mc^2, \qquad K = (\gamma - 1)mc^2, \qquad E^2 - c^2|\mathbf{p}|^2 = m^2c^4. \tag{8.3}$$

The observer energy and kinetic energy can change continuously while the invariant mass remains fixed. Section 7 gave the same lesson at formation: energy above threshold can enter relative motion, radiation, fields, or additional channels rather than making a continuously heavier electron. V13.4 calls this **root-surplus separation**. [2, §4; 3, §40]

That distinction lets the theory ask a clean question: what makes the branch stable while the encounter around it remains variable?

8.3 Persistence organizes the species problem

The V13 *Particle Reality Inventory* is useful here as a research map. It separates persistent building blocks, short-lived physical states, internal response sectors, and speculative or representational objects instead of assuming that every named object has the same ontological status. [4, §§1-4]

This chapter uses that organizing principle without inheriting the inventory's stronger blanket exclusions. A short-lived state can still be physically real. A field contribution can be indispensable inside a composite without existing as a freely isolated particle. A calculational representation does not become a new substance merely because it has a familiar name. The current standard is simpler: a proposed species or degree of freedom must establish its lawful state, physical source, observable role, and allowed transitions. [1, §§9, 12-13]

A lifetime is likewise not an identity theorem. Persistence, decay channels, and transition rates belong to the dynamics of the full state. The species problem is therefore broader than listing stable particles: it asks which complete configurations can exist, how long they persist, and what lawful processes connect them.

8.4 Three response directions are not yet three species

V13.4 obtains one exact local count around a timelike root. At an event p, let U be a timelike tangent and consider its metric-orthogonal normal space:

$$N_p = \{v \in T_p\mathcal{M} : g(U, v) = 0\}, \qquad \dim N_p = 4 - 1 = 3, \qquad H_\perp : N_p \longrightarrow N_p. \tag{8.4}$$

The orthogonality condition removes one direction from the admitted $3 + 1$ tangent space. In the source's reduced normal-response problem, $H_\perp$ is self-adjoint on this three-dimensional physical space and therefore has three spectral directions counting multiplicity. That is a result about the stated tangent response, not a count of every internal or field degree of freedom of an extended particle. [2, §16; 3, §44]

The retained cyclic example makes the three directions explicit:

$$P_3 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \quad P_3^3 = I_3, \quad \mathcal{D}_3 = I_3 - e^{i\beta} P_3, \quad \mathsf{U}_\perp = \tfrac{1}{2}\mathcal{D}_3^\dagger \mathcal{D}_3. \tag{8.5}$$

If $P_3 v_k = e^{2\pi i k/3} v_k$, then

$$\begin{aligned} &\mathsf{U}_\perp v_k = u_k(\beta) v_k, \\ &u_k(\beta) = \tfrac{1}{2}|1 - e^{i(\beta + 2\pi k/3)}|^2 = 1 - \cos\left(\beta + \frac{2\pi k}{3}\right), \quad k = 0, 1, 2. \end{aligned} \tag{8.6}$$

For generic $\beta \notin (\pi/3)\mathbb{Z}$ the three values are positive and distinct. Their common formula, rather than three independent fitted eigenvalues, produces the splitting. [3, §44]

Three response directions around one root are not automatically three particle generations. A tangent stiffness is not a rest mass, and nonlinear recurrent branches need not be in one-to-one correspondence with tangent modes. The exact count is useful because it narrows the next question: can the complete nonlinear theory turn those modes into distinct stable roots with genuinely family-sensitive loading? [2, §16; 3, §§44, 62]

8.5 Identity and exchange are different operations

Species identity says what kind of participant is present. Exchange statistics says how complete states of identical participants compose. The 4π return of one spinorial representative does not by itself choose the sign acquired when two complete particles are exchanged. V13.4 therefore introduces the composition problem separately. [2, §18; 3, §§32-33]

Let V be the admitted one-particle state space, W a pair space, $\mathcal{C}$ a bilinear composition, and X the exchange involution:

$$\mathcal{C} : V \times V \to W, \qquad \mathsf{X}^2 = I, \qquad \mathsf{X}\mathcal{C}(u, v) = \mathcal{C}(v, u). \tag{8.7}$$

Its parity projectors split the composition into symmetric and antisymmetric sectors:

$$\begin{aligned} &\mathsf{P}_\pm = \tfrac{1}{2}(I \pm \mathsf{X}), \qquad \mathcal{C} = \mathcal{C}_+ + \mathcal{C}_-, \qquad \mathcal{C}_\pm = \mathsf{P}_\pm \mathcal{C}, \\ &\mathcal{C}_\pm(v, u) = \pm \mathcal{C}_\pm(u, v), \qquad V \otimes V = \mathrm{Sym}^2 V \oplus \Lambda^2 V. \end{aligned} \tag{8.8}$$

This is an exact algebraic classification. Reversible exchange and identical roots still leave both parities available until the physical composition law selects one. [3, §§33.1-33.3]

That is the useful conclusion: the statistics problem has been localized. It belongs to **which compositions of complete roots are physically admissible**, not to a new force coefficient and not automatically to the rotational sign of one isolated particle.

8.6 The precise condition behind exclusion

Within a complex bilinear composition, suppose every complete state has zero self-composition. Applying that same condition to $u + v$ gives

$$\begin{aligned} 0 &= \mathcal{C}(u+v, u+v) \\ &= \mathcal{C}(u,u) + \mathcal{C}(u,v) + \mathcal{C}(v,u) + \mathcal{C}(v,v) \\ &= \mathcal{C}(u,v) + \mathcal{C}(v,u). \end{aligned} \tag{8.9}$$

Conversely, antisymmetry gives $2\mathcal{C}(u,u) = 0$. Over the complex numbers,

$$\boxed{\mathcal{C}(u,u) = 0 \text{ for every } u \quad \Longleftrightarrow \quad \mathcal{C}(u,v) = -\mathcal{C}(v,u).} \tag{8.10}$$

This is the source's **universal self-composition nullity equivalence**. It is stronger than requiring only a selected basis to self-annihilate, because every superposition must satisfy the same condition. V13.4 does not derive this physical premise from primitive-root non-scalability or from spinor rotation; the native composition law still has to earn it. [3, §§33.4-33.6, 38.1-38.3]

Once an exchange parity $\eta = \pm 1$ is supplied, two orthonormal complete one-particle states give

$$|\Psi_\eta\rangle = \frac{|u\rangle|v\rangle + \eta|v\rangle|u\rangle}{\sqrt{2}}, \qquad \langle u|v\rangle = 0. \tag{8.11}$$

The minus state vanishes when the *complete* one-particle states are identical. Two electrons may share the same spatial orbital while differing in spin; those are different complete one-particle states, so their alternating composition need not vanish. That distinction becomes essential in the atomic section. [3, §§33.3-33.5, 38.3]

Exchange also changes how a joint state samples the same interaction operator. For a Hermitian exchange-symmetric $\hat{V}$,

$$\begin{aligned} J = \langle uv|\hat{V}|uv\rangle, \quad K = \langle uv|\hat{V}|vu\rangle \in \mathbb{R}, \\ \langle \Psi_\eta|\hat{V}|\Psi_\eta\rangle = J + \eta K. \end{aligned} \tag{8.12}$$

The $J \pm K$ result therefore does not introduce a new force. The selected composition changes the joint state on which the existing interaction is evaluated. [3, §34]

8.7 A constituent can carry a fractional label without being an isolated particle

The complete-state principle becomes especially important in the strong-composite proposal. For a complete isolated compact charge/framing map, the integrated angular differential has integer degree,

$$n = \frac{1}{2\pi} \oint \mathfrak{h} \in \mathbb{Z}. \tag{8.13}$$

This is a statement about the declared complete map, not a claim that every gauge holonomy must be an integer. A projection inside a larger structure need not itself satisfy the boundary conditions of an isolated particle. V13.4 makes that distinction part of the composite construction. [2, §11; 3, §§20-22]

Its leading undressed winding model uses

$$S_{\text{form}}(n_Y) = A_Y n_Y^2, \quad A_Y > 0, \qquad Q_\pm = \pm\frac{1}{2} + \frac{n_Y}{6}, \qquad n_Y \in \Lambda_{\text{lawful}}. \tag{8.14}$$

where Q is in units of the common q_0 and n_Y belongs to the declared lawful class. Restricting the minimization to the relevant boundary class gives

$$\begin{aligned} Q_\pm \in \mathbb{Z} &\implies n_Y \equiv 3 \pmod 6, \quad \min|n_Y| = 3, \\ n_Y = -3: &\quad (Q_+, Q_-) = (0, -1), \\ Q_\pm \in \tfrac{1}{3}\mathbb{Z} &\implies n_Y \text{ odd}, \quad \min|n_Y| = 1, \\ n_Y = 1: &\quad (Q_+, Q_-) = \left(\tfrac{2}{3}, -\tfrac{1}{3}\right). \end{aligned} \tag{8.15}$$

The arithmetic constrains the doublet as a family: integer exterior charges require odd multiples of three, while the third-integer branch admits the familiar $2/3, -1/3$ pair at minimal $|n_Y|$. The other orientation supplies the conjugate family. This does not derive the absolute charge strength or the formation process that selects a boundary class. [3, §20]

The physical point is more general: **a real constituent of a complete configuration need not itself be an admissible isolated asymptotic particle.**

8.8 Completing the fractional structure

The source asks for the smallest cyclic fractional classification with distinct nonzero conjugate orientations. Two classes fail because the only nonzero residue is its own negative; three are sufficient:

$$\begin{aligned} &\mathbb{Z}_2: \quad 1 = -1 \pmod 2, \\ &\mathbb{Z}_3: \quad 1 \neq -1 = 2 \pmod 3, \qquad 1 + 2 = 0 \pmod 3, \qquad 1 + 1 + 1 = 0 \pmod 3. \end{aligned} \tag{8.16}$$

Within that declared classification, nonzero constituent residues can therefore combine to a complete zero residue. Let r_a denote those residues while X contains the actual currents, support, framing, connection, and geometry. The composite domain is

$$\mathcal{D}^{(0)}_{\mathbb{Z}_3} = \left\{ X : \sum_a r_a = 0 \pmod 3 \right\}, \qquad X \in \mathcal{D}^{(0)}_{\mathbb{Z}_3}, \qquad \delta S_J[X] = 0. \tag{8.17}$$

Zero residue means completeness in this internal bookkeeping, not zero electric charge. The physical state still has to solve the full joint problem. A transition into this domain forms the complete composite; it does not first require independently lawful fractional remnants that the same domain forbids. [3, §§22-23]

The whole can be lawful even when separately projected pieces are not lawful isolated objects. Binding strength, size, and the actual hadron spectrum remain properties of the nonlinear composite solution, not consequences of modular arithmetic alone. [2, §11]

8.9 Strong response must resolve the internal organization

A response depending only on total scalar density cannot distinguish different internal decompositions with the same sum:

$$\rho_{\text{tot}} = \sum_a \rho_a = \sum_a \tilde{\rho}_a \quad \implies \quad S[\rho_{\text{tot}}] = S[\tilde{\rho}_{\text{tot}}]. \tag{8.18}$$

Any claimed port or color response therefore needs variables that retain those distinctions. Renaming an undifferentiated scalar attraction cannot create the missing internal information. [3, §23.2]

V13.4 supplies a conditional matrix-valued response class on a complex three-component internal fiber. Removing the common Abelian direction gives a trace-free unitary connection of the $SU(3)$ type,

$$D_\mu = \partial_\mu - i\mathcal{A}_\mu, \qquad \mathcal{F}_{\mu\nu} = \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - i[\mathcal{A}_\mu, \mathcal{A}_\nu]. \tag{8.19}$$

with curvature action

$$S_s = -\frac{1}{2g_s^2}\int \mathrm{Tr}(\mathcal{F}\wedge *\mathcal{F}) + S_{\mathrm{comp}}[\mathcal{A}]. \tag{8.20}$$

One normalization $g_s(\mu_0)$ controls the curvature sector. Because the curvature already contains the noncommuting connection, its cubic and quartic self-interactions are not separately tunable additions if this action is retained. This establishes a coherent response class, not a solved hadron. [2, §12; 3, §§23-24]

The pure curvature term also cannot stabilize a finite static size by itself. Under the source's allowed spatial rescaling,

$$\begin{aligned} \mathcal{A}_i^{(\lambda)}(\mathbf{x}) &= \lambda \mathcal{A}_i(\lambda\mathbf{x}), \\ \mathcal{F}_{ij}^{(\lambda)}(\mathbf{x}) &= \lambda^2 \mathcal{F}_{ij}(\lambda\mathbf{x}), \qquad E_{\mathrm{curv}}(\lambda) = \lambda E_{\mathrm{curv}}(1). \end{aligned} \tag{8.21}$$

so the nonzero curvature energy scales linearly with λ and has no finite stationary point along that direction. Material, source, support, or other coupled terms must provide the missing balance. The recorded unstable spherical reduction therefore tells the program what a successful composite must add; it is not a program killer. [3, §§23.5, 45]

8.10 A complete proton can enter an effective atom

A lower-energy application does not need to resolve every internal proton variable each time it uses the proton's exterior mass and charge. V13.3 therefore treats the proton as a guarded effective participant in the first-atom calculation. Its selected constituent charge labels satisfy

$$Q_p = 2Q_u + Q_d = 2\left(\frac{2}{3}\right) - \frac{1}{3} = 1, \qquad Q_n = Q_u + 2Q_d = \frac{2}{3} - 2\left(\frac{1}{3}\right) = 0. \tag{8.22}$$

These sums reproduce the familiar uud and udd charge assignments. They do not construct a proton or neutron by adding three isolated quark solutions. [4, §2; 6, §§6-8]

The proton mass contains a much larger unresolved internal contribution in the source's seed bookkeeping:

$$E_{\mathrm{field},p}^{\mathrm{target}} = m_p c^2 - (2m_u + m_d)c^2, \qquad (2m_u + m_d)c^2 \simeq 9.1\ \mathrm{MeV}. \tag{8.23}$$

Using that paper's supplied proton anchor gives a remainder of about 929.17 MeV. This number is a target defined by subtraction, not a computed confinement energy and not a threshold for separating free quarks. [6, §8]

Once the effective proton is used, its quoted mass already contains that interior. Adding the remainder again would double count the same physical source. A later resolved proton model can replace the effective interior only by reproducing the same exterior observables at the scale where the effective description was valid. This is the same one-source accounting used for the electron's self-dressing and interaction field. [1, §9.4; 2, §§4-5]

8.11 Binding changes the whole

For constituents with a defined separated reference, binding compares two complete configurations. Let the isolated rest energies sum to E_{sep} and define positive binding energy B by

$$\begin{aligned} E_{\mathrm{sep}} &= \sum_i m_i c^2, \qquad B = E_{\mathrm{sep}} - E_{\mathrm{bound}} > 0, \\ M_{\mathrm{bound}} &= \sum_i m_i - \frac{B}{c^2}. \end{aligned} \tag{8.24}$$

The constituent masses and the joint energy must use one consistent approximation and reference convention. B is an energy difference, not another substance added to the bound state. Internal motion, shared fields, and support can all change when the bound configuration forms.

The leading hydrogen application then has

$$q_H = q_p + q_e = 0, \qquad M_H c^2 \simeq (m_p + m_e)c^2 - B_H, \qquad B_H \simeq 13.5983 \text{ eV}. \tag{8.25}$$

with the displayed binding inherited from the atomic calculation's supplied masses, electromagnetic normalization, and localization law. The proton and electron retain their constituent species while the atom has a new total mass and zero net charge. It is neither an annihilated pair nor a larger electron. [6, §§10-12]

Actual capture also needs somewhere for the released energy to go. An energy minimum identifies a bound configuration; the formation process must carry the difference into radiation, recoil, or another admitted channel. This is the same distinction already encountered between a maintained particle and the dynamics that forms it.

A composite can therefore become a new effective participant while its constituents retain their own identities and local histories. The whole adds new relations and new readouts; it does not erase the structures from which it is built.

8.12 What carries forward

The species-and-composite account now has several clean constraints. Structural identity proposes common intrinsic readouts across instances. Primitive-root non-scalability says multiplicity requires a real joint solution. Root-surplus separation lets energy and motion vary without continuously redefining the species. The composition theorem identifies the exact additional condition behind antisymmetry. Fractional-composite admissibility shows how a lawful whole can contain projected sectors that are not isolated particles. The strong-response analysis identifies what a port-resolved composite law must contain and why pure curvature is not enough. [1, §10; 2, §§4, 11-12, 16, 18]

These results do not compete with one another; they organize different parts of the same question. The remaining program is to identify the recurrent classes, derive their composition and statistics, and solve the complete sources that maintain larger structures.

An electron remains an electron by continuing as its maintained organization. A composite becomes a new participant through the organization its constituents maintain together. The complete source is still counted once, even when a simpler effective description becomes useful.

The next section follows the processes that can change identity: **weak transitions, neutral response, and the collective scales that couple to particle families.**

Sources and scope

[1] *SFT V13.13 — Live Ontology Constitution*, Revision B, especially §§4-5, 9-10, 12-13. Governs structural identity, instance language, the separation from exchange statistics, and one-source accounting. Structural identity remains a candidate explanation rather than a proved uniqueness theorem.

[2] Shaun Fosmark and GPT-5.6 SOL. *SFT V13.4-V2 — Relational Interactions: Persistent Particle Roots and Complete Joint Configurations*, Final v1.0, especially §§4-5, 11-12, 16, 18-21. Governing public interaction source.

[3] *SFT V13.4 Relational Interaction Update Blend v0.11*, controlling technical companion named by [2]. Used here for charge/composite structure, exchange and root multiplicity, root-surplus separation, normal-response modes, and the strong-root boundary. The normalized $J \pm K$ example only expands the stated consequence for orthonormal inputs; no older fitted force constants are imported.

[4] Shaun Fosmark. *SFT Particle Reality Inventory*, June 2026. Used only as a research-priority map distinguishing persistence, transient states, internal response, and speculative/representational categories. Its printed counts and blanket nonexistence claims are not promoted.

[5] *SFT V13.12 — The Maintained Electron*, public v0.2, and Section 3 of this reader. Supplies the concrete maintained reference and its parameter/scale boundary. The narrative companion *What the Particle Remembers* informs the structural question; [1] and the technical sources control uniqueness and statistics claims.

[6] *SFT V13.3 — The First Locally Real Atom*, R257A freeze, June 2026, especially §§6, 8, 10-13. Used here only for the guarded proton, charge sums, unresolved interior-energy target, and binding accounting. The localization derivation is reserved for Section 10.

Mathematical scope. All 25 displayed equations retain their V13 meanings. The three normal-response directions belong to one finite local tangent problem, not a universal species count. The bilinear exchange theorem classifies possible composition; it does not select electron statistics. Winding and $\mathbb{Z}_3$ statements depend on their declared boundary/composite representations. The $SU(3)$ response class is conditional and does not by itself produce a stable hadron. No new particle root, lifetime, mass spectrum, hadron, formation, atomic, or scattering solve is introduced here.

9. Weak transitions, neutral response, and collective scales

An interaction can change a particle's motion while preserving its identity. A transition asks something deeper: **which maintained organization continues out of the encounter?** In SFT, changing species is not a replacement of the world's physical stock. It is a change in the organization that the causal support maintains. [1, §§2, 13; 2, §§3-4]

V13.4 gives that proposal a conditional mathematical architecture. It separates four jobs that are easy to blur together: which sector may participate, how a shared collective response loads charged and neutral directions, how families can respond differently, and how an allowed channel acquires an actual transition current. The resulting model ties several leading quantities together without claiming that the frozen electron has already generated the full weak sector. [1, §§13-17, 19-21]

The progression is therefore **access → common loading → scale → radial response → family structure → open-channel current**. Most field formulas below use $c = \hbar = 1$; dimensional range and time-evolution formulas restore the constants explicitly.

9.1 Which part of the state can make the transition?

Chirality here is an orientation property of the spinorial response: proper Lorentz transformations preserve it while spatial parity reverses it. It is distinct from electric-charge sign and from a spin component measured along an apparatus axis. The source represents the relevant orientation by a Hermitian involution Ξ and asks for the leading access map built from it:

$$\Xi^\dagger = \Xi, \qquad \Xi^2 = I, \qquad \Pi = aI + b\Xi, \qquad a, b \in \mathbb{R}. \tag{9.1}$$

If access is projective, applying it twice must select the same sector. Squaring (9.1) gives

$$\Pi^2 = (a^2 + b^2)I + 2ab\Xi = \Pi \quad \Longrightarrow \quad a^2 + b^2 = a, \qquad 2ab = b. \tag{9.2}$$

For a nontrivial orientation-dependent projector, $b \neq 0$, so $a = 1/2$ and $b = \pm 1/2$:

$$\boxed{\Pi_\varepsilon = \frac{I + \varepsilon\Xi}{2}, \qquad \varepsilon = \pm 1.} \qquad \Pi_+\Pi_- = 0, \qquad \Pi_+ + \Pi_- = I. \tag{9.3}$$

This is the **chiral-projector classification**. Within this class there is no continuously adjustable degree of projective chirality. The algebra leaves three qualitatively different possibilities: no access, full vectorlike access, or one of the two complementary chiral sectors. Which physical transition actually uses them is a dynamical question. [1, §13; 2, §46]

The direction of an internal branch change is a separate operation. Let $T_\pm$ raise or lower that branch label while ε labels chirality:

$$\begin{aligned} \mathcal{W}^{\chi}_{\varepsilon,\pm} &= \Pi_\varepsilon \otimes T_\pm, & \mathcal{W}^{V}_{\pm} &= I \otimes T_\pm, \\ \mathcal{P}\Xi\mathcal{P}^{-1} &= -\Xi, & \mathcal{P}\Pi_\varepsilon\mathcal{P}^{-1} &= \Pi_{-\varepsilon}. \end{aligned} \tag{9.4}$$

The tensor factors keep chirality and branch transfer from becoming the same sign convention. If the complete action and preparation are parity symmetric, the mirrored access remains lawful. The remaining

physics is therefore not a fitted amount of chirality; it is the law or preparation that selects one access class. [2, §§47, 57.8]

9.2 One collective response loads the same two-branch space

The minimal V13.4 construction places the collective response in the same local two-component complex branch space, $E \simeq \mathbb{C}^2$, on which the internal connection acts. This is the source's *same-fiber* premise. It is not the two spatial spin directions of Section 5, even though both two-state algebras can be represented with Pauli matrices. [1, §14; 2, §49]

Let h be the positive Hermitian response metric. Invariance under every local $SU(2)$ basis rotation in the irreducible doublet gives

$$U^\dagger \mathsf{h} U = \mathsf{h} \quad \text{for every } U \in SU(2), \qquad \mathsf{h} > 0 \quad \Longrightarrow \quad \boxed{\mathsf{h} = K_L I_2, \qquad K_L > 0.} \tag{9.5}$$

The leading internal loading is therefore common in every direction of this minimal representation. The symmetry does not permit several unrelated stiffnesses to be assigned independently and then called one invariant response. [2, §49]

A nonzero collective orientation is chosen so that one electromagnetic direction remains unbroken. Using the charge dictionary from Section 8 and $T_3 = \frac{1}{2}\,\mathrm{diag}(1,-1)$,

$$Q = T_3 + \frac{n_Y}{6} I_2, \qquad Q\Phi_\star = 0, \quad \Phi_\star \neq 0 \quad \Longrightarrow \quad |n_Y| = 3. \tag{9.6}$$

One component must therefore be neutral. For the formulas below choose $n_Y = 3$ and the lower neutral orientation; the conjugate convention reverses that choice. This labels the collective configuration, not a change in the electron's own charge. [2, §49]

9.3 Charged response, neutral response, and the surviving light direction

Absorb K_L into a canonical amplitude $v > 0$ and use canonically normalized connection fields. In the source's plus-i convention,

$$D_\mu = \partial_\mu + i\left(g_W W_\mu^a T_a + g_Y B_\mu Y\right), \quad T_a = \frac{\sigma_a}{2}, \quad Y = \frac{I_2}{2}, \quad \Phi_\star = \frac{v}{\sqrt{2}}\begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{9.7}$$

Here the Pauli matrices act on the internal branch factor. g_W and g_Y are the two canonical connection couplings, and B_μ is the Abelian connection of this sector rather than the magnetic field.

For one real connection component, the positive loading norm is

$$\begin{aligned} (g_W W^a T_a + g_Y B Y)\Phi_\star &= \frac{v}{2\sqrt{2}}\begin{pmatrix} g_W(W^1 - iW^2) \\ -g_W W^3 + g_Y B \end{pmatrix}, \\ \mathcal{Q}_{\mathrm{load}} &= \frac{v^2}{8}\left[g_W^2\{(W^1)^2 + (W^2)^2\} + (g_W W^3 - g_Y B)^2\right]. \end{aligned} \tag{9.8}$$

Reading these coefficients in the canonical $\frac{1}{2}m^2$ convention gives equal charged loading and a rank-one neutral block:

$$m_W^2 = \frac{g_W^2 v^2}{4}, \qquad \mathsf{H}_0 = \frac{v^2}{4}\begin{pmatrix} g_W^2 & -g_W g_Y \\ -g_W g_Y & g_Y^2 \end{pmatrix}, \qquad \det \mathsf{H}_0 = 0. \tag{9.9}$$

One neutral linear combination is therefore unloaded while its orthogonal combination is massive. Write the two normalized orientations as

$$\begin{gathered} \tan\theta_W = \frac{g_Y}{g_W}, \qquad \mathbf{n}_A = \begin{pmatrix} \sin\theta_W \\ \cos\theta_W \end{pmatrix}, \qquad \mathbf{n}_Z = \begin{pmatrix} \cos\theta_W \\ -\sin\theta_W \end{pmatrix}, \\ \mathsf{H}_0\mathbf{n}_A = 0, \qquad \mathsf{H}_0\mathbf{n}_Z = m_Z^2\mathbf{n}_Z, \qquad m_Z^2 = \frac{v^2}{4}(g_W^2 + g_Y^2). \end{gathered} \tag{9.10}$$

A labels the surviving electromagnetic direction and Z the orthogonal neutral massive response. These are directions in the internal (W^3, B) space, not directions in ordinary space. Their eigenvalues give

$$\boxed{\frac{m_W}{m_Z} = \frac{g_W}{\sqrt{g_W^2 + g_Y^2}} = \cos\theta_W.} \tag{9.11}$$

This is the important shared-response result: **the charged and neutral scales are related because one collective configuration loads them both.** The relation follows from the minimal common norm; it does not by itself choose the numerical angle, overall scale, or unique physical realization. [1, §§13-14; 2, §§30, 48-49]

9.4 The electromagnetic normalization is used once

The massless direction in (9.10) must be the same electromagnetic sector that already dresses and couples the electron. In the source's geometric-connection convention the charge unit is absorbed into the connection, so write

$$\mathcal{K}_W = \frac{1}{g_W^2}, \qquad \mathcal{K}_Y = \frac{1}{g_Y^2}, \qquad \mathcal{K}_{\mathrm{EM}} = \mathcal{K}_W + \mathcal{K}_Y = \frac{1}{e^2}. \tag{9.12}$$

Relative to Section 5, $\mathcal{K}_{\mathrm{EM}} = K_{\mathrm{EM}}/q_0^2$ and $e = q_0/\sqrt{K_{\mathrm{EM}}}$, so $\alpha = e^2/(4\pi)$. This is the same physical normalization written in different coordinates, not an additional electromagnetic fit. [2, §51]

The orientation then fixes the two weak couplings once e is supplied:

$$\begin{aligned} r_{\mathrm{EW}} &= \frac{\mathcal{K}_W}{\mathcal{K}_Y} = \frac{g_Y^2}{g_W^2} = \tan^2\theta_W, \\ e &= g_W \sin\theta_W = g_Y \cos\theta_W, \\ g_W &= \frac{e}{\sin\theta_W}, \qquad g_Y = \frac{e}{\cos\theta_W}. \end{aligned} \tag{9.13}$$

At this leading scope, electric, weak-isospin, and hypercharge normalizations are therefore not three independent knobs. The electromagnetic normalization has already been paid for; choosing the relative orientation supplies the remaining ratio. [1, §14; 2, §51]

9.5 Shape does not select the absolute weak scale

The response matrix separates dimensionless geometry from its overall scale. Define $\mu_{\mathrm{EW}} = v/2$:

$$\mathcal{M}^2_{\mathrm{EW}} = \mu^2_{\mathrm{EW}} \widehat{\mathcal{M}}^2_{\mathrm{EW}}, \qquad \mu_{\mathrm{EW}} \mapsto s\mu_{\mathrm{EW}} \quad \Longrightarrow \quad m_i \mapsto s m_i, \qquad s > 0. \tag{9.14}$$

The rank, null direction, eigenvectors, and mass ratio remain unchanged under the scaling. The geometry therefore does not select the absolute weak scale. Relative to a maintained reference root of mass m_A, V13.4 can define the dimensionless target $\xi_{\mathrm{EW}|A} = \mu_{\mathrm{EW}}/m_A$; the present construction does not derive it. [2, §§48, 50]

Once a positive mass curvature exists, however, its static range follows immediately. Restoring dimensions,

$$\left[-\nabla^2 + \left(\frac{m_i c}{\hbar}\right)^2\right] w_i = j_i, \qquad \ell_i = \frac{\hbar}{m_i c}, \qquad m_i > 0. \tag{9.15}$$

The null electromagnetic direction lacks this mass-loading term, whereas the massive directions acquire finite characteristic ranges. The same canonical construction also gives $g_W^2/m_W^2 = 4/v^2$, so the leading low-energy strength is controlled by the common scale rather than by another independent normalization. Channel and generator factors still belong to the particular transition. [1, §13; 2, §§48, 51]

9.6 One radial amplitude remains

A complex doublet has four real components. Once a nonzero orientation and its three gauge-orientation directions are accounted for, one gauge-invariant radial amplitude remains. V13.4 interprets this as a **Higgs-like amplitude response** of the collective configuration rather than, by that fact alone, a newly derived fundamental scalar substance. [1, §15; 2, §52]

Let $\rho = \Phi^\dagger \Phi$ and write the radial displacement as χ_H:

$$\Phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + \chi_H \end{pmatrix}, \qquad \rho = \frac{(v+\chi_H)^2}{2}, \qquad \rho_\star = \frac{v^2}{2}, \qquad V'(\rho_\star) = 0. \tag{9.16}$$

For the canonical radial kinetic normalization used by the source, the curvature of the invariant potential gives

$$\begin{aligned} \frac{dV}{d\chi_H} &= (v+\chi_H) V'(\rho), \\ \left.\frac{d^2V}{d\chi_H^2}\right|_0 &= V'(\rho_\star) + v^2 V''(\rho_\star) = \boxed{m_H^2 = v^2 V''(\rho_\star)}. \end{aligned} \tag{9.17}$$

The equilibrium location and the radial stiffness are different information. The audit makes that independence explicit by deforming an already admitted potential without moving its minimum:

$$\begin{aligned} V_\delta(\rho) &= V_0(\rho) + \delta\lambda(\rho - \rho_\star)^2, \\ V_\delta'(\rho_\star) &= 0, \qquad V_\delta''(\rho_\star) = V_0''(\rho_\star) + 2\delta\lambda, \\ \zeta_H &= \frac{m_H^2}{m_W^2} > 0 \quad \text{on the strictly stable canonical branch.} \end{aligned} \tag{9.18}$$

The weak-loading geometry can therefore remain fixed while the radial mass changes. This is a successful exclusion of the stronger claim that the equilibrium alone predicts the scalar mass. One dimensionless radial-curvature input remains. [2, §§52-53]

Other consequences of the same loading are less free. The massive-direction coefficients change with the radial amplitude as

$$m_i^2(\chi_H) = m_i^2(0)\left(1+\frac{\chi_H}{v}\right)^2, \qquad \left.\frac{dm_i^2}{d\chi_H}\right|_0 = \frac{2m_i^2}{v}, \qquad i = W, Z. \tag{9.19}$$

Thus the model leaves the radial stiffness to determine, but it does not permit the leading radial response to W and Z loading to be tuned independently. [1, §15; 2, §52]

9.7 Families require something that distinguishes families

Section 8's three normal-response directions did not by themselves produce three masses. If a collective response acts identically on every family direction, it cannot generate a hierarchy:

$$H_{\text{blind}} = I_3 \otimes H_{\text{collective}}, \qquad Y_{gH} \not\propto I_3 \quad \text{is required for family-sensitive loading.} \tag{9.20}$$

Y_{gH} is the proposed family-collective cross-response. The physical content is simple: **a mechanism cannot distinguish families if its operator is blind to family.** Actual masses still require the on-shell roots and their coupling to that response. [1, §16; 2, §57.6]

Mixing asks whether the mass response and transition response distinguish the same combinations. If both are only functions of the same cyclic incidence matrix,

$$H_m = f(P_3), \qquad H_w = g(P_3), \qquad P_3^3 = I_3 \quad \Longrightarrow \quad [H_m, H_w] = 0. \tag{9.21}$$

For the self-adjoint operators considered here, they share an eigenbasis; relabeling it cannot create physical mixing. Let U_m and U_w collect the eigenvectors of the actual two response operators. Their relative basis matrix is

$$U_{\text{rel}} = U_m^\dagger U_w, \qquad U_{\text{rel}}^\dagger U_{\text{rel}} = I. \tag{9.22}$$

This displays the role a noncommuting second response could play in CKM- or PMNS-type mixing. It is not a prediction of the observed matrices. The angles and phases belong to the physical operators that actually produce the basis misalignment. [1, §16; 2, §57.7]

9.8 Neutrality, small mass, and weak access are different properties

A neutral maintained root can have rest energy and a local cadence while its net electromagnetic charge vanishes. Weak access is another property again. V13.4 therefore does not obtain a neutrino by setting the electron's electric coupling to zero and leaving the rest of its structure untouched. [2, §57.7]

The source keeps three possible mechanisms open: suppressed coupling to the collective response, a mismatch between the mass and weak-access bases, or a lawful self-conjugate neutral composition. These answer different questions. Suppression controls magnitude; basis misalignment controls which combinations participate; self-conjugacy concerns the composition of the neutral state itself.

No neutral mass, heavy-partner scale, or Dirac/Majorana choice follows from those possibilities yet. The positive point is that **neutrality, a small mass response, and weak participation need not have the same physical cause.** That separation gives the neutral-root program concrete things to derive rather than a single label called neutrino. [2, §57.7]

9.9 Permission is not a transition rate

The projector of §9.1 says which sector is accessible. It does not say how fast amplitude moves into that sector. V13.4 makes the missing dynamics explicit with a larger self-adjoint system. Restore $\hbar$ and let $p(t)$ denote retained response amplitudes and $q(t)$ an additional responding sector:

$$i\hbar\frac{d}{dt}\begin{pmatrix}p\\q\end{pmatrix}=\begin{pmatrix}H_{PP}&V_{PQ}\\V_{PQ}^{\dagger}&H_{QQ}\end{pmatrix}\begin{pmatrix}p\\q\end{pmatrix}. \tag{9.23}$$

The Hamiltonian blocks and V_{PQ} have energy units. The amplitudes are response variables of this declared model, not a list of already identified decay products. Their norm balance is exact:

$$\begin{gathered}\frac{d\|p\|^2}{dt}=\frac{2}{\hbar}\operatorname{Im}\langle p,V_{PQ}q\rangle,\qquad \frac{d\|q\|^2}{dt}=-\frac{2}{\hbar}\operatorname{Im}\langle p,V_{PQ}q\rangle,\\ \frac{d}{dt}(\|p\|^2+\|q\|^2)=0.\end{gathered} \tag{9.24}$$

What leaves one sector enters the other. A closed finite system may exchange response back and forth, so loss from p need not be monotonic. **A disappearing reduced amplitude is not yet a physical decay explanation until the receiving sector is part of the model.** [2, §7.1]

For constant H_{QQ}, the second block integrates to

$$q(t)=e^{-iH_{QQ}(t-t_0)/\hbar}q(t_0)-\frac{i}{\hbar}\int_{t_0}^{t}e^{-iH_{QQ}(t-t')/\hbar}V_{PQ}^{\dagger}p(t')\,dt'. \tag{9.25}$$

Substituting this into the retained equation produces dependence on earlier p and on the initial q preparation. The resulting memory belongs to the response of omitted physical variables; it is not a second substance inserted to represent delay. [1, §17; 2, §57.3]

9.10 Open channels give widths

At stationary outgoing-response scope, eliminating the open sector gives

$$\begin{aligned}\Sigma^{R}(E)&=V_{PQ}(E-H_{QQ}+i0)^{-1}V_{PQ}^{\dagger}=\Delta(E)-\frac{i}{2}\Gamma(E),\\ \Gamma(E)&=2\pi V_{PQ}\delta(E-H_{QQ})V_{PQ}^{\dagger}\succeq 0.\end{aligned} \tag{9.26}$$

Δ is the reactive part of the response and Γ the positive-semidefinite outgoing width. A closed off-resonant sector can shift a state while giving $\Gamma=0$, so *influencing a state* and *providing an open decay channel* are distinct physical roles. [2, §6]

For independently resolved outgoing channels, $\Gamma=\sum_i\Gamma_i$, and the stationary current is

$$J_i^{P\to Q} = \frac{1}{\hbar}\langle p, \Gamma_i p\rangle \geq 0, \qquad \sum_i J_i^{P\to Q} = -\frac{d\|p\|^2}{dt}. \tag{9.27}$$

The current has inverse-time units. Turning this stationary result into a species lifetime requires actual outgoing channels and the approximation that makes the rate interpretation valid. A finite closed spectrum is not automatically a decay bath, and the general time-dependent reduction can retain the memory of (9.25). [2, §§7.2-7.3]

This keeps three quantities separated cleanly. A recurrent particle supplies a clock unit. An open-channel coupling supplies a transition width. A detector supplies the further dynamics that turns one completed transition into a durable record.

9.11 What this construction actually connects

At its leading scope, V13.4 leaves the continuous input inventory

$$\alpha(\mu_0), \qquad g_s(\mu_0), \qquad r_{\mathrm{EW}}, \qquad \mu_{\mathrm{EW}}, \qquad \zeta_H. \tag{9.28}$$

The electromagnetic input has already appeared in Section 5 and the strong normalization in Section 8. The present construction adds the relative electroweak orientation, the common scale, and the radial curvature. The electron's formation/support normalization and dimensional anchor remain separately owned; family operators, neutral-root dynamics, and higher-order corrections are not hidden inside this five-entry list. [1, §19]

What has been gained is linkage. The same minimal collective configuration produces the charged/neutral mass relation, leaves one electromagnetic direction unbroken, carries the already normalized electromagnetic coupling into the weak connection, and fixes the leading radial response of the massive directions once the radial amplitude is specified. The open-channel construction then shows what an actual transition model must add: a receiving sector and a current into it.

Access, loading, mixing, and decay are different physical jobs. V13 has mathematical structures connecting them, without pretending that one label or one symmetry automatically performs all four. That is a stronger foundation for continued work than fitting each observed quantity independently. [1, §§20-22]

The weak and neutral sectors remain conditional collective/transition models rather than a complete finite-particle derivation from Electron V1. Their remaining scale selection, family operators, neutral roots, and microscopic transition dynamics are direct targets for V14 Part 2. [2, §73.11]

Next we use the participants already assembled to make something larger that persists: an atom. The particle identity, electromagnetic interaction, composition law, and complete energy accounting now meet in the first bound-state application.

Sources and scope

[1] Shaun Fosmark and GPT-5.6 SOL. *SFT V13.4-V2 — Relational Interactions: Persistent Particle Roots and Complete Joint Configurations*, final public replacement v1.0, August 2026. Primary scopes: §§13-17 (weak access, same-fiber response, radial amplitude, family response, open channels), §19 (input inventory), §§20-22 (scope and physical interpretation). This chapter preserves the paper's conditional electroweak/Higgs-like status; no measured weak mass or coupling is promoted to a prediction.

[2] *SFT V13.4 Relational Interaction Update Blend v0.11 — Electron/Force Integration*, exact controlling derivation identified by [1], SHA-256 cd561c041eab7c067a46a1d6c56f3670700aa862685be0d67cf81732cec1100e. Used scopes include §§3-7, §30, §§46-53, §§57.3 and 57.6-57.8, §73.11 and §74. The final public noncommuting-family correction in [1, §16] governs the flavor summary. The three neutral mechanism classes remain proposals; no specific seesaw matrix, neutral mass, or Dirac/Majorana selection is supplied.

Interpretive and normalization scope. Equations (9.7)-(9.11) use the source's fundamental-doublet normalization $T_a = \sigma_a/2$, $Y = I_2/2$, canonical amplitude v, and $\Phi_\star = (0, v)/\sqrt{2}$. The positive component loading form is not asserted to be the sign of a complete Lorentzian action. The geometric electromagnetic stiffness $\mathcal{K}_{\mathrm{EM}}$ absorbs the charge unit and is related explicitly to Section 5's K_{EM} and q_0. The common-scale convention is $\mu_{\mathrm{EW}} = v/2$. The radial mass readout assumes the source's canonical kinetic normalization. The chirality sign ε and the internal raising/lowering sign are independent operations.

Model boundaries. The three normal-response directions of Section 8 are not promoted to generations here. Equation (9.22) describes the relative eigenbasis of already supplied response operators and is not a numerical CKM or PMNS prediction. Equation (9.25) is the exact integrated second block of (9.23) for constant blocks, not a universal microscopic memory law. The stationary width and current in (9.26)-(9.27) are not promoted to time-local decay for arbitrary finite or nonstationary dynamics.

Final-edit scope. This pass changes exposition only. No weak-transition, neutral-root, scalar-resonance, particle-spectrum, thermal, or experimental solver is run, and no inherited scientific verdict is changed. New physical construction remains assigned to V14 Part 2.

10. The first locally real atom

A proton and an electron can remain distinct particles while becoming a physical whole that neither is alone: a neutral, bound atom. Their identities persist; what changes is the organization they maintain together. **The relationship acquires a finite scale, a lower energy, and a new total mass.** [1, §§10–12; 4, C11.S02]

V13.3 makes this step quantitative. A supplied effective proton contributes an attractive Coulomb exterior, while the admitted electronic support carries a positive localization cost. Those two terms select the reduced-mass Bohr scale and the leading hydrogen binding from one energy minimum. With the declared masses and electromagnetic normalization, the retained result is approximately **52.9465 pm and 13.5983 eV**. They are linked outputs, not separately fitted targets. [1, §§9–11; 2, R255A]

This section follows that one balance from the participants to the atom. The proton interior remains an effective input and the localization law remains the one supplied by V13.3. That is enough to establish the leading bound relationship without pretending that proton confinement, precision spectroscopy, or capture dynamics have already been solved.

10.1 The participants arrive with their identities

The atomic problem begins with a formed electron and an effective proton. The electron brings its positive rest mass, negative charge, and maintained identity. The proton brings its complete effective mass and positive exterior charge. With the charge convention already used for interactions,

$$Q_p = +1, \qquad Q_e = -1, \qquad q_i = q_0 Q_i, \qquad E_{\mathrm{sep}} = (m_p + m_e)c^2. \tag{10.1}$$

Q_i is the dimensionless charge label and q_0 the common charge unit. E_{msep} is the rest-energy reference of the separated pair with no relative kinetic energy. [1, §§8–11; 3, §§8, 10]

The proton is not reassembled here from free quarks. Its unresolved confinement and support are already contained in the supplied proton mass. At this scale the atom consumes that complete effective participant and its exterior response once. A future resolved proton must reproduce the same exterior rather than being added on top of it. [1, §8; 4, C11.S01–C11.S02]

The electron also keeps two roles distinct. Section 3 asked what maintains the electron itself. This section asks how that already maintained participant is distributed in a bound relation with the proton. **The atomic support scale is a property of the joint state, not a replacement for the electron's own internal support.** [1, §12]

Being complete as a particle therefore does not mean being independent in every configuration. The atom is the next physical whole.

10.2 The proton supplies an attractive exterior

The original atom paper describes the charged exterior through conserved directional flux. At the present interpretation, Section 5 supplies its electromagnetic current/connection meaning; the useful content here is the spherical flux law and Coulomb normalization. [1, §9; 3, §§8–10]

For a spherically symmetric exterior outside the source,

$$A(r) = 4\pi r^2, \qquad \Phi_Q = \int_{S_r} \mathbf{D}_Q \cdot d\mathbf{S} = 4\pi r^2 D_Q(r), \qquad D_Q(r) = \frac{\Phi_Q}{4\pi r^2}. \tag{10.2}$$

The area growth dilutes the radial response as r^{-2}. This is the exterior of a specified source and conserved flux; geometry alone does not choose the source law. [1, §9]

With interaction energy zero at infinite separation,

$$\begin{aligned} C_{\mathrm{EM}} &= \alpha\hbar c, \qquad V_{pe}(r) = \frac{C_{\mathrm{EM}} Q_p Q_e}{r} = -\frac{\alpha\hbar c}{r}, \\ F_r &= -\frac{dV_{pe}}{dr} = -\frac{\alpha\hbar c}{r^2} < 0, \qquad r > 0. \end{aligned} \tag{10.3}$$

Since increasing r points outward, $F_r < 0$ is attractive. The same electromagnetic response that raises the energy of equal charges lowers the energy of this opposite-charge pair as they approach. [1, §§9–10; 4, C11.S02]

The mutual contribution is the cross-energy of the shared field relative to the isolated references, not a third electromagnetic stock added to two dressed particles. Attraction explains why smaller separations are favored. It does not yet explain why contraction stops.

10.3 Both participants move

The proton is heavy, not infinite. In the center-of-momentum description the two translational momenta are $\mathbf{p}$ and $-\mathbf{p}$, so the relative kinetic energy uses the reduced mass:

$$K_{\mathrm{rel}} = \frac{|\mathbf{p}|^2}{2m_e} + \frac{|\mathbf{p}|^2}{2m_p} = \frac{|\mathbf{p}|^2}{2\mu}, \qquad \frac{1}{\mu} = \frac{1}{m_e} + \frac{1}{m_p}, \qquad \mu = \frac{m_e m_p}{m_e + m_p}. \tag{10.4}$$

This is the mechanical meaning of the reduced-mass input. It concerns the translation of the complete particles relative to one another, not the locally null support inside either particle. A nonrelativistic atomic reduction therefore does not say that the electron's internal causal propagation has slowed. [1, §10; 4, C11.S02]

Because $\mu < m_e$, finite proton recoil slightly enlarges the preferred atomic scale and slightly lowers the binding. Both changes will follow from the same factor rather than from a second correction parameter.

10.4 Localization supplies the competing cost

Write the atomic support length as $a > 0$, reserving r for pointwise separation in the exterior potential. V13.3 supplies the one-length energy family

$$\boxed{E(a) = K(a) + V(a) = \frac{\hbar^2}{2\mu a^2} - \frac{\alpha\hbar c}{a}, \qquad a > 0.} \tag{10.5}$$

$E(a)$ is measured relative to the separated pair. The positive term penalizes tighter localization of the admitted electronic support; the negative term is Coulomb attraction in the same family. The coefficient of the positive term is an input of this effective atom model, not a new derivation from Electron V1. No rigid shell or classical circular orbit is required. [1, §10; 2, R255A; 4, C11.S02]

The competition is already visible in its limits:

$$\lim_{a\to 0^+} E(a) = +\infty, \qquad \lim_{a\to\infty} E(a) = 0 \text{ from below.} \tag{10.6}$$

Tight localization drives the positive cost upward faster than attraction can lower the energy; at large support the energy approaches separation from below. The small-a limit is a property of the stated model, not a claim that its nonrelativistic exterior remains valid inside the proton.

The atom is not held open by a wall. A tighter joint configuration costs energy. That positive cost is what competes with attraction.

10.5 The preferred scale follows from one balance

Differentiate the same energy with respect to a:

$$\begin{aligned} \frac{dE}{da} &= -\frac{\hbar^2}{\mu a^3} + \frac{\alpha\hbar c}{a^2}, \\ \frac{dE}{da} = 0 \quad &\Longrightarrow \quad \boxed{a_\mu = \frac{\hbar}{\mu c\alpha}.} \end{aligned} \tag{10.7}$$

No separate radius parameter is introduced after the fact. The stationary point is selected by quantities already present in the energy law. Its curvature is positive:

$$\left.\frac{d^2E}{da^2}\right|_{a_\mu} = \left.\left(\frac{3\hbar^2}{\mu a^4} - \frac{2\alpha\hbar c}{a^3}\right)\right|_{a_\mu} = \frac{\hbar^2}{\mu a_\mu^4} > 0. \tag{10.8}$$

So this one-length family has a local minimum against the variation being tested. Substituting the selected scale gives

$$\begin{aligned} B_H = -E(a_\mu) &= \frac{\mu c^2\alpha^2}{2} > 0, \\ K(a_\mu) = B_H, \qquad V(a_\mu) &= -2B_H, \qquad 2K(a_\mu) + V(a_\mu) = 0. \end{aligned} \tag{10.9}$$

The Coulomb term contributes twice the magnitude of the positive localization cost; their difference leaves the atom below the separated reference. The result is therefore stronger than "collapse is prevented." **A definite finite scale is energetically preferred.** [1, §10; 2, R255A]

10.6 The complete energy curve exposes the result

The whole minimization can be written as a square:

$$\boxed{E(a) + B_H = \frac{\hbar^2}{2\mu}\left(\frac{1}{a} - \frac{1}{a_\mu}\right)^2 \geq 0.} \tag{10.10}$$

Equality occurs only at $a = a_\mu$, so within this positive one-length family the stationary point is the unique global minimum. The dimensionless form makes the shape equally transparent. With $x = a/a_\mu$,

$$\frac{E(a)}{B_H} = \frac{1}{x^2} - \frac{2}{x} = \left(\frac{1}{x} - 1\right)^2 - 1. \tag{10.11}$$

At half the preferred scale the relative energy has already risen to zero; at twice the scale it remains bound at $-3B_H/4$. Compression and expansion are not symmetric because the two terms scale differently.

A small fractional displacement gives

$$E(a) - E(a_\mu) = B_H \frac{\varepsilon^2}{(1+\varepsilon)^2} = B_H\varepsilon^2 + O(\varepsilon^3). \tag{10.12}$$

This is an energy penalty, not yet an oscillation frequency. A breathing frequency would require the inertia and dynamics of that collective mode. Likewise, either term alone fails to produce the atom: attraction alone has no finite minimum, while localization alone produces no binding. **The bound scale belongs to their competition.** [1, §10]

10.7 The numerical hydrogen result

The preserved V13.3 package evaluates the minimum with these declared anchors:

$$\begin{aligned} \hbar c &= 197.3269804 \text{ MeV fm}, \qquad \alpha = \frac{1}{137.035999084}, \\ m_e c^2 &= 0.51099895 \text{ MeV}, \qquad m_p c^2 = 938.2720813 \text{ MeV}. \end{aligned} \tag{10.13}$$

They are the source calculation's inputs. In particular, the atomic value of α is more precise than the rounded 1/137 used in Electron V1; this section preserves the atom calculation as the effective application it actually was rather than silently pretending the two numerical preparations were identical. [1, Appendix A; 2, quick reproduction script; 6, §1.8]

Writing the reduced rest energy as $\mu_E = \mu c^2$ keeps the calculation in MeV and fm:

$$\mu_E = \frac{(m_e c^2)(m_p c^2)}{m_e c^2 + m_p c^2}, \qquad E(a) = \frac{(\hbar c)^2}{2\mu_E a^2} - \frac{\alpha \hbar c}{a}. \tag{10.14}$$

The stored evaluation gives

$$\begin{aligned} \frac{\mu}{m_e} &\simeq 0.999455679421, \\ \boxed{a_\mu \simeq 52.946540966 \text{ pm}}, &\qquad \boxed{B_H \simeq 13.598287264 \text{ eV}}. \end{aligned} \tag{10.15}$$

The digits identify the archived arithmetic, not an uncertainty claim. At the minimum the two contributions are approximately +13.5983 eV and −27.1966 eV, satisfying the same virial balance. [2, R255A]

The infinite-proton result shifts in exactly the way the reduced mass predicts:

$$\frac{a_\mu}{a_0} = \frac{m_e}{\mu} = 1 + \frac{m_e}{m_p}, \qquad \frac{B_H}{B_\infty} = \frac{\mu}{m_e}, \qquad \boxed{B_H a_\mu = \frac{\alpha \hbar c}{2}}. \tag{10.16}$$

Thus finite recoil makes the scale slightly larger and the binding slightly smaller, while their product remains independent of reduced mass in this model. **The two headline numbers move together because they come from one balance.** [1, §10; 2, R255A]

10.8 Neutrality and binding belong to the whole atom

Charge and energy combine differently. The charges cancel while the bound energy lowers the total rest energy:

$$\boxed{Q_H = Q_p + Q_e = 0, \qquad M_H c^2 = m_p c^2 + m_e c^2 - B_H.} \tag{10.17}$$

With the same anchors, $M_H c^2 \simeq 938.783066651713\,\mathrm{MeV}$. The electron's isolated mass has not been changed; the complete joint state lies below the two separated rest energies. Electrical neutrality likewise does not erase the charged constituents or their interaction. [1, §11; 2, R256A]

The binding is tiny compared with the particle rest energies,

$$\frac{B_H}{m_e c^2} \simeq 2.66 \times 10^{-5}, \qquad \frac{B_H}{M_H c^2} \simeq 1.45 \times 10^{-8}. \tag{10.18}$$

but the relevant comparison for binding is the separated threshold, not the entire rest energy required to have the participants in the first place. [2, R256A]

This also gives a concrete one-count rule across descriptive scales. At constituent resolution we retain proton, electron, and binding once. At coarse resolution the effective atom already carries M_H. Adding that atom mass to the same resolved constituents would duplicate one physical configuration. [4, C11.S01–C11.S02]

The atom is more organized than its separated constituents, not made from an additional kind of substance. Neutrality and atomic mass are properties of the complete organization.

10.9 Particle size and atomic size are different questions

The nuclear source, the electron's internal reference scales, and the atomic support length must not be collapsed into one radius. V13.3 first checks the nuclear hierarchy using its supplied proton reference:

$$a_\mu \simeq 52946.5\ \mathrm{fm}, \qquad \frac{r_p}{a_\mu} \simeq 1.5865 \times 10^{-5}. \tag{10.19}$$

The atomic scale is vastly larger, supporting the leading point-source treatment at its characteristic radius while leaving precision finite-size effects to the resolved source problem. [1, §12; 4, C11.S02]

The Compton comparisons make the same distinction from the electron side:

$$\begin{aligned} \bar{\ell}_{C,e} &= \frac{\hbar}{m_e c}, & \ell_{C,e} &= \frac{h}{m_e c} = 2\pi \bar{\ell}_{C,e}, \\ \frac{a_\mu}{\bar{\ell}_{C,e}} &= \frac{m_e}{\mu \alpha}, & \frac{a_\mu}{\ell_{C,e}} &= \frac{m_e}{2\pi \mu \alpha}. \end{aligned} \tag{10.20}$$

The atomic scale is roughly 137 **reduced** electron Compton lengths, not 137 full cycle lengths. Neither quantity becomes a hard electron surface merely because it is useful as a unit.

The historical first-atom report also used

$$r_{\mathrm{spin,e}} = \frac{\hbar}{2m_e c} \simeq 193.08 \text{ fm}, \qquad \frac{a_\mu}{r_{\mathrm{spin,e}}} = \frac{2m_e}{\mu\alpha} = \frac{2}{\alpha}\left(1 + \frac{m_e}{m_p}\right) \simeq 274.22. \tag{10.21}$$

as an internal spin-radius diagnostic. The reduced-mass correction remains explicit. In the current account this is preserved as the source's historical internal reference, not promoted to the later V13.12 support width or to a scattering radius. The durable result is the hierarchy itself: **an atom can change its electronic support scale without redefining the intrinsic electron by the same factor.** [1, §12 and Appendix B; 4, C11.S02]

10.10 Binding and capture are different calculations

A lower-energy atom can exist only if a real encounter carries away the difference between the incoming and final configurations. Let K_{in} be incoming relative kinetic energy and E_{out} include final recoil, radiation, or any other admitted outgoing channel. Then

$$\begin{aligned} (m_p + m_e)c^2 + K_{\mathrm{in}} &= M_H c^2 + E_{\mathrm{out}}, \\ E_{\mathrm{out}} &= K_{\mathrm{in}} + B_H. \end{aligned} \tag{10.22}$$

This is energy bookkeeping, not a capture-rate law. The minimum tells us which configuration is energetically favored once specified; the coupling and outgoing channels determine whether a particular encounter reaches it. [4, C11.S02; 3, §17]

This distinction also keeps two uses of *stability* separate. Electron V1 tests whether the particle's recurrent structure survives its own transverse disturbances. The atom calculation tests whether the relative proton-electron support has a finite energetic minimum. A completed dynamical atom must contain both jobs, but neither inherits the other's certificate merely because both concern persistence.

Formation makes the bound relationship. Binding determines the energy of that relationship once it exists. V13.3 establishes the second at leading scope.

10.11 The first atom opens spectroscopy and chemistry

A preferred ground-state scale is not yet a complete spectrum. Spectroscopy requires an atomic operator, its admitted states, and transition dynamics:

$$H_{\mathrm{atom}}\psi_j = E_j\psi_j, \qquad \Delta E_{ul} = E_u - E_l > 0. \tag{10.23}$$

This display states the next problem; it does not claim that V13.3 already produced fine structure, hyperfine structure, the Lamb shift, transition strengths, or the complete radiation/recoil map. [1, §§13–14; 4, C11.S02]

Adding electrons then adds another joint-composition problem: accessible states, exchange, screening, and interaction energy have to be specified together. V13.6 carries the existing program into shell capacities, qualitative filling, and shared-support bonding. [5, §§7–9]

The first-atom result is already a major bridge in the V13 account. A maintained electron and an effective proton enter with their own identities; one shared electromagnetic attraction and one supplied

localization law create a finite minimum; that minimum simultaneously fixes the leading length, binding, and bound-state mass accounting. [1, §15; 2, R255A–R257A]

The electron's persistence makes a participant. Its bound relationship with the proton makes an atom. The next section asks what happens when more electrons occupy the available states and neighboring atoms share electronic support: **occupied states and the first chemical bond.**

Sources and scope

[1] Shaun Fosmark, *Solvency Field Theory V13.3: The First Locally Real Atom*, R257A freeze, June 2026. Sections 8–12 supply the guarded proton exterior, energy balance, hydrogen budget, and scale hierarchy; §§13–15 delimit the leading nonrelativistic scope.

[2] *SFT V13.3 R251–R257 Results and Scripts QuickPack*, June 2026. The archived R254A–R257A reports, tables, summaries, and `scripts/v13_3_quick_repro.py` own the numerical hydrogen evaluation. The reproduction script evaluates declared anchors; it is not a native atom solver.

[3] Shaun Fosmark and GPT-5.6 SOL, *SFT V13.4-V2: Relational Interactions*, final v1.0, with *Relational Interaction Update Blend v0.11*. Supplies the electromagnetic normalization, joint-source subtraction, and open-channel accounting used consistently here.

[4] *SFT Composed Manuscript v0.1*, under V13.13 Revision B, C11.S01–C11.S02. Used only for already-established source/accounting clarifications: reduced mass, atomic support versus intrinsic electron width, the supplied localization law, and the separation of minimum, spectrum, and capture. It contributes no additional native atomic result.

[5] Shaun Fosmark, *SFT V13.6: From Quantum Settlement Rules to Atomic Structure and Chemical Bonding*, §§7–9. Used only for the next section's existing occupied-state and bonding program; its historical exchange-from-rotation claim is not imported.

[6] *SFT V13.12: The Maintained Electron*, public draft v0.2, §1.8. Supplies the rounded 1/137 electromagnetic normalization used by the frozen electron, retained here only to make the numerical-preparation difference explicit.

Section scope. The leading atomic energy law, numerical anchors, reduced-mass correction, and archived hydrogen outputs retain their original ownership. No proton interior, native localization coefficient, full spectrum, precision correction, capture rate, or new electron-proton solver is added in this edit.

11. Occupied states and the first chemical bond

A bound atom gives an electron a relationship with one nucleus. Chemistry begins when the same kind of maintained participants can occupy several allowed states or organize their electronic support across more than one nucleus. **A bond is not a new substance between atoms; it is a lower-cost organization of the particles already present.** [1, §9; 2, §11]

V13.6 reaches this scale through three linked results: finite occupation capacity, a qualitative filling order, and the first shared electronic support pattern for a covalent bond. The state counts are exact under the declared atomic labels and supplied antisymmetric composition. The ordering is a qualitative energetic rule. The H2 work establishes interference, overlap, and the energy conditions a bond must satisfy; it does not yet produce a molecular energy curve. [1, §§7–9; 2, §§9–11]

The later V13 composition result controls the interpretation: antisymmetric electron composition is used here as an application rule, not derived from a single electron's 4π rotation. Older *settlement addresses* are read as complete state labels rather than locations in a physical register. With that retyping fixed, the chemistry arithmetic can stand on its own. [4, §18; 5, §§7, 9]

11.1 An occupied state describes participation in the atom

Section 10 selected a preferred ground-state scale but did not generate a full orbital spectrum. V13.6 adopts the central-potential labels used by its atomic application:

$$\begin{gathered} q_{\mathrm{at}} = (n, \ell, m_\ell, m_s), \qquad n = 1, 2, \dots, \\ \ell = 0, 1, \dots, n-1, \qquad m_\ell = -\ell, -\ell+1, \dots, \ell, \qquad m_s = \pm\tfrac{1}{2}. \end{gathered} \tag{11.1}$$

The four labels distinguish radial class, angular class, orientation within that class, and spin branch in the chosen basis. They describe the electron's participation in the atomic state. An orbital is therefore not the electron's microscopic recurrent route, and m_s is not a second spatial coordinate. Two electrons can share one spatial orbital while occupying different complete one-electron states. [1, §7; 2, §9]

The electron's intrinsic identity belongs to the maintained particle; the atomic labels belong to the larger configuration in which it participates.

11.2 Exclusion constrains the joint state

For this application, use the antisymmetric two-electron composition already isolated in Section 8. For orthonormal complete one-electron states u and v,

$$\Psi_{uv}(1,2) = \frac{u(1)v(2) - v(1)u(2)}{\sqrt{2}}, \qquad \Psi_{uv}(2,1) = -\Psi_{uv}(1,2). \tag{11.2}$$

Setting $u = v$ makes the numerator vanish. The consequence is exact once antisymmetry is supplied; the physical law selecting that exchange class remains a separate problem. [2, §8; 4, §18]

What is excluded is duplicate occupation of one complete state, not overlap in ordinary space. A second electron can occupy the same spatial region or spatial orbital when another degree of freedom makes its complete state distinct. Exclusion is therefore not an extra repulsive potential to be added to the energy.

11.3 The state labels give finite capacities

For fixed n and ℓ, there are $2\ell+1$ allowed m_ℓ values and two spin branches. Thus

$$\boxed{N_{n\ell} = 2(2\ell+1).} \qquad \begin{array}{c|cccc} \text{subshell} & s & p & d & f \\ \ell & 0 & 1 & 2 & 3 \\ N_{n\ell} & 2 & 6 & 10 & 14 \end{array} \tag{11.3}$$

Summing all admitted ℓ values in shell n gives

$$\begin{aligned} N_n &= \sum_{\ell=0}^{n-1} 2(2\ell+1) = 4\frac{n(n-1)}{2} + 2n = \boxed{2n^2}, \\ (N_1, N_2, N_3, N_4) &= (2, 8, 18, 32). \end{aligned} \tag{11.4}$$

The result is pure counting once the state domain and composition rule are fixed. The archived R273A script reproduces these capacities. [2, §9; 3, R273A]

This solves a multiplicity question: the lowest complete state cannot be duplicated indefinitely. It does **not** yet decide which available states are energetically occupied first.

11.4 Capacity and filling order are different results

V13.6 proposes a qualitative ordering rule that combines radial-band cost with an angular penetration/barrier cost:

$$\mathfrak{t}_{n\ell} = n + \ell, \qquad \text{order key} = (n+\ell, n), \quad \text{smaller first.} \tag{11.5}$$

The score ranks states; it is not an energy in electronvolts. The source's two comparisons expose the rule and tie-break directly:

$$\begin{array}{lll} 4s : (n+\ell, n) = (4,4), & 3d : (5,3), & 4s \prec 3d, \\ 3d : (5,3), & 4p : (5,4), & 3d \prec 4p. \end{array} \tag{11.6}$$

Applying the same ordering gives

$$\begin{aligned} &1s \to 2s \to 2p \to 3s \to 3p \to 4s \to 3d \to 4p \to 5s \\ &\to 4d \to 5p \to 6s \to 4f \to 5d \to 6p \to 7s \to 5f \to 6d \to 7p. \end{aligned} \tag{11.7}$$

The R273B generator implements this sort. Grouping after the p closures yields

$$\begin{aligned} &2; \quad 2+6; \quad 2+6; \quad 2+10+6; \quad 2+10+6; \\ &\quad 2+14+10+6; \quad 2+14+10+6 \\ &\Longrightarrow \ (2, 8, 8, 18, 18, 32, 32). \end{aligned} \tag{11.8}$$

and the familiar skeleton $(2, 8, 8, 18, 18, 32, 32)$. These group lengths differ from the single-shell capacities $2n^2$ because the ordering interleaves principal shells. **Capacity says how many states exist in a class; filling order proposes which classes are used first.** [7, §§4–5; 2, §§9–10]

Exact configurations, Hund structure, screening, and exchange/correlation shifts remain outside this qualitative rule. That boundary identifies the physics needed to turn the skeleton into a quantitative spectrum without weakening the counting result.

11.5 Electronic support need not end at a hard edge

The chemistry picture requires neighboring atomic regions to become one problem. V13.6 models a costly static barrier with an evanescent solution rather than a wall that deletes the amplitude:

$$\frac{d^2\psi}{dx^2} = \kappa^2\psi, \qquad \psi(x) = Ae^{-\kappa x} + Be^{\kappa x}, \qquad \psi_{\mathrm{dec}}(d) = \psi_{\mathrm{dec}}(0)e^{-\kappa d}. \tag{11.9}$$

For the selected decaying branch, support falls exponentially across the barrier. Layer factors make the same point:

$$\frac{A_N}{A_0} = \prod_{k=1}^{N} r_k, \qquad \frac{I_N}{I_0} = \prod_{k=1}^{N} r_k^2; \qquad I_{\mathrm{far}} \propto e^{-2\int \kappa(x)\,dx} \quad \text{in the stated attenuation approximation.} \tag{11.10}$$

The squared factors describe attenuation of intensity in this approximation. They are not by themselves detector probabilities or a complete transmitted-current calculation. [2, §7]

The durable physical point is simpler: **electronic support can extend outside one locally favored region.** Once a second nucleus is present, the complete surrounding configuration can reorganize the state.

11.6 Two atomic supports form one phase-sensitive pattern

Take the first V13.6 bond example, two hydrogen centers A and B. Let ψ_A and ψ_B be the effective atomic support amplitudes in one common phase convention. Their two linear combinations are

$$\tilde{\phi}_{\pm} = \psi_A \pm \psi_B, \qquad |\tilde{\phi}_{\pm}|^2 = |\psi_A|^2 + |\psi_B|^2 \pm 2\,\mathrm{Re}(\psi_A^*\psi_B). \tag{11.11}$$

Writing the amplitudes by magnitude and phase makes the cross term explicit:

$$\psi_j = |\psi_j|e^{i\varphi_j}, \qquad 2\,\mathrm{Re}(\psi_A^*\psi_B) = 2|\psi_A||\psi_B|\cos(\varphi_B - \varphi_A). \tag{11.12}$$

Only the relative phase matters. At the symmetric midpoint of the source's simplest real toy,

$$\tilde{\phi}_+ = 2a, \qquad |\tilde{\phi}_+|^2 = 4a^2, \qquad \tilde{\phi}_- = 0, \qquad |\tilde{\phi}_-|^2 = 0. \tag{11.13}$$

so the plus branch reinforces support between the nuclei while the minus branch has a node. The R274A midpoint script checks these arithmetic values. These are **bonding and antibonding support patterns**, not yet molecular energies. [1, §9; 2, §11; 3, R274A]

The mathematical change is one cross term. Physically, it means the electronic state can belong to the two-center configuration rather than being assigned independently to one atom or the other.

11.7 Normalization keeps the electronic content fixed

Raw addition changes the norm, so the two-center states must be normalized before they are compared. For unit-norm profiles with real overlap S,

$$\begin{aligned} &\langle \psi_A, \psi_A \rangle = \langle \psi_B, \psi_B \rangle = 1, \qquad S = \langle \psi_A, \psi_B \rangle \in \mathbb{R}, \qquad |S| < 1, \\ &\|\psi_A \pm \psi_B\|^2 = 2(1 \pm S), \qquad \boxed{\phi_\pm = \frac{\psi_A \pm \psi_B}{\sqrt{2(1 \pm S)}}.} \end{aligned} \tag{11.14}$$

The corresponding intensities satisfy

$$\begin{aligned} |\phi_\pm|^2 &= \frac{|\psi_A|^2 + |\psi_B|^2 \pm 2\,\mathrm{Re}(\psi_A^* \psi_B)}{2(1 \pm S)}, \\ \int |\phi_\pm|^2 d^3x = 1, \qquad |\phi_+(\mathrm{mid})|^2 &= \frac{2a^2}{1+S}, \qquad |\phi_-(\mathrm{mid})|^2 = 0. \end{aligned} \tag{11.15}$$

Interference therefore redistributes a fixed normalized state; it does not manufacture extra electronic content. Increased support between the nuclei may matter to the energy, but $|\phi|^2$ is not itself an energy. [2, §11; 6, C11.S03]

11.8 The archived overlap falls with separation

The second R274A toy uses identical $1s$ Slater profiles with a supplied reference length:

$$\rho = \frac{R}{a_{\mathrm{ref}}}, \qquad \boxed{S(\rho) = e^{-\rho}\left(1 + \rho + \frac{\rho^2}{3}\right)}, \qquad a_{\mathrm{ref}} = 52.9177210903 \text{ pm}. \tag{11.16}$$

This is a correspondence toy with its own fixed scale, not a precision molecular solve. Differentiating its overlap gives

$$\frac{dS}{d\rho} = -\frac{\rho(1+\rho)}{3} e^{-\rho} < 0 \quad (\rho > 0), \qquad S(0) = 1, \qquad \lim_{\rho \to \infty} S(\rho) = 0. \tag{11.17}$$

so the overlap decreases continuously from unity at coincident centers to zero at infinite separation. The script samples several chosen separations; none of those sample points is selected as a bond length. No molecular energy is minimized in this calculation. [3, R274A overlap toy]

What the two R274A tests establish is exactly what they contain: phase-sensitive reinforcement/cancellation and a concrete separation-dependent overlap.

11.9 Two electrons can share the same spatial organization

A molecular spatial pattern is not yet a complete two-electron state. Let $\chi_\uparrow$ and $\chi_\downarrow$ be orthonormal spin states and give both electrons the same normalized ϕ_+ spatial factor:

$$u(\mathbf{r}, \sigma) = \phi_+(\mathbf{r})\chi_\uparrow(\sigma), \qquad v(\mathbf{r}, \sigma) = \phi_+(\mathbf{r})\chi_\downarrow(\sigma), \qquad \langle u, v \rangle = 0. \tag{11.18}$$

These are distinct complete one-electron states because their spin factors differ. Their antisymmetric composition is

$$\Psi(1,2) = \phi_+(\mathbf{r}_1)\phi_+(\mathbf{r}_2)\frac{\chi_\uparrow(\sigma_1)\chi_\downarrow(\sigma_2) - \chi_\downarrow(\sigma_1)\chi_\uparrow(\sigma_2)}{\sqrt{2}}. \tag{11.19}$$

The spatial factor is symmetric while the spin factor is antisymmetric. This is an explicit trial realization of the source's opposite-spin shared-support picture, not a solved H2 ground state. [2, §11; 4, §18]

Its electron-number density is

$$n_e(\mathbf{r}) = 2|\phi_+(\mathbf{r})|^2, \qquad \int n_e(\mathbf{r})\, d^3r = 2. \tag{11.20}$$

so there are still two electrons. Interference changes where their joint spatial support is concentrated; it does not merge them into one double-amplitude primitive particle.

Sharing a spatial organization and duplicating a complete electron state are different operations. That distinction is what lets the bond picture coexist with exclusion.

11.10 The bond is decided by the complete energy

The decisive molecular question is not whether the overlap is positive, but whether the complete two-center configuration lies below the separated atoms. Define

$$\Delta E(R) = E_{\mathrm{H_2}}(R) - 2E_{\mathrm{H}}. \tag{11.21}$$

V13.6 writes the intended contributions schematically as

$$\Delta E(R) = -\mathcal{G}_{\mathrm{shared}}(R) + U_{pp}(R) + U_{ee}(R) + \mathcal{T}_{\mathrm{loc}}(R) + \mathcal{T}_{\mathrm{node/pair}}(R). \tag{11.22}$$

These names identify physical roles in the eventual common energy. They are not permission to add several independently fitted potentials if one completed operator already contains the same kinetic or interaction contribution. [1, §9; 2, §11]

In particular, S is dimensionless. Turning it into an attractive energy with an arbitrary coefficient would insert the result being sought. Constructive overlap is an ingredient of the candidate organization; the full energy must determine whether its benefit exceeds proton-proton repulsion, electron-electron repulsion, localization, and the other costs.

A bound radial equilibrium therefore has three separate requirements:

$$\boxed{\Delta E(R_\star) < 0, \qquad \left.\frac{dE_{\mathrm{H_2}}}{dR}\right|_{R_\star} = 0, \qquad \left.\frac{d^2E_{\mathrm{H_2}}}{dR^2}\right|_{R_\star} > 0, \quad R_\star > 0.} \tag{11.23}$$

Below-threshold energy, stationarity, and positive curvature do different jobs. The same relaxed electronic branch must be followed with R. These conditions state the molecular target without pretending the overlap toy has already evaluated it. [1, §9; 2, §11]

11.11 A true minimum makes the separation restoring

If a future molecular energy branch satisfies (11.23), define $B_{\mathrm{mol}} = -\Delta E(R_\star) > 0$ and $k_R = E''_{\mathrm{H}_2}(R_\star) > 0$. Near equilibrium,

$$\begin{aligned} \Delta E(R) &= -B_{\mathrm{mol}} + \tfrac{1}{2} k_R (\delta R)^2 + O((\delta R)^3), \\ F_R &= -\frac{dE_{\mathrm{H}_2}}{dR} = -k_R \delta R + O((\delta R)^2), \qquad k_R > 0. \end{aligned} \tag{11.24}$$

so a small stretch produces an inward restoring force and a small compression an outward one. This Taylor expansion is a consequence of the equilibrium conditions, not a newly calculated H2 potential or vibrational spectrum. [2, §11; 6, C11.S03]

The eventual energy calculation must use compatible atomic references, nuclear motion, electronic states, and interaction terms on both sides of the comparison. The present V13 sources support **no numerical bond energy or equilibrium distance** from the overlap results alone.

Once a complete molecular binding is obtained, the mass accounting is already fixed:

$$M_{\mathrm{H}_2} c^2 = 2 M_{\mathrm{H}} c^2 - B_{\mathrm{mol}}. \tag{11.25}$$

The chemical interaction is counted once, either in the resolved joint energy or in the coarse molecular mass. Actual formation must also carry the released energy and momentum into radiation, recoil, or the environment. A lower-energy state does not make capture instantaneous.

11.12 Chemistry is persistent organization at a larger scale

The V13 chemistry chain is now clear. Atomic labels plus antisymmetric composition give finite occupation capacities. The radial-plus-angular rule supplies a qualitative filling skeleton. Relative phase produces a shared bridge or a node, and the overlap toy shows how that shared support fades with separation. **The complete energy—not the overlap alone—decides whether the organization is a bond.** [2, §§9–11]

The exact shell counts and the qualitative bond ingredients therefore have different evidence status. The overlap tests are not failed molecular predictions; they successfully test the narrower structures they were written to test. Quantitative screening, correlation, valence structure, and other bond classes remain the next chemistry calculations. [2, §14; 3, R273A–R274A]

An electron's recurrence maintains the particle. An atomic configuration binds it to a nucleus. A molecular configuration can organize electronic support across several nuclei and maintain their separation together. Chemistry adds persistent relationships, spacings, and collective response without adding a new substance or a new origin of time. [5, §§6–7; 6, C01.S01]

The next section moves from particular atoms and bonds to the general response of larger configurations: **collective response and the common action.**

Sources and scope

[1] *SFT V13.6 — From Quantum Settlement Rules to Atomic Structure and Chemical Bonding*, full writeup, July 2026. §§7–9 supply the atomic labels, occupation counts, qualitative filling rule, H2 support combinations, and complete-energy criterion.

[2] *SFT V13.6 — Quantum-to-Chemistry Full Derivation.* §§7–11 supply the static attenuation model, shell capacities, ordering, and qualitative first-bond account; later sections preserve the uncomputed quantitative chemistry boundary.

[3] *SFT_V13_6_REPRO_SCRIPTS_ONLY (1).zip.* The original R273A shell-capacity, R273B ordering, R274A midpoint-interference, and R274A Slater-overlap scripts own the archived arithmetic examples. They enumerate supplied state counts/order and overlap formulas; none computes an H2 energy.

[4] Shaun Fosmark and GPT-5.6 SOL, *SFT V13.4-V2 — Relational Interactions*, final v1.0, §18. Governs the exchange boundary: antisymmetric electron composition is used here conditionally and is not derived from isolated spin rotation.

[5] *SFT V13.13 — Live Revision B.* Governs persistent structure, relational ownership, and the retirement of a physical write/address register.

[6] *SFT Composed Manuscript v0.1*, C11.S03 and complete-source clarifications. Used only for the already-recorded normalization, overlap-versus-energy, common-reference, and source-accounting distinctions; no additional molecular Hamiltonian is imported.

[7] *SFT V13.6 Narrative.* Used for the period grouping in (11.8). Its stronger exchange and fully derived filling claims do not override the later technical boundaries above.

Section scope. All 25 numbered displays and the source models they summarize retain their original ownership. No exchange derivation, quantitative molecular Hamiltonian, bond energy, bond length, spectroscopy, or new chemistry solver is added in this edit.

12. Collective response and the common action

A molecule can be described by a separation coordinate, a solid by a displacement, or a larger body by only a few collective variables. But the internal structure does not stop participating because we stop writing all of it down. Its fields, supports, and constituent motions can adjust, react back, and continue responding after the visible disturbance has changed. **A reduced description is physically adequate only when the response of what was omitted is carried into what remains.** [1, §§61, 69–71; 2, §6]

V13's common-action program makes that statement precise. Internal variables can be solved and eliminated without inventing a second participant-side force law. Their adjustment changes the effective stiffness of the retained variables; a causal dynamical elimination can leave a retarded memory kernel. These are different reductions of one specified joint physical problem. [1, §§61, 69, 71]

The architecture is common, not the constitutive law. Each sector still has to say what responds, which configurations are lawful, how the response is generated, and which normalizations are supplied. What V13 contributes here is the rule for carrying that response consistently across scales.

12.1 The inside of the object still participates

Let x denote the collective variables we keep and y the additional physical response we do not want to display explicitly. The complete state is $X = (x, y)$. The split is descriptive: variables placed in y remain part of the physical configuration and can change what x does. [1, §69.2]

The joint action is written

$$X = (x, y) \in \mathcal{D}_J, \qquad \delta S_J[x, y] = 0. \tag{12.1}$$

$\mathcal{D}_J$ contains the lawful states and variations of the specified problem, including its constraints, preparation, and boundaries. The notation does not say that electromagnetism, chemistry, matter response, and gravity all share one numerical kernel. It says that once a joint problem is specified, all of its participating response must be varied consistently. [1, §§3–4, 69.9]

This is already familiar from the atom and molecule. An energy curve for moving nuclei depends on whether the electronic state is frozen or allowed to relax. Those are different preparations of the same physical ingredients.

12.2 Solve the response, then read the participant equation

On a smooth conservative branch, suppose the internal variables satisfy their own stationary equation for every retained configuration. Define the reduced action by inserting that solved response:

$$\frac{\delta S_J[x, y]}{\delta y} = 0 \quad \Longrightarrow \quad y = y_\star[x], \qquad S_{\text{eff}}[x] = S_J[x, y_\star[x]]. \tag{12.2}$$

The star marks the solution obtained with the same constraints and boundary data as the parent problem. It is not an independently chosen favorable pattern. [1, §71.2; 2, §6]

Varying the reduced action produces

$$\begin{aligned} \delta S_{\mathrm{eff}} &= \left\langle \frac{\delta S_J}{\delta x}, \delta x \right\rangle + \left\langle \frac{\delta S_J}{\delta y}, D y_\star[\delta x] \right\rangle , \\ \boxed{\frac{\delta S_{\mathrm{eff}}}{\delta x}} &= \boxed{\left.\frac{\delta S_J}{\delta x}\right|_{y=y_\star[x]}} . \end{aligned} \tag{12.3}$$

The second term vanishes on the solved internal equation. This is the branchwise envelope result: **once the internal configuration is solved, its effect on the retained variables is already contained in the same on-shell action.** There is no need to append a second force representing the reaction we just eliminated. [1, §71.2]

Constraints participate in the same reduction. For regular equality constraints,

$$\mathscr{A}_J = S_J + \lambda^a \mathcal{C}_a, \qquad \mathcal{C}_a = 0, \qquad z = (y, \lambda), \qquad D S_{\mathrm{eff}} = D_x \mathscr{A}_J|_{z=z_\star[x]} . \tag{12.4}$$

The multiplier reaction is part of the derivative. Eliminating a constrained support coordinate therefore does not erase the stress through which that support acts. [1, §71.3]

12.3 Relaxation changes the effective stiffness

Near a solved state, let the second variation on the physical perturbation space be

$$H = \begin{pmatrix} H_{xx} & H_{xy} \\ H_{yx} & H_{yy} \end{pmatrix}, \qquad H_{yx} = H_{xy}^\dagger . \tag{12.5}$$

The cross blocks describe reciprocal response: the same mixed variation that lets x disturb y also carries the reaction of y back into the x equation. Gauge redundancies and constraints belong to the definition of this perturbation space. [1, §69.2]

Differentiating the solved internal equation gives

$$H_{yx}\delta x + H_{yy}\delta y = 0, \qquad \delta y = -H_{yy}^{-1} H_{yx} \delta x, \qquad D y_\star = -H_{yy}^{-1} H_{yx} . \tag{12.6}$$

and therefore

$$\boxed{H_{\mathrm{eff}} = H_{xx} - H_{xy} H_{yy}^{-1} H_{yx} .} \tag{12.7}$$

This Schur complement is the small-deformation form of the nonlinear reduction above. H_{xx} is the stiffness seen when the interior is held fixed; the second term is what changes when it is allowed to respond. [1, §§69.3, 71.4]

For a static energy with positive internal block,

$$K_{yy} > 0 \quad \Longrightarrow \quad \langle \delta x, (K_{xx} - K_{\mathrm{eff}}) \delta x \rangle = \| K_{yy}^{-1/2} K_{yx} \delta x \|^2 \geq 0. \tag{12.8}$$

Stable internal relaxation can only lower or leave unchanged the tested energy cost relative to freezing that internal motion. It does not guarantee stability of the whole: the relaxed stiffness can itself vanish or become negative.

12.4 A two-coordinate example makes the reaction visible

Take one retained displacement x and one internal displacement y with energy

$$V(x,y) = \frac{1}{2}ax^2 + bxy + \frac{1}{2}dy^2, \qquad a,d>0, \quad b\in\mathbb{R}. \tag{12.9}$$

This is an explanatory model, not a fitted material. Solving the internal stationarity equation and completing the square gives

$$\begin{aligned} y_\star &= -\frac{b}{d}x, & k_{\mathrm{eff}} &= a - \frac{b^2}{d}, \\ V(x,y) &= \frac{d}{2}\left(y + \frac{b}{d}x\right)^2 + \frac{k_{\mathrm{eff}}}{2}x^2, & V_{\mathrm{eff}}(x) &= \frac{k_{\mathrm{eff}}}{2}x^2. \end{aligned} \tag{12.10}$$

Freezing $y=0$ gives stiffness a; letting it relax reduces that stiffness by b^2/d. The same energy produces both forces:

$$F_x = -ax - by, \qquad F_y = -bx - dy, \qquad F_x\big|_{y_\star} = -\left(a - \frac{b^2}{d}\right)x. \tag{12.11}$$

The same coupling b appears in both directions. Separate coefficients for action and reaction would describe a different model. Reciprocal response need not mean equal acceleration because the coordinates can carry different inertias. [1, §§69.2, 69.5]

12.5 Lawful response can be constrained

Topology, charge, boundaries, or support conditions can remove directions that a naive coordinate list would otherwise permit. For reduced real coordinates u, positive static Hessian K, source f, and independent linear constraints $\mathsf{C}u = 0$, write

$$\mathcal{Q}(u;f) = \tfrac{1}{2}u^TKu - f^Tu, \qquad Ku + \mathsf{C}^T\lambda = f, \qquad \mathsf{C}u = 0. \tag{12.12}$$

Eliminating the multipliers gives

$$\begin{aligned} u &= R_{\mathsf{C}}f, \\ R_{\mathsf{C}} &= K^{-1} - K^{-1}\mathsf{C}^T(\mathsf{C}K^{-1}\mathsf{C}^T)^{-1}\mathsf{C}K^{-1}. \end{aligned} \tag{12.13}$$

This is a constrained static response, not an instantaneous propagation law. Defining $P = R_{\mathsf{C}}K$ gives

$$\mathsf{C}R_{\mathsf{C}} = 0, \qquad P^2 = P, \qquad \mathsf{C}P = 0, \qquad P^TK = KP. \tag{12.14}$$

So P projects onto the allowed perturbations in the energy metric. Its rank counts available directions; it does not assign a fixed physical fraction of the response to each one. **The projector represents the lawful subspace. The physical constraints are what create that subspace.** [1, §4.2; 2, §4]

12.6 Stationarity is not yet dynamical stability

To give the two-coordinate example real motion, add positive inertias:

$$L = \tfrac{1}{2}m_x\dot{x}^2 + \tfrac{1}{2}m_y\dot{y}^2 - V(x,y), \qquad \mathcal{E} = \tfrac{1}{2}m_x\dot{x}^2 + \tfrac{1}{2}m_y\dot{y}^2 + V(x,y). \tag{12.15}$$

Variation yields

$$\begin{gathered} m_x\ddot{x} + ax + by = 0, \qquad m_y\ddot{y} + bx + dy = 0, \qquad \frac{d\mathcal{E}}{dt} = 0, \\ (a - m_x\omega^2)(d - m_y\omega^2) - b^2 = 0. \end{gathered} \tag{12.16}$$

If $ad > b^2$, the potential is positive definite and the conserved energy bounds this illustrative motion. If $ad < b^2$, one squared mode frequency is negative; equality leaves a zero mode. All three cases satisfy stationary equations. Stability needs the restoring shape, inertia, and actual dynamical evolution together. [5, C01.S03]

Likewise, replacing $y(t)$ by its static minimizer $y_\star[x(t)]$ is exact only when the dynamical assumptions justify that reduction. Static relaxation and full two-inertia evolution are different physical approximations.

12.7 A singular reduction can signal a change of organization

The regular reduction ends when the internal response cannot be uniquely inverted. Several lawful internal branches may also coexist:

$$y_\star^{(a)}[x], \qquad S_{\mathrm{eff}}^{(a)}[x] = S_J[x, y_\star^{(a)}[x]], \qquad \det H_{yy} = 0 \text{ at a singular finite-dimensional reduction.} \tag{12.17}$$

In a field problem the corresponding zero modes and operator domain must be examined directly. V13.4 identifies such points as possible bifurcations, instabilities, or openings and closings of response channels. [1, §§71.6–71.7]

A reduced branch can therefore be valid without explaining how a transition chooses another branch. Failure of the reduced inverse does not by itself invalidate the parent physics; it tells us that the eliminated variables have become dynamically important again. This is the collective analogue of the earlier distinction between maintenance and formation.

12.8 Eliminated dynamics can reappear as memory

If the internal sector has its own time evolution, its state can continue responding after the retained coordinate changes. Section 9's enlarged self-adjoint system gives the exact retained equation

$$i\hbar\dot{p}(t) = H_{PP}p(t) + \xi_q(t) + \int_{t_0}^{t} \Sigma^R(t-t')p(t')\,dt'. \tag{12.18}$$

with

$$\boxed{\Sigma^R(\Delta t) = -\frac{i}{\hbar}\Theta(\Delta t)V_{PQ}e^{-iH_{QQ}\Delta t/\hbar}V_{PQ}^\dagger,} \qquad \xi_q(t) = V_{PQ}e^{-iH_{QQ}(t-t_0)/\hbar}q(t_0). \tag{12.19}$$

Θ imposes retarded support. The initial term records the continuing influence of the actual initial internal state; it is not assumed random noise. [1, §61]

The full system is local in the displayed variables p and q. Once q is removed, the same physics appears as dependence on earlier p. **Memory in the reduced equation is the continuing response of physical variables we chose not to follow explicitly.** Equal retained coordinates can therefore evolve differently when their internal preparations differ. [1, §§57.3, 61]

For causal open systems, the retarded prescription belongs to the parent dynamics. The resulting reduced equation need not come from an ordinary instantaneous local action for p alone; this is why V13 keeps causal open-system elimination distinct from the conservative envelope theorem.

12.9 Slow collective behavior can inherit the same kernel

When the response is analytic at low frequency and long wavelength, the retarded kernel can be expanded as

$$\widetilde{\Sigma}^{R}(\omega,\mathbf{k}) = \Sigma_0 + c_k|\mathbf{k}|^2 - i\eta_{\mathrm{mem}}\omega + c_\omega\omega^2 + \cdots . \tag{12.20}$$

The terms separate static response, spatial-gradient response, and dependence on the rate and acceleration of a disturbance. Their values, tensor structure, and range of validity come from the parent kernel. With genuine open channels, the dissipative part is constrained by the positive channel structure developed in Section 9; eliminating a coordinate from a closed finite system does not automatically create irreversible damping. [1, §§6–7, 61]

This gives V13 a route toward pressure-like, shear-like, and other material responses without treating those macroscopic words as new substances. The actual microscopic variables still have to supply the appropriate scalar or tensor response. A response time alone is not a heat law or a finite-speed theorem.

12.10 Symmetry and source accounting survive the reduction

If g is a symmetry of the parent action and lawful domain, and the chosen internal branch transforms with it,

$$\begin{aligned} y_\star[gx] &= g y_\star[x], \\ S_{\mathrm{eff}}[gx] = S_J[gx, g y_\star[x]] &= S_J[x, y_\star[x]] = \boxed{S_{\mathrm{eff}}[x]}. \end{aligned} \tag{12.21}$$

The reduced theory then inherits the symmetry rather than receiving it through an additional rule. A branch that is not equivariant cannot borrow this conclusion. [1, §71.5]

The same logic controls the source passed into gravity. For an admitted covariant physical-sector action with signature $(-,+,+,+)$ and $c=1$,

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}}\frac{\delta S_{\mathrm{phys}}}{\delta g^{\mu\nu}}, \qquad \delta_g S_{\mathrm{phys}} = -\frac{1}{2}\int T_{\mu\nu}\delta g^{\mu\nu}\, dV_g, \qquad dV_g = \sqrt{-g}\, d^4x. \tag{12.22}$$

S_{phys} contains the counted matter, field, support, and constraint contributions of that physical sector. A support term can vanish on the solved state while its metric variation still contributes to stress. This is the covariant source correspondence retained by the assembly; it is not a claim that every V13 source has already been derived microscopically. [3, §9.4; core §3.3]

For a closed covariant sector,

$$\nabla_\mu T^\mu{}_\nu = 0. \qquad \nabla_\mu T^\mu{}_\nu = f^{\text{ext}}_\nu \quad \text{when an imposed source exchanges momentum.} \tag{12.23}$$

The second line keeps the external exchange explicit. Local conservation does not by itself define a unique global energy in arbitrary geometry; boundaries and spacetime symmetries still matter. [5, C01.S03]

On a regular reduced branch, the eliminated response is already present in the on-shell source. Adding the fully resolved internal source again would double count it. The atom's mass subtraction and the electromagnetic cross term were earlier examples of the same rule.

12.11 One architecture does not mean one undifferentiated law

V13 summarizes a physical response sector by

$$\mathfrak{C}_s = (\mathcal{Y}_s, \mathcal{D}_s, \mathcal{K}_s, \lambda_s). \tag{12.24}$$

$\mathcal{Y}_s$ says what responds; $\mathcal{D}_s$ defines the lawful configurations; $\mathcal{K}_s$ gives the constitutive and cross-response dynamics; and λ_s contains genuinely supplied scales or normalizations. These are ingredients of a sector, not four universal fields. [1, §69.6]

The electromagnetic pair, the conditional electroweak response, and the effective chemistry applications already illustrate different realizations of this architecture. Material and gravitational sectors must likewise identify their own physical response rather than inherit a coefficient because a familiar mathematical operator happens to appear.

What survives the change of scale is the physical response of the same configuration. In a conservative reduction it appears through the on-shell action and the Schur complement; in a causal dynamical reduction it can appear as a retarded kernel. Internal organization has not vanished. We have changed which variables we display.

The next section carries that source into gravity: **one shared geometry must change trajectories, clock comparisons, and light propagation together.** The clocks already exist. Gravity changes how their continuing histories compare.

Sources for this section

[1] *SFT V13.4 Relational Interaction Update Blend v0.11*, especially §§3–4, 6–7, 57.3, 61, 69, 71, and 73.11. Primary source for elimination, reciprocal response, constraints, memory, and sector architecture.

[2] *SFT V13.8-V2: There Is No Write Operator*, final v1.0, §§2–6. Governs lawful evolution and the non-ontic use of projectors and bookkeeping.

[3] *SFT V13.13: Live Ontology Constitution, Revision B*, §§8–10. Governs collective interpretation and source accounting.

[4] *SFT V13.7: The Master Action II*. Architectural ancestry only; superseded write machinery and selectors are not restored.

[5] Existing *SFT Composed Manuscript*, C01.S03. Used only for the two-coordinate illustration, stability distinction, and covariant source convention.

Execution scope. The displayed equations develop or restate the V13 response identities under their stated premises. No particle, material, gravity, or continuum solver is rerun, and no universal constitutive coefficients are introduced. Microscopic material closures and new empirical extensions belong to V14 Part 2.

13. Gravity, clocks, and shared geometry

A clock is a physical process. If elapsed history is made by maintained recurrence, gravitational time dilation must change the continuation of that process rather than slow a separate substance called time. The same gravitational configuration must also govern freely moving bodies and light. **One source and one geometry must account for clocks, trajectories, and null propagation together.** [2, §§7.2, 8.1–8.3]

V13 develops that claim in two complementary gravitational strands. V13.1-V3 supplies a selected covariant Einstein-form structural branch. V13.1-V4 builds a constructive clock-and-spatial-response route through an exact spherical exterior, many-source weak fields, pressure, moving sources, and radiation. V4 uses explicit response hypotheses and one gravitational calibration; those premises remain attached to the results. The present particle program has not yet derived the full exterior directly from Electron V1. [1, §§1, 4–9; 3, §§1–8]

Revision B supplies the interpretation used here. Geometry is physical without being a rubber medium, and literal writable addresses are not part of the ontology. V4's continuation and timing variables are therefore read as a response model for how physical structures continue in a shared geometry, not as a ledger that blocks later events. [1, §§4.1–4.2, 13; 2, §§3.2, 8.1–8.3]

13.1 The source is the complete thing

A persistent particle can possess intrinsic source content and local closure response before another body compares it. Mutual acceleration, redshift, and deflection are relational readouts of the larger configuration. The particle's internal closure geometry is not automatically its already-solved distant exterior. [2, §§8.2–8.3]

At every scale, the gravitational source must be the complete physical configuration represented at that scale. Binding energy, field stress, pressure, and support cannot be counted once in a resolved description and then added again after coarse graining. Section 12's common-action accounting therefore matters directly to gravity. [2, §9.4; core §§8, 10, 12]

V3 states its selected covariant target as

$$G_{\mu\nu}[g(\Sigma)] = \frac{8\pi G_{\mathrm{eff}}}{c^4}\,\mathcal{W}_{\mu\nu}. \tag{13.1}$$

Here $G_{\mu\nu}$ is the Einstein curvature tensor, $g(\Sigma)$ the metric associated with the paper's symmetric response carrier, and $\mathcal{W}_{\mu\nu}$ the complete source in energy-density units. G_{eff} is the gravitational coupling. V3's status revision records the required composition, covariance, and source-conservation work for its selected regular branch. [3, §§1–6]

That result does not remove the remaining source map: the present maintained electron still has to be connected explicitly to the source and exterior variables of the gravitational construction. What follows develops the V13 gravitational relationships without hiding that boundary. [2, §8.3; 3, §§1–7]

13.2 The weak field connects two very different lengths

For a source of mass $M > 0$, define its gravitational and reduced Compton lengths. V3's area-to-coupling dictionary gives

$$r_g = \frac{GM}{c^2}, \qquad \bar{\ell}_C = \frac{\hbar}{Mc}, \qquad r_g \bar{\ell}_C = \frac{\hbar G}{c^3} = \ell_P^2, \qquad G_{\text{eff}} = \frac{a_\phi c^3}{\hbar}. \tag{13.2}$$

The product $r_g \bar{\ell}_C = \ell_P^2$ follows directly from the definitions: the inverse mass dependence of the Compton scale cancels the mass dependence of the gravitational scale. The area a_ϕ supplies the corresponding gravitational normalization; the dimensionless relation alone does not determine its absolute value. [1, §3; 3, §8]

V4's weak source-depth construction is

$$f(d) = \frac{\omega_B (d/c) a_\phi}{\kappa d^2} = \frac{\omega_B a_\phi}{\kappa c\, d}, \qquad \omega_B = \frac{E_B}{\hbar}, \qquad d > 0. \tag{13.3}$$

The model combines a source rate, a propagation interval d/c, and dilution over area κd^2. Their product gives a dimensionless depth proportional to $1/d$, while its gradient later supplies the inverse-square acceleration. The accumulation interval is therefore doing real work: a passing flux by itself has the wrong distance dependence. [1, §3]

After calibration, the depth used below is $f = r_g/d$. V3's separate numerical-G branch used a different optical-normalization postulate and remains a separate conditional calculation. The common achievement is a source-proportional weak potential depth with an explicit normalization problem. [1, §§3, 13; 3, §§9–13]

13.3 A clock factor is only half the geometry

Use signature $(-,+,+,+)$ and a static spherical chart with areal radius r:

$$ds^2 = -N(r)^2 c^2 dt^2 + S_r(r)^2 dr^2 + r^2 d\Omega^2, \qquad N, S_r > 0, \qquad N(\infty) = S_r(\infty) = 1. \tag{13.4}$$

A stationary clock reads $d\tau = N dt$, while radial proper distance is $d\ell = S_r |dr|$. Radial null propagation then gives

$$\frac{d\ell}{d\tau_{\text{static}}} = c, \qquad \left|\frac{dr}{dt}\right| = c\frac{N}{S_r}, \qquad S_r = N^{-\gamma_g}. \tag{13.5}$$

γ_g is V4's reciprocal radial-response exponent, not a Lorentz factor. The selected correspondence member has $\gamma_g = 1$. Local light speed c does not select this exponent: many positive lapse and spatial-scale pairs preserve the same locally measured null speed. [1, §4]

A maintained loop shows how the geometry reaches the process making a clock. In V4's static fixed-route approximation,

$$P_\infty = \oint \frac{d\ell}{cN(\mathbf{x})}, \qquad N \text{ uniform on the route}: \quad P_\infty = \frac{L}{cN}, \qquad P_{\text{local}} = \frac{L}{c}. \tag{13.6}$$

The support still propagates locally; what changes is the relation between its local traversal and the reference-time description. The point is not that gravity deletes cycles or makes a carrier wait. It changes the duration of the continuing physical process. [1, §4.2; core §§3–4]

13.4 Availability and cadence need a timing law

V4 introduces a normalized continuation measure W. It is not a fraction of a photon, a fraction of rest energy, or an amount of time. Under uniform attenuation in a dimensionless exposure $\mathcal{H}$,

$$\frac{dW}{d\mathcal{H}} = -W, \qquad W(0) = 1, \qquad W = e^{-\mathcal{H}}, \qquad N = W^{\tau_0}, \qquad \Phi = -\ln N = \tau_0 \mathcal{H}. \tag{13.7}$$

The exponential law for W follows from the stated attenuation premise. The map $N = W^{\tau_0}$ is the additional **exponential-cadence hypothesis**. Neither follows merely from irreversible history. [1, §§4.1–4.2]

The source's renewal example makes the timing assumption explicit:

$$N = \frac{1}{1 + \tau_f(\mathcal{H})(e^{\mathcal{H}} - 1)}, \qquad \tau_f(\mathcal{H}) = \frac{e^{\tau_0 \mathcal{H}} - 1}{e^{\mathcal{H}} - 1} \quad \Longrightarrow \quad N = e^{-\tau_0 \mathcal{H}}. \tag{13.8}$$

The ratio has the continuous limit $\tau_f(0) = \tau_0$. This example shows what kind of duration law can generate the lapse; it does not require every physical clock to be a renewal counter. The later continuation tests matter because coherent clocks can realize the same lapse scaling by a different microscopic route. [1, §4.2]

13.5 The radial feedback has an exact solution

The clock response and radial stretch change the dwell accumulated through the source environment. V4 closes its spherical model with

$$\Phi(d) = \int_d^\infty \frac{A_g}{r^2} e^{(1+\gamma_g)\Phi(r)}\, dr, \qquad f = \frac{A_g}{d}, \qquad \Phi(\infty) = 0. \tag{13.9}$$

The exponential is S_r/N, so the recursion contains both clock and spatial response. It is a supplied constitutive equation, not a consequence of spherical symmetry alone. Differentiating the lower limit and defining $w = e^{-(1+\gamma_g)\Phi}$ gives

$$\Phi'(d) = -\frac{A_g}{d^2} e^{(1+\gamma_g)\Phi}, \qquad w'(d) = \frac{(1+\gamma_g)A_g}{d^2}, \qquad w(d) = 1 - \frac{(1+\gamma_g)A_g}{d}. \tag{13.10}$$

The boundary at infinity fixes the integration constant. The positive lapse branch is therefore

$$\boxed{N(f) = \left[1 - (1+\gamma_g) f\right]^{1/(1+\gamma_g)}, \qquad S_r = N^{-\gamma_g}, \qquad 0 \le f < \frac{1}{1+\gamma_g}.} \tag{13.11}$$

For the selected member $\gamma_g = 1$ and $A_g = r_g = GM/c^2$,

$$\boxed{ds^2 = -\left(1 - \frac{2r_g}{r}\right) c^2 dt^2 + \frac{dr^2}{1 - 2r_g/r} + r^2 d\Omega^2, \qquad r > 2r_g.} \tag{13.12}$$

The declared response problem recovers the Schwarzschild exterior exactly, not only its first weak-field term. The numerical static module independently integrates the feedback and checks

the lapse, proper radial flux, and null travel time against this solution. The result remains conditional on the continuation and reciprocal-response hypotheses that produced it. [1, §4; 5, static module]

13.6 The same lapse gives attraction and different clock histories

Expanding the exact lapse at weak depth gives

$$N = 1 - f - \frac{\gamma_g}{2} f^2 + O(f^3), \qquad \phi_N = -\frac{GM}{r}, \qquad N = 1 + \frac{\phi_N}{c^2} + O(f^2). \tag{13.13}$$

The usual Newtonian potential appears as $\phi_N = -GM/r$, and its gradient gives the leading acceleration $\mathbf{a} = -GM\hat{\mathbf{r}}/r^2$. The test body's mass drops out. The $1/r$ potential and $1/r^2$ acceleration are two readouts of the same source dependence. [1, §§3–4; 3, §7]

For a clock moving through the static geometry,

$$\frac{d\tau}{dt} = \frac{N}{\gamma_{\mathrm{loc}}}, \qquad \frac{dN_C}{dt} = \nu_C \frac{N}{\gamma_{\mathrm{loc}}}, \qquad \gamma_{\mathrm{loc}} = \left(1 - \frac{v_{\mathrm{loc}}^2}{c^2}\right)^{-1/2}. \tag{13.14}$$

The maintained branch keeps its proper-time cadence ν_C; the shared geometry and local motion change how that cadence compares with the reference coordinate. A stationary clock can therefore exhibit gravitational dilation with $\gamma_{\mathrm{loc}} = 1$. Gravity is not being reduced to extra local spatial speed. [1, §4; 2, §7.2; core §4.8]

Two stationary clocks exchanging pulses through the same solved geometry satisfy

$$\frac{\nu_{\mathrm{rec,B}}}{\nu_{\mathrm{emit,A}}} = \frac{N_A}{N_B}. \tag{13.15}$$

The emitter, receiver, and signal are one comparison problem; no universal present is required. [core §4.8; 1, §4.2]

For radial fall from rest at infinity, the same exterior gives

$$\frac{v_{\mathrm{loc}}^2}{c^2} = \frac{2r_g}{r}, \qquad \frac{d\tau}{dt} = \sqrt{1 - 2r_g/r}\sqrt{1 - 2r_g/r} = 1 - \frac{2r_g}{r}. \tag{13.16}$$

At the horizon, the static-coordinate comparison tends to zero while the infalling clock continues its own local history. A failure of the particle's maintained support would be a separate physical transition, not a consequence of the distant-coordinate rate. [1, §§4–4.2]

13.7 Light sees the spatial response too

A lapse alone cannot determine the optical prediction. In weak isotropic coordinates, V4 obtains

$$\begin{gathered} r = \varrho + \gamma_g r_g + O(r_g^2/\varrho), \qquad U = \frac{r_g}{\varrho}, \\ N^2 = 1 - 2U + (1 + \gamma_g)U^2 + O(U^3), \qquad h_{ij} = (1 + 2\gamma_g U)\delta_{ij} + O(U^2). \end{gathered} \tag{13.17}$$

Matching $g_{00} = -N^2$ in this convention gives $\beta = (1+\gamma_g)/2$, hence $\beta = 1$ for the selected $\gamma_g = 1$ member. The radial-coordinate coefficient by itself is not the same post-Newtonian quantity. [1, §5]

The weak null interval can be written as a coordinate optical index,

$$c\,dt = n_{\rm opt}|d\mathbf{x}|, \qquad n_{\rm opt} = \frac{\sqrt{1+2\gamma_g U}}{N} = 1 + (1+\gamma_g)U + O(U^2). \tag{13.18}$$

This packages coordinate travel time; it is not a material refractive medium and does not change locally measured light speed. At first order, clock and spatial response contribute together. [1, §4; 2, §8.1]

The corresponding leading deflection and Shapiro delay are

$$\begin{aligned} |\delta\theta| &\simeq \frac{2(1+\gamma_g)r_g}{b}, \\ \Delta t_{\rm Shapiro} &\simeq \frac{(1+\gamma_g)r_g}{c}\ln\!\left(\frac{4r_A r_B}{b^2}\right), \qquad r_A, r_B \gg b. \end{aligned} \tag{13.19}$$

At $\gamma_g = 1$, the deflection is $4GM/(bc^2)$. **Getting the lapse right while omitting the spatial response would miss half of this leading bending.** Redshift, light bending, and delay are therefore tests of one shared metric rather than separately adjustable stories. [1, §4; 3, §7]

13.8 Many sources share a geometry, not a collection of separate lapses

A one-body exterior cannot simply be copied source by source into a nonlinear many-body geometry. V4's successful static extension conserves proper response flux in the common spatial metric:

$$j^i = c\,h^{ij}\partial_j N, \qquad D_i j^i = \rho_j, \qquad D_h^2 N = \frac{\rho_j}{c}, \qquad D_h^2 N = \frac{1}{\sqrt{h}}\partial_i(\sqrt{h}\,h^{ij}\partial_j N). \tag{13.20}$$

D_i is the derivative compatible with h_{ij}; j^i is gravitational constraint-response flux, not electric current. Path independence of the static clock rapidity and proper-flux conservation are premises of this construction. The equation belongs to the whole configuration. [1, §5]

Its vacuum first-post-Newtonian check gives

$$\begin{aligned} N &= 1 - U + \tfrac{1}{2}U^2 + O(U^3), \\ N^2 &= 1 - 2U + 2U^2 + O(U^3), \qquad 2U^2 = 2\sum_i U_i^2 + 4\sum_{i<j} U_i U_j. \end{aligned} \tag{13.21}$$

The cross terms $U_i U_j$ are properties of the common state. A tested coordinate-flux alternative misses the required second-order structure, which is why nonlinear gravity cannot be obtained by stacking independent one-body lapses. The proper-flux result fixes this static order; full binary dynamics and second-order spatial geometry remain further calculations. [1, §5]

Many sources modify one gravitational response through which their already-existing clocks continue. They do not manufacture a new layer of time.

13.9 Pressure and radiation belong in the source

Rest mass alone is incomplete when momentum and stress are present. V4's local observer-frame source rule is

$$\frac{E+\mathbf{p}\cdot\mathbf{v}}{h}, \qquad \frac{T^{00}+\sum_i T^{ii}}{h}. \tag{13.22}$$

The combination is deliberately different from the proper recurrence derivative $E-\mathbf{p}\cdot\mathbf{v}$ of Section 4. Equation (13.22) is a gravitational source prescription, not a photon or particle clock. [1, §6]

For isotropic pressure and radiation it yields

$$\epsilon+\sum_i T^{ii}=\epsilon+3p, \qquad \text{radiation: } \epsilon+3p=2\epsilon, \qquad D_h^2 N=\frac{4\pi G}{c^4}N(\epsilon+3p). \tag{13.23}$$

Radiation gravitates even though it has no massive rest tally. The relevant source is its energy and momentum transport. The apparent factor two for free radiation is resolved only by counting the stresses that make a complete stationary system. In the corresponding static asymptotically normalized branch,

$$M_{\text{active}}=\frac{1}{c^2}\int(\epsilon+3p)N\sqrt{h}\,d^3x=M_{\text{ADM}}, \tag{13.24}$$

For a radiation box, the positive radiation stress and negative confining-wall stress combine so that the complete active source matches the total asymptotic mass rather than doubling it. This is the same one-source discipline used for bound particles and materials. [1, §6]

V4 also records pressure-modified static alternatives. Their differing lapse and spatial masses are properties of those supplied static equations; they are not, by themselves, binary-dynamics tests. The durable result here is that pressure and support stress belong in the gravitational source that the physical configuration actually carries. [1, §§6, 10]

13.10 Motion requires the tensor source

The energy-plus-stress rule has a compact weak-field tensor form. With $c=1$ and $\eta_{\mu\nu}=\text{diag}(-1,1,1,1)$,

$$\mathcal{C}^{\mu\nu}=2T^{\mu\nu}-\eta^{\mu\nu}\mathcal{T}, \qquad \mathcal{C}^{00}=T^{00}+\sum_i T^{ii}. \tag{13.25}$$

$\mathcal{C}^{\mu\nu}$ is twice the trace-reversed source of the selected linearized response. Pressure's factor three, radiation's factor two, and the moving-particle $E+\mathbf{p}\cdot\mathbf{v}$ term are evaluations of the same tensor prescription. [1, §§6–7]

Boosting the first-order static perturbation produces mixed time-space components. For a slowly moving source,

$$\delta g_{0i}=-4U\frac{v_i}{c}+\text{higher-order terms}. \tag{13.26}$$

The leading factor four is not a second rotational coupling; it follows from transforming both the temporal and spatial response. V4's direct numerical boost check gives $\delta g_{0x}/[U(v/c)]\simeq -4.000004$ at $v/c=10^{-3}$. A rotating source still requires its distributed current and solved metric, but the mixed response belongs to the same tensor geometry. [1, §7]

Tensor form alone does not establish the absence of preferred-frame observables. Those require the actual native mechanism and its mapping to experiment. [1, §7]

13.11 Geometry can carry a changing response outward

Static geometry does not determine its own radiative degrees of freedom. V4 therefore supplies a conserved, massless spin-two dynamical completion using the same calibrated G and outgoing-wave conditions. In its declared weak-field gauge,

$$\Box\,\delta g_{\mu\nu} = -\frac{16\pi G}{c^4}\left(T_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\mathcal{T}\right), \qquad \Box = -\frac{1}{c^2}\partial_t^2 + \nabla^2. \tag{13.27}$$

The characteristics propagate at c. Four harmonic constraints and four residual gauge directions reduce the ten symmetric metric amplitudes to two transverse-traceless polarizations. That count belongs to the supplied dynamical completion, not to the static exponent or propagation speed alone. [1, §8]

For a slow localized binary, the outgoing quadrupole power and mechanical reaction work are

$$P_\infty = \frac{G}{5c^5} I_{ij}^{(3)} I_{ij}^{(3)}, \qquad \dot{E}_{\mathrm{mech}} = -\frac{G}{5c^5} I_{ij}^{(1)} I_{ij}^{(5)}. \tag{13.28}$$

The reaction coefficient is not adjusted to force agreement with the far-zone flux. Instantaneously, the two powers need not cancel because the near-zone response can store and return energy. V4's Schott term completes the balance:

$$E_{\mathrm{Schott}} = \frac{G}{5c^5}\left(I_{ij}^{(1)} I_{ij}^{(4)} - I_{ij}^{(2)} I_{ij}^{(3)}\right), \qquad \boxed{\dot{E}_{\mathrm{mech}} + P_\infty + \dot{E}_{\mathrm{Schott}} = 0.} \tag{13.29}$$

Differentiating the two products in E_{Schott} gives exactly the difference required by (13.28); for a periodic orbit its mean derivative vanishes. **Source, near field, mechanical motion, and outgoing radiation therefore remain in one energy account.** [1, §8]

The archived synthetic slow-binary tests give mean-power enhancement factors approximately 1.000000, 1.776243, and 10.227919 for eccentricities 0, 0.3, and 0.6. Independent angular-flux and reaction-work calculations agree within the module's numerical tolerance. The static and radiation scripts are correspondence tests of the declared equations, not a new microscopic gravitational-wave source. [1, §8; 5]

13.12 What the continuation tests teach the clock picture

V4's later continuation work approaches gravitational timing from the particle side. Continuous-support models can deform, exchange energy reciprocally, and remain active for many circuits; some ring coherently rather than settling into memoryless attempts. Those tests make a crucial distinction visible: an acceptance fraction is not yet an elapsed-time rate until the physical duration law is specified. [1, §12]

The same program supplies a useful counterexample to making the renewal picture mandatory. A coherent oscillator evolving under a uniform supplied lapse, $dz/dt = NL_0(z)$, runs at frequency ratio N even when its tick is read from internal phase. The reported two-oscillator checks agree at about 7×10^{-15}. Supplying the lapse does not derive its source, but it shows that the gravitational clock proposal can be broader than one state-blind counting mechanism. [1, §12, W9]

The physical target is therefore clear: the gravitational source must alter the lawful continuation of the same maintained structure while preserving the current, stress, and local causal behavior that make the structure a particle. The exact exterior and radiation results tell us what that particle-level mechanism must reproduce. [1, §§12–13; 2, §8.3]

13.13 Gravity carries the physical account across scales

The V13 gravity program now forms a connected chain. A calibrated source depth and two declared response hypotheses give an exact spherical exterior. The same clock and spatial factors produce Newtonian attraction, gravitational redshift, light bending, and Shapiro delay. Proper-flux conservation supplies the many-source static cross terms. Complete stress brings pressure and radiation into the source, tensor transformation supplies the leading moving-source response, and the spin-two completion carries changing geometry outward with a checked reaction-energy balance. [1, §§4–9, 13]

V3 retains its covariant structural branch and its interior, compact-source, rotating, and cosmological correspondences under their stated assumptions. V4 does not replace that work, and neither branch yet makes Electron V1 a completed universal gravitational source. The remaining source-to-response and source-to-exterior map is a concrete V14 Part 2 task, not a reason to weaken the results already obtained. [1, §§1, 13; 2, §8.3; 3, §§1–8]

Matter is active; its clock is part of that activity; gravity changes how the complete configuration continues. More sources and more organization change the shared gravitational response through which those clocks, bodies, and signals relate. Geometry can be real without being a material medium, and a relation can be physical without becoming a separate force substance. [2, §§7–8]

The next section carries that picture to an evolving large-scale geometry. V13's expansion proposal contains useful mathematics and physical intuitions, but Revision B has removed the literal write/address mechanism on which part of its original explanation relied. We therefore separate what survives from what must be rebuilt before the cosmological account becomes a current prediction. [2, §8.4]

Sources and scope

[1] *SFT V13.1-V4 — Gravity as a Relation Between Two Magnitudes*, v1.4, 8 September 2026. Governs the constructive clock/spatial-response branch, its static and many-source calculations, pressure and motion rules, radiation completion, continuation tests, and v1.4 corrections.

[2] *SFT V13.13 — Live Ontology Constitution*, Revision B, especially §§3, 7–9. Governs complete sources, local history, gravitational readouts, and removal of literal write machinery.

[3] *SFT V13.1-V3 Gravity*, provenance-audited status revision v1.1 and revised derivation canon. Supplies the selected covariant structural branch and its stated premise discharges; it is not treated as a derivation from Electron V1.

[4] The earlier composed SFT manuscript, C14 and source notes, is used only for already established interpretation and scope boundaries.

[5] *SFT Gravity Constructive Repair v0.1*, static and radiation modules within the V4 v1.4 reproduction archive. Their original scripts are preserved as the numerical correspondence checks cited above.

Section scope. Equations (13.1)–(13.29) are retained from the V13 gravity assembly. The lower-limit differentiation in (13.10), convention-explicit weak expansions and optical readout, and the product-rule check of (13.29) are explanatory developments of the cited source equations. No new microscopic gravity, star, native-source, or cosmological solver is introduced here.

14. Expansion and the larger physical history

A maintained particle makes a local history. A universe contains many such particles, open radiation, bound structures, fields, and a geometry relating them. Expansion asks a different-scale question: **how does that shared geometry change as the whole physical configuration continues?** It does not require a second kind of time. It requires a law connecting complete large-scale sources to evolving large-scale geometry. [1, §§1–3; 3, §8.4]

V13.2 attacked that problem with a quantitative rate–capacity construction. It distinguished present activity from accumulated history, weighted active channels by energy divided by Planck action, and compared that rate with an area-based capacity. The resulting model has definite matter/radiation scalings, a spherical density coefficient, selected-radius and compactness relations, and sharp consistency tests. Those calculations remain useful because they make the proposed cosmological dependencies explicit. [1, §§5–12]

Revision B changes the interpretation. Literal writable addresses and compulsory fresh-address demand are no longer part of SFT's ontology. We therefore keep V13.2 as a **defined historical source–capacity model whose mathematics can be tested**, not as a microscopic expansion mechanism rescued by renaming its writes. The current task is to identify what a real source-to-geometry dynamics would have to preserve, replace, or explain. [3, §8.4]

14.1 Expansion does not require another kind of time

The clock account already supplies local elapsed history. Cosmology can compare a specified family of observers and ask how their separations evolve. In the homogeneous isotropic correspondence, let $a(t) > 0$ be dimensionless and let t be the proper time of comoving observers. For fixed comoving separation χ and reference volume V_0,

$$H = \frac{\dot{a}}{a}, \qquad d(t) = a(t)\chi, \qquad \dot{d} = Hd, \qquad V(t) = V_0 a^3, \qquad \frac{\dot{V}}{V} = 3H. \tag{14.1}$$

χ and d have length units and H has inverse-time units. These are operational definitions: they identify which physical clocks and separations are being compared. They do not install a master clock outside the universe. [4, C15.S01]

Nor does $a(t)$ force every bound structure to expand. Electron support, atomic separation, and molecular binding are maintained by their own local configurations. A population can separate on cosmological scales while bound members retain their local dimensions. [1, §9; 2, §5; 4, C15.S03]

What becomes collective is the source and the geometry through which local histories are compared. The remaining question is the law that determines that geometry.

14.2 The historical rate–capacity proposal

V13.2's central rate is energy weighted. Keeping its original notation makes the historical construction auditable:

$$\mathcal{R}_E \equiv \dot{N}_{\mathrm{write}} = \sum_a \chi_a \frac{E_a}{h}, \qquad I_{\mathrm{write}} = \eta_I \frac{A_{\mathrm{boundary}}}{A_{\mathrm{write}}}, \qquad \boxed{H = \frac{\mathcal{R}_E}{I_{\mathrm{write}}}.} \tag{14.2}$$

E_a is the source's accessible channel energy, χ_a its dimensionless participation factor, h Planck action in cycles, A_{write} a supplied area scale, and η_I a capacity convention. The numerator has inverse-time units and the denominator is dimensionless. Positive inputs select an expanding sign in this model. [1, §§4–7]

The original interpretation made I_{write} a count of writable addresses. That ontology is retired. What survives here is the mathematical proposal: a present source rate is compared with a boundary-scaled capacity. A current theory must identify the physical boundary and responding state that actually play those roles. [3, §8.4]

V13.2 also integrates the rate:

$$S_{\mathrm{write}} = \int_{t_0}^{t} \mathcal{R}_E(t')\, dt'. \tag{14.3}$$

As written, S_{write} is dimensionless accumulated production. It is not automatically an energy or a variational action for the metric. Its useful content is the distinction between **what is happening now** and **how much has accumulated over an interval**. [1, §5; 3, §§8.4, 9.1–9.2]

14.3 Energy weighting, matter, and radiation

Equal accessible energy gives equal instantaneous weight in (14.2), but energy frequency is not always a massive particle's own recurrence rate. For a formed particle moving with Lorentz factor γ,

$$\nu_C = \frac{mc^2}{h}, \qquad \frac{dN_C}{dt} = \frac{\nu_C}{\gamma}, \qquad \frac{E_{\mathrm{lab}}}{h} = \gamma \nu_C. \tag{14.4}$$

The middle quantity follows the particle along its path; the last is its laboratory energy divided by h. They agree in the rest frame and differ in motion. Thus $\mathcal{R}_E$ is an energy-weighted source rate, not simply the total count of proper-clock ticks. Radiation can contribute energy without possessing a massive rest clock. [4, C02.S01; core §4.6]

That distinction makes the source's matter/radiation scalings easy to state. For a fixed comoving population of nonrelativistic maintained roots with no conversion and negligible changing kinetic contribution,

$$N_m = \text{constant}, \qquad E_{m,\mathrm{one}} = mc^2, \qquad \rho_{E,m} = \frac{N_m mc^2}{V_0 a^3} \propto a^{-3}. \tag{14.5}$$

For freely propagating radiation, the source additionally assumes the usual metric redshift of photon energy,

$$N_\gamma = \text{constant}, \qquad E_\gamma(a) \propto a^{-1}, \qquad \rho_{E,\gamma} = \frac{N_\gamma E_\gamma(a)}{V_0 a^3} \propto a^{-4}. \tag{14.6}$$

Matter therefore dilutes as volume, while radiation carries one additional power from the decrease of photon energy. Equal present energy weighting does not imply identical evolution. [1, §9; 2, §5]

14.4 Turnover exposes what the boundary model must mean

If a finite remaining capacity is consumed at a fixed rate without replenishment, its exhaustion time is

$$t_{\text{exhaust}} = \frac{I_{\text{remaining}}}{\mathcal{R}_E}, \qquad I_{\text{remaining}} = I_{\text{write}} \quad \Longrightarrow \quad t_{\text{turn}} = \frac{I_{\text{write}}}{\mathcal{R}_E} = \frac{1}{H}. \tag{14.7}$$

The identity $t_{\text{turn}} = 1/H$ is exact under those premises. It becomes a physical deadline only if the capacity is literally finite, consumable, and unreplenished. Revision B no longer supplies that premise through event-address exhaustion. [1, §6; 3, §8.4]

Area growth also shows why "capacity production" cannot be left verbally ambiguous. If the capacity scales with spherical area and the radius itself expands at fractional rate H,

$$\begin{aligned} I_{\text{write}} \propto R^2 \quad &\Longrightarrow \quad \frac{\dot{I}_{\text{write}}}{I_{\text{write}}} = 2\frac{\dot{R}}{R}, \\ \frac{\dot{R}}{R} = H = \frac{\mathcal{R}_E}{I_{\text{write}}} \quad &\Longrightarrow \quad \dot{I}_{\text{write}} = 2\mathcal{R}_E. \end{aligned} \tag{14.8}$$

The factor two is ordinary area geometry. Active capacity, remaining capacity, and newly produced capacity cannot all be treated as the same object without an explicit boundary dynamics. This is a successful clarification of the original model, not a new coefficient. [4, C15.S03]

14.5 The boundary–bulk coefficient is explicit—and testable

The main V13.2 spherical calculation evaluates rate and capacity on one homogeneous sphere. With the positive assignment $R_H = c/H$ and full accessibility,

$$R_H = \frac{c}{H}, \qquad E_H = \frac{4\pi}{3}R_H^3\rho_E, \qquad \mathcal{R}_E = \frac{E_H}{h}, \qquad I_{\text{write}} = \eta_I \frac{4\pi R_H^2}{A_{\text{write}}}. \tag{14.9}$$

Substitution gives

$$H = \frac{A_{\text{write}}\rho_E R_H}{3\eta_I h} \quad \Longrightarrow \quad \boxed{H^2 = \frac{A_{\text{write}} c \rho_E}{3\eta_I h}}. \tag{14.10}$$

With the source's frozen $\eta_I = 1/4$ and $h = 2\pi\hbar$,

$$\begin{aligned} \eta_I = \frac{1}{4} : \qquad & \boxed{H^2 = \frac{2A_{\text{write}} c \rho_E}{3\pi\hbar}}, \\ G_A = \frac{A_{\text{write}} c^3}{\hbar}, \quad \rho_E = \rho_M c^2 \quad &\Longrightarrow \quad H^2 = \frac{2G_A\rho_M}{3\pi}. \end{aligned} \tag{14.11}$$

Here ρ_E is energy density in the chosen source frame and $\rho_M = \rho_E/c^2$ its mass-equivalent density. G_A packages the supplied area scale as a gravitationally dimensioned coupling. The calculation is valuable precisely because every normalization is visible. [1, §§7–8]

It also makes the mismatch with the separately supplied flat GR matter coefficient impossible to hide:

$$H_{\mathrm{cap}}^2 = \frac{2G_A\rho_M}{3\pi}, \qquad H_{\mathrm{GR}}^2 = \frac{8\pi G_A\rho_M}{3}, \qquad \boxed{\frac{H_{\mathrm{cap}}^2}{H_{\mathrm{GR}}^2} = \frac{1}{4\pi^2}.} \tag{14.12}$$

At fixed G_A and density, the two are not the same background law. V13.2 itself states that its result is not standard Friedmann equivalence. The factor $1/(4\pi^2)$ is therefore a real matching problem for any successor theory—not something removed by a change of vocabulary. [1, §7; 2, §4]

14.6 Radius and compactness make the normalization visible again

The model's selected radius comes from equating its capacity rate with c/R:

$$\frac{EA_{\mathrm{write}}}{4\pi\eta_I hR^2} = \frac{c}{R} \quad \Longrightarrow \quad R_{\mathrm{set}} = \frac{EA_{\mathrm{write}}}{4\pi\eta_I hc}. \tag{14.13}$$

For $E = Mc^2$ and the same area-to-coupling dictionary,

$$R_{\mathrm{set}} = \frac{G_A M}{8\pi^2\eta_I c^2}, \qquad \eta_I = \frac{1}{4}: \quad R_{\mathrm{set}} = \frac{r_s}{4\pi^2}, \qquad r_s = \frac{2G_A M}{c^2}. \tag{14.14}$$

This is a radius selected by the declared rate–capacity construction. It is not automatically a black-hole horizon; the cosmological sphere and an isolated Schwarzschild exterior are different physical problems. [1, §8]

The related compactness diagnostic starts from $C_A = G_A M/(c^2 R)$ and a homogeneous background law $H^2 = b_H G_A \rho_M$:

$$M = \frac{4\pi}{3}\rho_M R^3, \qquad H^2 = b_H G_A \rho_M \quad \Longrightarrow \quad \boxed{C_A = \frac{4\pi}{3b_H}.} \tag{14.15}$$

The two supplied coefficients then give

$$\begin{aligned} b_H &= \frac{8\pi}{3}: \quad C_{A,\mathrm{GR}} = \frac{1}{2}, \\ b_H &= \frac{2}{3\pi}: \quad C_{A,\mathrm{cap}} = 2\pi^2, \\ \frac{C_{A,\mathrm{cap}}}{C_{A,\mathrm{GR}}} &= 4\pi^2, \qquad \frac{C_{A,\mathrm{cap}}}{4\pi^2} = \frac{1}{2}. \end{aligned} \tag{14.16}$$

The recurring $4\pi^2$ factor is not an independent confirmation; it is the same normalization appearing in another algebraic readout. Its value is that the model's boundary choice remains inspectable. [1, §12]

14.7 A Hubble radius is not automatically a causal front

Defining $R_H = c/H$ does not say how that radius moves. Differentiation gives

$$R_H = \frac{c}{H}, \qquad HR_H = c, \qquad \boxed{\dot{R}_H = -\frac{c\dot{H}}{H^2}.} \tag{14.17}$$

If one additionally imposes $\dot{R}_H = c$, then the entire rate history is fixed:

$$\dot{H} = -H^2, \qquad H = \frac{1}{t - t_*}, \qquad R_H = c(t - t_*), \qquad a \propto t - t_*, \qquad t > t_*. \tag{14.18}$$

This is a coasting branch, $a \propto t - t_*$. It is not a generic consequence of the definition $R_H = c/H$. [1, §8; 4, C15.S03]

The point becomes sharper for a separately conserved homogeneous fluid with $p = w\epsilon$ and constant positive coefficient in $H^2 \propto \epsilon$:

$$\dot{\epsilon} = -3H(1 + w)\epsilon, \qquad \dot{H} = -\frac{3}{2}(1 + w)H^2, \qquad \boxed{\frac{\dot{R}_H}{c} = \frac{3}{2}(1 + w).} \tag{14.19}$$

Dust gives $\dot{R}_H = 3c/2$, radiation gives $2c$, and an always-c Hubble-radius speed requires $w = -1/3$ under these premises. This is a useful **compatibility exclusion**: those assumptions cannot all be imposed together. It does not forbid some other physically defined front. [4, C15.S03]

Actual light propagation makes the distinction concrete. In the flat FLRW correspondence,

$$ds^2 = -c^2 dt^2 + a^2(t)(d\chi^2 + \chi^2 d\Omega^2), \qquad R = a\chi, \qquad \dot{R}_{\text{null}} = HR \pm c. \tag{14.20}$$

At R_H, outgoing and ingoing areal rates are $2c$ and 0, while the local null signal still propagates at c. Hubble radius, particle horizon, event horizon, and a formation boundary therefore have different definitions and must not be merged merely because each is a length. [4, C15.S03]

14.8 Source evolution must stay consistent with geometry

An expanding volume changes the energy inventory even when no new source is created. For homogeneous density ϵ in physical volume V,

$$E = \epsilon V, \qquad \dot{E} = V\dot{\epsilon} + \epsilon\dot{V}. \tag{14.21}$$

With perfect-fluid pressure work and an explicit transfer rate Q_E,

$$\dot{\epsilon} + 3H(\epsilon + p) = Q_E, \qquad Q_E = 0 : \quad d(\epsilon V) = -p\, dV. \tag{14.22}$$

Cold matter and radiation recover the scalings above under their respective pressure and transfer assumptions. A moving horizon-selected volume also needs its boundary flux. Stock, flow, and volume change are distinct pieces of the same energy account. [4, C15.S02; C01.S02]

The forward direction of realized history does not itself select expansion. A contracting configuration can continue to have new events. Equation (14.2) chooses $H > 0$ through its positive rate/capacity prescription; a squared-rate law alone admits both signs. Expansion requires a physical branch-selection law. [3, §8.4; 4, C15.S01]

Acceleration adds another requirement. In the flat zero-Λ Einstein correspondence of Section 13,

$$H^2 = \frac{8\pi G}{3c^2}\epsilon, \qquad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2}(\epsilon + 3p). \tag{14.23}$$

This is a separate declared geometric law, not an extra equation to append silently to (14.11). Positive ongoing activity does not by itself imply negative effective pressure or accelerated expansion. A current cosmology needs one complete source, one conservation account, and one consistent geometric evolution. [4, C02.S04; C15.S01]

14.9 Recombination reorganizes matter and radiation

V13.2's older phrase *ledger crystallization* is best read physically as a change in the coupled matter–radiation medium. Electrons and protons already exist as clock-bearing participants before neutral atoms form. Recombination changes binding and scattering; it does not create local time for the first time. [1, §11; 4, C15.S01]

The source keeps optical depth as the schematic indicator:

$$\begin{aligned} \tau_{\mathrm{opt}} \gg 1 &: \quad \text{strong scattering,} \\ \tau_{\mathrm{opt}} \sim 1 &: \quad \text{last-scattering regime,} \\ \tau_{\mathrm{opt}} \ll 1 &: \quad \text{free-streaming regime.} \end{aligned} \tag{14.24}$$

The actual transition history requires the scattering rates and their evolution. Last scattering is therefore a historical emission/interaction regime, not automatically the present expansion boundary used by (14.2). [1, §11]

The radiation observed later carries both current propagating energy and information about earlier interactions. Those are two descriptions of the same photons, not two source stocks. The permanence of past events does not freeze the later observed sky against propagation or intervening interactions. CMB spectra, BAO, and late-time parameter fits remain outside the V13.2 freeze. [1, §§2, 11, 14; Appendix B]

14.10 Compact sources remain gravitational sources

For black holes, V13.2 introduced a separate expansion-accessibility factor,

$$\mathcal{R}_{\mathrm{BH,exp}} = f_{\mathrm{access}} \frac{Mc^2}{h}, \qquad 0 \le f_{\mathrm{access}} \le 1. \tag{14.25}$$

The function is left open. It weights one proposed expansion channel; it does **not** replace the conserved exterior gravitational mass by $f_{\mathrm{access}} M$. The complete-source rule from Sections 12–13 still applies, and horizon area or entropy cannot be added as a second copy of the mass already represented in the geometry. [1, §10.1; 3, §9.4]

The useful surviving distinction is stock versus flow versus boundary capacity. Thermodynamic entropy, accumulated history, present transfer, and geometric area can be related in a model, but they are not identical by definition. A future compact-source cosmology must specify the physical exchange that makes any such relation dynamical.

14.11 Local-to-global feedback remains a concrete target

V13.2 also proposed a possible meeting point between recurrent local support and global response. In its historical per-cycle notation,

$$\Delta \ln S_i = L_{\mathrm{eff}} u_i - \Phi_{\mathrm{relax}}, \qquad S_i > 0. \tag{14.26}$$

S_i is the source's settlement-amplitude variable, $L_{\text{eff}}u_i$ its dimensionless reinforcement, and Φ_{relax} a shared relaxation term. The microscopic relaxation operator was never solved. Section 12 shows how eliminating real internal variables can produce memory in a reduced description, but it does not derive this specific term. [1, §13; core §12.8]

If the relaxation contribution is genuinely common at the same step, subtraction gives

$$\boxed{\Delta \ln\left(\frac{S_i}{S_j}\right) = L_{\text{eff}}(u_i - u_j).} \tag{14.27}$$

The Δ matters. The algebra fixes the **change** in the logarithmic ratio. An absolute ratio requires an initial condition and accumulated evolution; identifying that ratio with particle masses requires an independently defined mass readout. [1, §13; 2, §8]

The interesting possibility survives: local persistence and global geometry may be coupled responses of the same world. The equation tells us what a successful common law would have to explain without pretending the hierarchy or expansion history has already been selected.

14.12 What survives, and what must be rebuilt

The V13 cosmological ambition remains coherent: maintained matter, propagating radiation, bound structures, gravity, and large-scale geometry should belong to one continuing physical history. Local recurrence supplies clocks. Complete sources determine gravitational response. Expansion asks how the geometry relating those clocks evolves. These are different questions about one world, not separate substances or separate energy accounts. [1, §§1–3; 3, §§8.4, 9.4]

The rate–capacity model remains valuable because it is explicit. Its dimensions close; the spherical coefficient is reproducible; matter and radiation expose their different transport assumptions; the radius, compactness, and front tests reveal exactly where boundary choices enter; and the $4\pi^2$ mismatch gives a concrete normalization problem rather than a hidden one. Those are real results of V13.2 even though its literal address mechanism has been retired. [1, §§5–14; 3, §8.4]

What must now be rebuilt is the causal source-to-geometry dynamics: the physical state that carries any history dependence, the complete source and boundary exchange, branch selection, and an evolving geometry consistent with conservation and observation. That successor law must then be tested through radiation transport, structure formation, and cosmological data. V13.2 did not already supply those fits. [1, §§2, 13–14; Appendix B]

The question survives the scaffold that first expressed it. V13 leaves both a physical target and a set of quantitative constraints for the next cosmological construction. V14 Part 2 owns that rebuild. The final technical section of this volume returns to the systems through which any such theory is tested: propagation, uncertainty, and physical readouts.

Sources and scope

[1] *SFT V13.2 — Expansion as Global Time Production and Boundary-Capacity Growth from Irreversible Writes*, R250E mechanism freeze, 26 June 2026. Primary source for the historical rate–capacity construction and its stated limits.

[2] *SFT V13.2 — Why the Universe Expands: Boundary Growth from Irreversible Writes*, narrative companion, June 2026. Used only where consistent with [1] and the current ontology.

[3] *SFT V13.13 — Live Ontology Constitution*, Revision B, especially §§3.2, 7, 8.4, and 9. Governs local history, complete source ownership, removal of literal write hardware, and the expansion target's current status.

[4] Earlier assembled clarifications C01.S02, C02.S01/C02.S04, and C15.S01–C15.S03. Used only for clock, conservation, volume, and boundary-consistency distinctions.

Section scope. Equations (14.1)–(14.27) are retained from the V13 expansion assembly. Explanatory algebra introduces no new cosmological coefficient. No native expansion, thermal-history, CMB, BAO, or cosmological-fit solver is introduced here.

15. Propagation, uncertainty, and physical readouts

A prepared electron enters an apparatus, continues through a responding environment, and may leave a detector record. Those are stages of one physical encounter. **The particle, the spread of available readouts, and the record that finally appears belong to different parts of the same problem.** [1, §§3–4; 3, §§4.2, 6–7]

V13.6 develops an uncertainty proposal around unresolved response, assigns the candidate diffusion scale $\hbar/(2m)$, and connects that scale to a Fisher-information bound. Its barrier work supplies evanescent attenuation. V13.9 then keeps the detector step separate by adding an explicit completion-rate law. Read together, these results give a useful order: propagation first, response structure next, event completion after that. [1, §§3–5; 2, §§5–7; 4, §14]

Earlier sources sometimes called elementary continuation a *next write*. Under V13.8-V2, that is now bookkeeping language for lawful evolution, not a physical append operator or address grid. The questions here are therefore ordinary physical ones: what propagates, what remains unresolved, what distribution follows, and what interaction produces a record? [5, §§3–7]

15.1 A propagating phase is not a clock riding on the ray

A signal can carry phase without carrying a massive rest clock. In a local inertial chart $X^\mu = (ct, \mathbf{x})$ with signature $(-,+,+,+)$, use

$$\Theta = \mathbf{k}\cdot\mathbf{x} - \omega t, \qquad k_\mu = \partial_\mu \Theta = (-\omega/c, \mathbf{k}), \qquad p_\mu = \hbar k_\mu, \qquad \omega(U) = -k_\mu U^\mu. \tag{15.1}$$

Θ is dimensionless, $\mathbf{k}$ has inverse-length units, and U^μ is the receiving observer's four-velocity with $U_\mu U^\mu = -c^2$. The contraction gives the observed angular frequency of the future-directed signal. A detector samples that phase along its own timelike history. No photon rest tally is required. [6, C02.S01]

For the supplied free positive-energy massive correspondence,

$$\begin{gathered} E^2 = c^2|\mathbf{p}|^2 + m^2c^4, \qquad \omega^2 = c^2|\mathbf{k}|^2 + \left(\frac{mc^2}{\hbar}\right)^2, \\ \mathbf{v}_g = \nabla_{\mathbf{k}}\omega = \frac{c^2\mathbf{k}}{\omega} = \frac{c^2\mathbf{p}}{E}, \qquad v_{\mathrm{ph}} = \frac{\omega}{|\mathbf{k}|}, \qquad v_{\mathrm{ph}}|\mathbf{v}_g| = c^2 \quad (|\mathbf{k}| > 0). \end{gathered} \tag{15.2}$$

The group velocity follows a narrow packet in this dispersion model; phase velocity follows a surface of constant phase. For a massive branch they need not equal one another, and neither statement licenses superluminal signaling. Causal propagation belongs to the complete equations and their characteristics. The massless limit gives both speeds c. [6, C02.S01, C02.S03]

Propagation, internal recurrence, and reception can therefore carry different rates without describing different substances. Each rate belongs to the operation that defines it.

15.2 The next continuation begins from the state already reached

V13.6 separates self-response, controlled external response, and unresolved response:

$$\begin{aligned}\Sigma_{\text{total}} &= \Sigma_{\text{self}} + \Sigma_{\text{controlled}} + \delta\Sigma_{\text{unresolved}},\\ x_{n+1} &= F(x_n, u_n, \Sigma_{\text{total}}(x_n, t_n))\,,\\ x_{n+1} &= x_{n+1}^{\text{clean}} + \xi_n.\end{aligned} \tag{15.3}$$

F stands for the physical continuation law to be supplied; the index n labels successive samples in this description and imposes no universal time lattice. The useful point is causal: an unresolved influence changes not only a future increment but potentially the state from which later evolution begins. [2, §5; 5, §§3, 6]

Statistics require more than the statement that the full continuation is definite. They require an ensemble, unresolved variables, their coupling, and their correlations. If additive transported deviations are summarized by $X_N = \sum_n \xi_n$,

$$\delta\xi_n = \xi_n - \langle\xi_n\rangle, \qquad \text{Cov}(X_N) = \sum_{n=1}^{N}\sum_{m=1}^{N} \langle\delta\xi_n \delta\xi_m^T\rangle\,. \tag{15.4}$$

The off-diagonal terms preserve correlations between different steps. Neglecting them is an extra physical assumption, especially when eliminated response has memory. **Past dependence is not yet a diffusion law.** [2, §§5–6; core §12.8]

15.3 A concrete candidate scale for unresolved fluctuations

R270R proposes the diffusion normalization

$$\begin{gathered} r_{\text{spin}} = \frac{\hbar}{2mc}, \qquad \boxed{D_{\text{eff}} = c\, r_{\text{spin}} = \frac{\hbar}{2m} = \frac{c\ell_C}{4\pi}},\\ \ell_C = \frac{h}{mc}, \qquad [D_{\text{eff}}] = L^2T^{-1}.\end{gathered} \tag{15.5}$$

This fixes D_{eff} once the half-reduced-Compton diagnostic and the product with c are supplied. It does not derive that length from the current Electron V1 geometry or identify it with a scattering radius. [1, §4; 2, §6]

The stochastic statement needed to turn that scale into an effective isotropic diffusion is stronger:

$$\left\langle \delta\xi_i \delta\xi_j \right\rangle = 2D_{\text{eff}}\delta_{ij}\Delta t = \frac{\hbar}{m}\delta_{ij}\Delta t. \tag{15.6}$$

Δt is the time coordinate of the effective ensemble model. Interpreting the left side as covariance requires centered increments, or an equivalent zero-mean premise. V13.6 leaves the microscopic derivation of this covariance open; its archived script checks the coefficient identity, giving $D_{\text{eff}} \simeq 5.78838 \times 10^{-5}\,\text{m}^2/\text{s}$ for its stored electron inputs. [1, §4; 7, R270R]

A continuum diffusion equation would be an approximation to the underlying causal dynamics, not its replacement. Its long tails do not define the signal cone.

15.4 The uncertainty scale follows once the statistical premises are stated

Let $\rho(x) > 0$ be smooth and normalized on the line, with finite variance and decaying boundary terms. Define

$$\int_{\mathbb{R}} \rho\, dx = 1, \qquad \bar{x} = \int x\rho\, dx, \qquad (\Delta x)^2 = \int (x-\bar{x})^2 \rho\, dx,$$
$$I_F = \int \rho \left(\partial_x \ln \rho\right)^2 dx < \infty. \tag{15.7}$$

Integration by parts gives

$$\int (x-\bar{x})\rho\, \partial_x \ln \rho\, dx = \int (x-\bar{x})\partial_x \rho\, dx = -1. \tag{15.8}$$

and Cauchy–Schwarz then yields

$$1 \le \left[\int (x-\bar{x})^2 \rho\, dx\right] \left[\int \rho (\partial_x \ln \rho)^2 dx\right], \qquad \boxed{(\Delta x)^2 I_F \ge 1.} \tag{15.9}$$

To express the source's diffusion scale as a momentum-valued fluctuation coordinate, add the explicit definition

$$p_F = m D_{\text{eff}} \partial_x \ln \rho, \qquad \langle p_F \rangle = 0, \qquad (\Delta p_F)^2 = m^2 D_{\text{eff}}^2 I_F,$$
$$\boxed{\Delta x \Delta p_F \ge m D_{\text{eff}} = \frac{\hbar}{2}}. \tag{15.10}$$

Under these premises the factor $1/2$ follows directly from $D_{\text{eff}} = \hbar/(2m)$. But p_F is a *score momentum*: a variable built from the spatial density gradient. The mathematics does not by itself establish that a detector measures this quantity. [1, §4; 2, §6]

15.5 A conventional momentum observable adds another layer

To connect the Fisher term to the ordinary momentum spread, specify a normalized amplitude and the conventional momentum operator:

$$\psi(x) = \sqrt{\rho(x)} e^{iS(x)/\hbar}, \qquad \hat{p} = -i\hbar \partial_x, \qquad \hat{p}\psi = \left[S'(x) - \frac{i\hbar}{2}\partial_x \ln \rho\right] \psi. \tag{15.11}$$

Direct evaluation gives

$$\langle \hat{p} \rangle = \int \rho S'\, dx,$$
$$(\Delta p)^2 = \mathrm{Var}_\rho(S') + \frac{\hbar^2}{4} I_F \ge \frac{\hbar^2}{4} I_F, \tag{15.12}$$
$$\boxed{\Delta x \Delta p \ge \frac{\hbar}{2}}.$$

The full momentum variance contains both phase-gradient variation and the Fisher amplitude-gradient term. The standard uncertainty product therefore follows within the stated representation, while the physical origin of the preparation ρ, S remains a dynamical question. [1, §4; 6, C08.S03]

This distinction matters. A fluctuation-scale argument, an operator correspondence, and an actual detector coupling are related steps, not interchangeable names for the same mechanism.

15.6 A Gaussian shows the limiting tradeoff

For a Gaussian position profile with constant phase gradient,

$$\begin{aligned} \rho(x) &= \frac{1}{\sqrt{2\pi}\sigma}\exp\left[-\frac{(x-x_0)^2}{2\sigma^2}\right], \\ \partial_x \ln\rho = -\frac{x-x_0}{\sigma^2}, &\qquad I_F = \frac{1}{\sigma^2}, \qquad \Delta x = \sigma, \qquad \Delta p = \frac{\hbar}{2\sigma}. \end{aligned} \tag{15.13}$$

the bound is saturated. Narrowing this family in position enlarges its momentum spread in the precise reciprocal way above. The Gaussian is a calibration example, not a uniquely selected SFT source profile. [6, C08.S03]

Nor are all quantities called *width* the same. Aperture size, standard deviation, and diffraction-zero spacing are different observables. The finite-variance domain matters: an ideal discontinuous top-hat has a divergent continuum momentum second moment and therefore does not furnish a finite counterexample to (15.12).

15.7 Interference belongs to the complete route comparison

Two alternatives brought to one detector channel must include any physical markers that distinguish them. Let

$$\begin{aligned} r &= a_A e_A + a_B e_B, \\ \gamma &= \langle e_A, e_B\rangle, \qquad |\gamma| \le 1, \\ \mu &= \|r\|^2 = |a_A|^2 + |a_B|^2 + 2\,\mathrm{Re}(a_A^* a_B \gamma). \end{aligned} \tag{15.14}$$

The route dynamics determines a_A, a_B; the marker interaction determines γ. If the marker states become orthogonal, their direct interference term vanishes. Merely separating the routes in space does not force that result. [4, §§6–11; core §6]

For balanced amplitudes,

$$\mu_{\max} = 2a^2(1+|\gamma|), \qquad \mu_{\min} = 2a^2(1-|\gamma|), \qquad \mathcal{V} = \frac{\mu_{\max}-\mu_{\min}}{\mu_{\max}+\mu_{\min}} = |\gamma| \quad (a>0). \tag{15.15}$$

so the visibility is exactly the marker-overlap magnitude in this ideal comparison. Loss of contrast is therefore a distinguishability statement, not yet the selection of one completed event.

The phase-bearing configuration determines how alternatives combine. The detector dynamics determines how one alternative becomes a record.

15.8 A barrier changes propagation before a detector counts anything

For the rectangular-barrier correspondence used by V13.6,

$$V_0 > E, \qquad \kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}, \qquad \psi'' = \kappa^2 \psi, \qquad \psi_{\mathrm{dec}}(x) \propto e^{-\kappa x}. \tag{15.16}$$

The decaying branch gives the leading interior attenuation

$$\mathcal{A}_{\mathrm{bar}}(d) \equiv e^{-2\kappa d}, \qquad \ln \mathcal{A}_{\mathrm{bar}} = -2\kappa d, \qquad \mathcal{A}_{\mathrm{bar}}(d_1 + d_2) = \mathcal{A}_{\mathrm{bar}}(d_1)\mathcal{A}_{\mathrm{bar}}(d_2). \tag{15.17}$$

This is deliberately an attenuation factor, not a full transmitted-current probability. Complete transmission requires boundary matching and current normalization. The native SFT program has not derived κ from the maintained particle itself. [1, §5; 2, §7; 7, R272Q]

A time-dependent material can modulate the barrier response, but static attenuation does not require a literal temporary gap. And a small far-side amplitude is still a propagation statement. Turning it into a rate of far-side records requires the detector layer.

15.9 Relative response and absolute event rate are different data

V13.9 retains an explicit bridge from response capacities to first-completion hazards. For eligible channels,

$$\lambda_i(u) = g_D(u)\mu_i, \qquad M_\mu = \sum_i \mu_i > 0, \qquad P(i \mid \text{completion}) = \frac{\mu_i}{M_\mu}. \tag{15.18}$$

The common profile g_D controls timing; the dimensionless capacities control the conditional proportions. Multiplying every rate by the same factor changes waiting times without changing the normalized outcome ratios. [4, §14]

A finite observation window makes non-completions explicit:

$$\begin{aligned} P(\emptyset_W) &= e^{-M_\mu G_W}, \\ P(i \text{ in } W) &= \frac{\mu_i}{M_\mu}\left(1 - e^{-M_\mu G_W}\right), \\ P(\emptyset_W) + \sum_i P(i \text{ in } W) &= 1. \end{aligned} \tag{15.19}$$

The missing probability is not silently discarded. A no-event window, an ineligible event, a late event, and an unregistered event can correspond to different apparatus histories.

If channels have different fixed efficiencies within the hazard model,

$$\lambda_i(u) = g_D(u)\eta_i\mu_i \quad \Longrightarrow \quad P(i \mid \text{registered completion}) = \frac{\eta_i\mu_i}{\sum_j \eta_j\mu_j}, \tag{15.20}$$

the registered proportions change. Thus the Born-like normalized table requires its detector assumptions to remain attached. Post-event record loss would be yet another instrument model, not the same modification. [4, §14; 6, C08.S02/C08.S05]

15.10 State spread and instrument error must not be merged

The uncertainty relation concerns a preparation and an observable representation. Instrument resolution is an additional physical limitation. For the clock-style phase readout $\Phi = \omega_0 \Delta\tau$,

$$\delta(\Delta\tau) \simeq \frac{\delta\Phi}{\omega_0} - \Delta\tau\frac{\delta\omega}{\omega_0}, \qquad \sigma_\tau^2 \simeq \frac{\sigma_\Phi^2}{\omega_0^2} + (\Delta\tau)^2\frac{\sigma_\omega^2}{\omega_0^2}. \tag{15.21}$$

The variance line assumes independent centered phase and calibration errors; drift, covariance, or missed whole cycles require additional terms. This is an apparatus-error model, not a fundamental time-energy uncertainty law. [6, C02.S02]

The apparatus itself is physical. Its standards are maintained configurations, its couplings are interactions, and its records are present states that preserve distinctions about previous events. **A realized event can be definite while its prediction is statistical, its record incomplete, or its reported value inaccurate.** [3, §§4.2, 5.5, 7]

15.11 From continuation to record

The pieces now have a definite order. A preparation establishes a state. Propagation and environmental response change its continuation. The chosen representation determines the available spread and interference. A detector interaction supplies an event law, and the apparatus can retain a record.

V13 contributes concrete mathematics at several points in that chain: a candidate $\hbar/(2m)$ fluctuation scale, a Fisher/Cauchy uncertainty route, evanescent barrier attenuation, phase-marker interference, and a conditional completion bridge. It does **not** yet derive the microscopic covariance in (15.6) or the physical actualization of one detector record. [1, §§3–5; 2, §§5–7; 4, §§14, 16–17]

V13.6 leans toward unresolved deterministic influences; V13.9 leaves fundamental versus emergent stochasticity open. The assembled account preserves that question rather than deciding it by wording. [1, §3; 4, §14]

We have returned to the systems doing the observing. The same physical participants that persist, interact, form atoms, molecules, and gravitational sources also build the apparatus through which those relations become numbers. The prediction and the measurement belong to one world.

Next is the closing synthesis: the V13 physical account, the results it actually carries, and the boundary with **V14 Part 2**.

Sources and scope

[1] *SFT V13.6 — Quantum Settlement Rules to Chemistry*, full writeup, especially §§3–5. Supplies the R270Q unresolved-response proposal, R270R diffusion normalization and Fisher/Cauchy claim, and R272Q barrier scope.

[2] *SFT V13.6 — Quantum-to-Chemistry Full Derivation*, especially §§5–7. Preserves the uncomputed covariance, native barrier coefficient, and detector dependencies.

[3] *SFT V13.13 — Live Ontology Constitution*, Revision B, especially §§4.2–7. Governs constituted apparatus, local clocks, relational response, and the distinction between realized history and present records.

[4] *SFT V13.9-V3 Working Relational Model v1.1 — SUPER FREEZE*, especially §§6–17. Governs phase-bearing response, distinguishing marks, the conditional completion bridge, and the open microscopic detector problem.

[5] *SFT V13.8-V2 — There Is No Write Operator*, final v1.0, §§3–7. Governs the reading of next-write language as lawful continuation rather than a storage/update machine.

[6] Earlier composed-manuscript clarifications C01.S01–C01.S03, C02.S01–C02.S03, and C08.S02–C08.S05 are used only for already-established phase/observer conventions, causal-versus-effective propagation, finite-variance momentum scope, and apparatus/readout distinctions.

[7] Original V13.6 scripts-only archive, R270R coefficient check and R272Q rectangular-barrier attenuation check. The stored scripts test the stated coefficient identities and supplied attenuation cases; neither is a native covariance, trajectory, transmission-current, or detector solver.

Mathematical scope. Equations (15.1)–(15.21) retain the original displayed relations. The Fisher, Cauchy–Schwarz, momentum, Gaussian, marker-overlap, hazard, and error-propagation calculations are explanatory deductions under their stated premises. No new physical coefficient is fitted, no native detector or uncertainty solver is introduced, and the preceding fourteen sections remain unchanged.

16. One continuing physical world

We began with a persistent thing and asked what makes it the thing it is. We then introduced another participant, the relations between them, and the larger structures those relations can maintain. By the end of the account, some of those structures have become clocks, rulers, emitters, and detectors. **The objects doing the observing belong to the same physical world as the objects being observed.** That is the circle this volume closes. [1, §§3–7, 18]

SFT proposes that open propagation, maintained matter, interaction, and geometry can be understood through complete physical configurations of common causal support. The recurring distinction is organization: what propagates openly, what returns persistently, what belongs intrinsically to one participant, and what exists only in a relation or larger configuration. The preceding sections give that proposal concrete models, conditional theorems, effective applications, and explicit open connections. This closing section gathers them without adding a new microscopic law. [1, §§1–3, 14]

16.1 Persistence is an achievement of the physical state

The maintained particle is not an inert object with a separate machine attached to keep it alive. Its propagation, finite support, source, reaction, framing, and geometry are parts of the organization that persists. A clock, current, magnetic moment, or mass readout is a way that organization appears in a specified comparison, not another object doing the particle's work. V13.12 makes that claim concrete for one frozen electron-class state. [2; Section 3]

For each specified model, the common order is

$$X_\star \in \mathcal{D}_J, \qquad \delta S_J[X]\big|_{X_\star} = 0, \qquad O_a = R_a[X_\star]. \tag{16.1}$$

The lawful domain $\mathcal{D}_J$ states which configurations and variations are admitted, S_J supplies the dynamics, $X_\star$ is the solved state, and R_a defines a readout. This notation does not assert one already-completed universal action. It enforces a simpler discipline: solve the physical state before assigning its properties. [3, §§69–71; 9]

That discipline matters in the electron. A round scalar support can coexist with nontrivial framing; vanishing ordinary orbital angular momentum does not mean inactive local support; charge dressing, recurrence, and spin response belong to the same maintained configuration. The particle is the organization, not a carrier plus an independently installed list of attributes. [2; Section 3]

Recurrence is likewise a return of organization, not a repetition of an event:

$$[\mathcal{C}(s+P)]_{\mathcal{G}} = [\mathcal{C}(s)]_{\mathcal{G}}, \qquad e_{n+1} \neq e_n. \tag{16.2}$$

$P > 0$ is the return interval in the stated continuation parameter and $\mathcal{G}$ contains only genuine equivalences of that model. Successive returns can be equivalent while belonging to distinct events. The particle can remain the same structure while its history grows. [1, §§2.2–3; Section 4]

The transverse stability result gives this idea teeth. For the reaction-completed return map M,

$$G_\perp = G_\perp^T > 0, \qquad M^T G_\perp M = G_\perp, \qquad I_\perp(\mathbf{z}_n) = I_\perp(\mathbf{z}_0), \qquad \mathbf{z}_n = M^n \mathbf{z}_0. \tag{16.3}$$

The positive invariant bounds the tested transverse departures. The frozen audit follows the map through 100,000 cycles, while the overdriven saddle and reaction-removed controls demonstrate that the calculation can distinguish bounded maintenance from growth or missed return. This is a real stability result at its stated scope, not a proof about every nonlinear field mode. [2, C17A and C17G16; Section 3]

Formation then has a sharply defined job: produce the organization that maintenance can sustain. V13.14 obtains a physical first turn, nascent supporting response, and a reduced connection to the maintained geometric class. The uninterrupted birth of two self-maintaining descendants remains unfinished. There is no physical handoff to a second electron machine once a loop closes; the same support must acquire and continue the complete state. [5; Section 7]

16.2 Local history supplies the clock and the ruler

A repeatable physical process can supply a local unit and a tally. In the formed fixed-mass correspondence,

$$T_C = \frac{h}{mc^2}, \qquad \nu_C = \frac{mc^2}{h}, \qquad \Delta\tau = T_C \Delta N_C, \qquad \frac{dN_C}{dt} = \frac{\nu_C}{\gamma} \quad \text{in the flat inertial comparison.} \tag{16.4}$$

The proposal is not that every measurable interval contains an integer number of turns. It is that recurrence supplies a physical local standard from which elapsed duration can be read. Different paths can accumulate different proper histories without a universal master clock or a reservoir of missing time. [9; Sections 2 and 4]

Open null propagation does a different job. A photon carries no massive rest clock, yet it connects emission and reception and acquires an observed frequency only relative to a receiving structure. Likewise, a Compton-associated length can supply a local unit without becoming the literal surface of an electron. Extended rulers require maintained relations among several persistent structures. [1, §§4–5; Sections 2, 4, 15]

This keeps the scale hierarchy coherent. An atom need not create a new kind of time above the electron; a material can use its own accessible transition or collective mode; a cosmological observer family can define the comparison used to describe expansion. **Clocks remain local. What grows with structure is the network of comparisons.** [1, §§4–5; Sections 4 and 14]

16.3 Relations are physical parts of the complete configuration

When two persistent structures interact, the physical problem changes. Their isolated states remain useful references, but the joint solution contains what they do together. On a conservative branch,

$$\Delta E_{AB}(R) = E[X^\star_{AB}(R)] - E[X^\star_A] - E[X^\star_B], \qquad F_R = -\frac{d\Delta E_{AB}}{dR}. \tag{16.5}$$

Force is therefore a participant-side readout of how the solved joint energy changes with separation. It is not an extra substance inserted between the participants. Changing and radiating systems require the corresponding dynamical exchange account, but the principle is the same: the interaction belongs to the complete configuration. [3; Sections 1 and 5]

The electromagnetic sector is the clearest particle-to-interaction match in V13. The frozen electron's charge and Pauli moment, together with one electromagnetic normalization, determine the leading

Coulomb, spin-spin, motion-dependent, and spin-orbit classes at long range. Those are not four independently tuned agreements. The self-dressing and the pair interaction use the same electromagnetic language. [3, §73; Section 5]

Joint structure can also survive after substantial mechanical coupling has ended. The binary relational model captures this with two exact normalized identities,

$$P^2 + V^2 + C^2 = 1, \qquad C_0^2 = A_{AB}^2 + X_{AB:E}^2. \tag{16.6}$$

The first relates local bias, local phase accessibility, and nonfactorizability. The second tracks how direct pair accessibility is redistributed when an environment physically distinguishes alternatives. Neither is an energy budget. Together they show that a local description of each participant can omit real structure of the whole. [4, §§7–11; Section 6]

The same declared binary interface reproduces the singlet angular law, the quantum-strength CHSH response, and setting-independent local marginals. A separate completion law connects normalized capacities to conditional detector frequencies. SFT therefore rejects Bell-local factorization without requiring a mechanically transmitted command between spacelike-separated analyzers. The settings act locally; the relevant source relation is joint. Individual event realization remains a distinct dynamical problem. [4, §§12–17; Sections 6 and 15]

Separation is not an eraser. A relationship changes when a physical process changes it.

16.4 Complete participants can build new complete participants

Species identity and composite identity answer different questions. An electron can change motion, orientation, or environment while remaining an instance of the same maintained class. Two electrons are not obtained by multiplying one nonlinear root's amplitude. Their joint state needs its own lawful composition. [3; Section 8]

The first atom shows how this becomes quantitative before every constituent interior is solved. With a guarded proton, supplied masses and electromagnetic normalization, and the stated localization law,

$$\begin{aligned} E(a) &= \frac{\hbar^2}{2\mu a^2} - \frac{\alpha\hbar c}{a}, \qquad \mu = \frac{m_e m_p}{m_e + m_p}, \\ a_\mu &= \frac{\hbar}{\mu c\alpha}, \qquad B_H = -E(a_\mu) = \frac{\mu c^2 \alpha^2}{2}. \end{aligned} \tag{16.7}$$

One balance selects both the reduced-mass atomic scale and the leading binding energy: approximately 52.9465 pm and 13.5983 eV with the archived inputs. The localization coefficient remains an effective input, but the radius and binding are not fitted separately. [6; Section 10]

The energy of the larger participant then follows one accounting rule:

$$E_{\text{bound}} = E_{\text{sep}} - B, \qquad M_{\text{bound}} = \sum_i m_i - \frac{B}{c^2}, \qquad B > 0. \tag{16.8}$$

Binding is a property of the complete state. Once the bound object is represented by its total mass, its constituents and binding energy are not added again as a second source. This same rule carries from atoms to molecules, materials, and gravitational source accounting. [3; Sections 8, 10–12]

Chemistry continues the pattern. Under the supplied state and antisymmetric-composition rules, finite shell capacities follow exactly; the filling rule supplies a qualitative energetic skeleton; two-center phase comparison produces bonding and antibonding support patterns. But overlap is not energy. A molecular bond requires a complete energy branch with a lower minimum at finite separation. The existing V13.6 calculations establish ingredients of that problem without pretending to have solved the full H_2 curve. [6; Section 11]

The strong, weak, and neutral sectors make the same distinction at other scales. Fractional labels can describe constituents of a lawful whole without implying freely isolated fractional particles. Chiral access, collective loading, and open-channel width describe different jobs in a transition. The minimal electroweak construction relates several leading responses through one parent norm while leaving scale, family dependence, and physical realization explicit. [3; Sections 8–9]

16.5 The inside still acts when we describe only the whole

A coarse description succeeds by solving or averaging internal response, not by making the interior unreal. On a regular conservative branch,

$$S_{\mathrm{eff}}[x] = S_J[x, y_\star[x]], \qquad H_{\mathrm{eff}} = H_{xx} - H_{xy}H_{yy}^{-1}H_{yx}. \tag{16.9}$$

The Schur term is the reaction of the variables no longer displayed explicitly. The nonlinear envelope theorem gives the same lesson at first variation; causal elimination can instead leave a retarded memory kernel. Constraints, branch choice, boundary conditions, and initial internal state remain part of the problem. [3, §§61, 69–71; 9; Section 12]

This is the bridge from microscopic organization to collective response. Internal variables can soften a displacement, store and return energy, remember an earlier disturbance, or open an outgoing channel. A particular material law still needs its actual variables and coefficients, but the response does not have to be appended as a second mechanism after the object is built. [3; Section 12]

Gravity makes source completeness unavoidable. Pressure, field stress, binding, support, and motion belong to the source of the geometry when they are physically present. That geometry then governs clocks, massive trajectories, and null propagation together. In the static spherical convention,

$$\begin{aligned} ds^2 &= -N(r)^2c^2dt^2 + S_r(r)^2dr^2 + r^2d\Omega^2, \\ \frac{d\tau}{dt} &= \frac{N}{\gamma_{\mathrm{loc}}}, \qquad \left|\frac{dr}{dt}\right|_{\mathrm{null}} = c\frac{N}{S_r}. \end{aligned} \tag{16.10}$$

One metric supplies these readouts. V13.1-V4's declared timing and radial-response hypotheses close to the exact Schwarzschild exterior, extend to the tested many-source static order, and join a specified spin-two radiative completion with a checked near-zone/far-zone energy balance. V13.1-V3 retains a distinct selected covariant branch under its own source assumptions. The present electron-to-exterior source match is still open; the existing gravitational constructions are not silently rebranded as a fresh Electron V1 derivation. [7; Section 13]

That boundary does not diminish the main requirement. **A clock explanation that works gravitationally must survive the trajectory and optical tests of the same geometry.** SFT cannot choose one geometry for clocks and another for light because each happens to fit a different observation.

Expansion is the corresponding large-scale question. V13.2 supplies explicit rate-capacity, density, radius, matter/radiation, and boundary-consistency calculations. Revision B retires the literal writable-address mechanism that originally motivated them. The surviving question is whether complete sources and their continuing response can produce the evolving geometry those calculations were trying to describe. Section 14 leaves that as a defined construction problem, not an already-fitted cosmology. [8; Section 14]

16.6 The observer is part of the same account

The apparatus is not exempt from the ontology. Its components are maintained matter; its clock is a physical standard; its fields couple to the incoming system; and its final configuration can preserve a distinction about what occurred. Errors, thresholds, missed events, and calibration limits are physical properties of that process. [1, §§5–7; Sections 6, 15]

That immediately separates several questions that are often collapsed together. A prepared state's spread is not the same thing as an instrument's error. A response capacity is not yet an absolute event rate. A scattering radius belongs to a specified interaction and readout, not automatically to an internal coordinate width. A conditional distribution among registered events must retain the selection by which those events were registered. [Sections 3–6, 15]

The same discipline applies to computation. A positive invariant can prove bounded motion for a declared map. Replaying an archived script can confirm that its stated inputs still produce its stated output. An algebraic identity can be exact. None of those is by itself an experiment showing that nature uses the model. Their value is that they make the next comparison sharper: state, law, preparation, and readout are already named. [1, §§14–15; Sections 3, 6, 10, 13]

Putting the observer inside the theory raises the standard of explanation. The same physical definitions used to predict a result must also describe how the apparatus makes it observable.

16.7 What V13 establishes together

Taken together, V13 is no longer merely a list of intuitions. It contains a maintained electron model with bounded transverse response; a matched long-range electromagnetic interaction hierarchy; a binary relational model with exact accessibility identities and singlet correlations; a concrete first-turn formation program; structured species, composition, and transition constraints; an effective hydrogen minimum; a first-bond architecture; exact common-action elimination results; and gravitational constructions joining clocks, space, sources, and radiation within their declared branches. [2–7]

Those results do not all have the same evidential status, and they should not. Some are exact algebra under stated premises. Some are numerical tests of a defined model. Some are effective applications with disclosed empirical inputs. Some are candidate mechanisms whose decisive dynamical connection remains open. Keeping those categories visible is what makes the successes reusable and the failures informative. [1, §§14–15]

The controls matter for the same reason. Removing the electron's reaction shows that free propagation does not maintain the return. Overdriving the saddle verifies that the stability machinery detects growth. The formation forward-return failure identifies the need for daughter-local feedback after the shared source fades. The Hubble-radius compatibility test rules out an incompatible combination of assumptions. These are not embarrassments when they answer the question they were designed to test. [Sections 3, 7, 14]

A supplied coefficient is not a prediction, but it can still participate in nontrivial consequences. A conditional theorem is not a universal law, but it can still constrain every model satisfying its premises. A coherent account should neither inflate nor erase those distinctions.

16.8 Part II begins where this account genuinely ends

This volume completes the V13 physical account. It does not claim to complete every dynamical connection in SFT. That distinction is now a feature rather than an unfinished manuscript problem. Part I states what we think the theory presently says; Part II is where the next physics is built.

The most consequential next work follows the existing dependencies. Formation must produce the maintained descendant without using the finished route and scale as its cause. The complete particle source must connect to a controlled exterior and many-particle gravitational description. The expansion program must replace the retired address mechanism with an actual responding state, conserved source, boundary exchange, and evolving geometry. [1, §12; 5; 7–8]

Other open joints remain equally concrete: native species and composition, family-sensitive neutral response, quantitative molecular binding, and the physical dynamics that realizes an individual detector record. None needs to be hidden in this volume, and none needs to be solved retroactively for the V13 results at narrower scope to remain what they are.

Earlier SFT remains a source of candidate mechanisms, constraints, and solved special cases. It enters the current account only through explicit physical compatibility. Preserving that history allows old work to be reused without forcing the reader to confuse chronological ancestry with present ontology. [1, §§13–16]

The theory remains answerable to future calculations and observations. A successful result can strengthen a connection; a contradiction can require revision. Preserving this volume protects the meaning of the present claims from drift, not from discovery. [1, §§13–14]

Conclusion — The same world at every step

SFT asks what familiar physical things are doing to be the things they are. In the present account, open propagation and maintained recurrence are different organizations of common causal support. Persistent structures can possess intrinsic magnitudes. When structures coexist, new physical relationships become possible. Larger structures can then preserve those relationships long enough to become atoms, molecules, materials, clocks, rulers, sources, and detectors.

The mathematical work makes that progression more than a metaphor. It supplies a maintained-state model, stability and response theorems, matched interaction classes, exact relational identities, effective atomic and chemical constructions, and gravitational and cosmological calculations whose assumptions remain visible. The unfinished connections define the next research program; they do not require us to pretend the established pieces are either more or less complete than they are. [2–8]

The central commitment is continuity. The particle that makes the clock is the particle that enters the interaction. The structure counted as a gravitational source must include the response that maintains it. The apparatus that reports a result is built from the same physical world as the system it measures. Mathematical simplification is useful only while it remains a faithful description of that continuing world. [1, §§7–9]

One thing can possess magnitudes. Two things have relationships. Many things can maintain a world of structures that compare one another.

A structure returns. Another event occurs. A signal reaches another structure. A record becomes possible. The task is to explain that continuing world precisely enough that the explanation can fail, survive, or improve under comparison with nature.

This is the V13 account we are putting forward. [1, §18]

The companion resources

The **complete reference assembly** preserves the longer derivations, assumptions, tests, and model scopes behind this reader. The **historical SFT archive** preserves the original papers, earlier versions, and research packages; its final public contents and links remain a release task.

Part I states the current theory. The reference supports technical examination. The archive preserves the path that produced it. **V14 Part II is the next investigation, not an invisible prerequisite for understanding Part I.**

Sources and scope

[1] *SFT V13.13 — Live Ontology Constitution*, Revision B, §§1–9 and 12–18. Physical interpretation, claim-level inheritance, local standards, open connections, and amendment rule. Original authorship and source credits remain in the preserved primary PDF.

[2] *SFT V13.12 — The Maintained Electron*, public v0.2; C17A optical companion; C17G16 stability audit. Their finite-state and transverse-map results are developed in Section 3, without extending the certificate.

[3] *SFT V13.4-V2 — Relational Interactions*, final v1.0, and controlling *Relational Interaction Update Blend v0.11*, especially §§61, 69–71, 73. Developed in Sections 1, 5, 8–9, and 12.

[4] *SFT V13.9 — How Two Things Become Independent*, replacement v1.1, and *V13.9-V3 Working Relational Model v1.1 SUPER FREEZE*. The binary, pure-state, marking, analyzer, and detector domains remain those of Section 6.

[5] *SFT V13.14 — Formation*, public v0.2, with the response, physical-turn, local-return, and first-mean-loop reports identified in Section 7. First turn and full maintained return retain their separate status.

[6] *SFT V13.3 — First Locally Real Atom* and its R251–R257 package; *SFT V13.6 — Quantum Settlement Rules to Chemistry* and its scripts. Sections 10–11 retain their effective inputs and atomic versus overlap-only scopes.

[7] *SFT V13.1-V3 Gravity*, status revision v1.1 and derivation canon; *SFT V13.1-V4 — Gravity as a Relation Between Two Magnitudes*, v1.4 and reproduction archive. Their distinct branches and source-matching boundary are developed in Section 13.

[8] *SFT V13.2 — Expansion as Global Time Production and Boundary-Capacity Growth from Irreversible Writes*, R250E mechanism freeze, under Revision B and Section 14's explicit interpretations. The retired address mechanism is not restored here.

[9] *SFT V13.10 — Time*; *V13.11 — The Cones and the Ruler*; *V13.8-V2 — There Is No Write Operator*. As developed and qualified in Sections 1–4 and 12.

Closing scope. Equations (16.1)–(16.10) collect existing relations without new physical premises, fits, or solver results. Closing checks cover selected restated algebra, source mapping, preservation of the previous manuscript, and publication integrity. A completed V13 account is not independent review or final publication approval.